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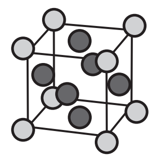
FACE: 1. The front part of the head, bordered by the hairline and the chin and consisting of the eyes, nose, mouth, ears, forehead, cheeks and jaw. The face is particularly concerned with sensibility and communication. It bears the special sense organs of vision, hearing, smell, taste and balance, as well as an especially sensitive skin. Voice and facial expression (fine facial movements) are mediated via the face. Neck mobility allows these organs to be directed quickly and easily toward an object without turning the body. 2. Any of the plane surfaces of a geometrical solid bounded by its edges. In other words, any of the flat surface of a three-dimensional figure, for example, a cube has six square faces. The plane angle formed by adjacent edges of a polygonal angle in space is called a **face angle**. 3. Any subset F of a convex set C that is extreme in the sense that any closed line segment in C with midpoint in F actually lies in F. For example, any subset of a set at which a linear functional achieves its maximum over the given set is a face and is said to be exposed by the functional. A zero-dimensional face is an extreme point. 4. Any of the areas of a connected plane graph, bounded by edges and incident with no edges other than those that bound it. The number of such faces is given in terms of the numbers of edges and vertices by Euler's formula.

FACE ANGLE: The plane angle formed by adjacent edges of a polyhedral angle in space.

FACE-CENTRED CUBIC, F.C.C. (CUBIC CLOSE-PACKED, CCP): A lattice whose unit cells are cubes, with lattice points at the centre of each face of the cube, as well as at the vertices.



Face-centred cube

FACEBOOK: An online social networking website (www.facebook.com) launched in February 2004 that was originally designed for college students, but is now open to anyone, who is 13 years of age or above. Facebook users can create and customise their own profiles with photos, videos and information about themselves. An interesting feature of Facebook is the profile that has a "wall," where friends can post comments. Facebook is the biggest social media network on the Internet, in terms of total number of users. As at August 2, 2018, Facebook had 2.230 billion active monthly users.

FACET: 1. A flat face of a crystal. 2. A lens segment in the compound eye of an insect or other arthropod. 3. A smooth flat area on a hard surface such as a bone or a tooth.

FACIAL: Not appearing on the sides or edges but rather on the face.

FACIAL ARTERY (EXTERNAL MAXILLARY): An artery that originates from the external carotid and branches into the ascending palatine, tonsillar, submental, inferior labial, superior labial, septal, lateral nasal, angular and glandular arteries. The facial artery branches into many smaller blood vessels around the face and oral cavity. These include the tonsillar and glandular branches, as well as the ascending palatine artery, the submental artery and many others. The facial artery delivers oxygenated blood to the regions it serves. Together with the anterior or posterior facial veins, they assist in draining deoxygenated blood from areas of the face, returning the blood to the lungs for oxygenation.

FACIAL NERVE (SEVENTH CRANIAL NERVE): The nerve which inervates all of the facial muscles as well as the taste buds of the tongue. The facial nerve emerges from the posterior border of the pons and reaches the facial muscles through various branches. The facial nerve is important for making facial expressions and to the sense of taste.

FACIAL SHIELD: 1. Face shields are safety devices designed to protect the face from impact hazard such as flying objects and other debris, chemical splashes, etc.; ensuring that visibility and mobility are not hindered. Face shields are commonly used in metalworking, welding, grinding, or cutting machinery applications. 2. In zoology, it is a hard structure found on the forehead of some birds such as the Eurasian coot, which serves as a display ornament.

FACIES: Any body of mineral rock or fossil features with specified characteristics that reflect a particular process or environment in which it is formed.

FACILITATED DIFFUSION: An absorption process in which the transport of molecules across the outer membrane of a living cell involves a specific carrier located within the cell membrane but does not require the expenditure of energy by the cell.

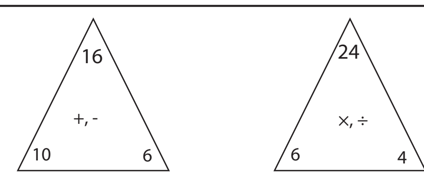
FACILITATION: 1. Also known as **neural facilitation** or **paired pulse** facilitation, it is the effect of successive stimuli on the postsynaptic membrane, which results in the generation of an excitatory postsynaptic potential (EPSP). 2. The phenomenon observed during succession in which the presence of one species increases the likelihood or speed of colonisation by a second species. It is categorised as **mutualisms**, in which both species benefit or **commensalisms**, in which one species benefits and the other is unaffected.

FACSIMILE TRANSMISSION MACHINE (FAX MACHINE): A device used for sending or receiving pictures and texts using a telephone line. Its principle to create a bitmap operation is to scan the document, digitize and transmit the bitmap on standard telephone line. At the receiving end, the bitmap is decoded to recreate the original document.

FACT EXTENSIONS: Calculations involving larger numbers based on knowledge of basic arithmetic facts. For example, knowing the addition fact of 5 × 10 = 50 makes it easier to solve problems such as 50 × 10 = 500 and 500 ÷ 10 = 50. Fact extensions apply to all the four basic arithmetic operations.

FACT FAMILY (NUMBER FAMILY): In mathematics, a set of three numbers that can be added and subtracted, or divided and multiplied together to link two inverse operations. For example, in the arithmetic operations of 10 + 6 = 16, 6 + 10 = 16; or 16 - 6 = 10 and 16 - 10 = 6 is an addition/subtraction fact family. Also, 4 × 6 = 24 or 6 × 4 = 24, 24 ÷ 6 = 4, or 24 ÷ 4 = 6 are a multiplication/division fact family. In the addition/subtraction fact family, the three set of numbers in operation are 10, 6 and 16. In the multiplication/division fact family the three set of numbers in operation are 4, 6 and 24. A fact family helps to understand the inverse relationship, or opposite relationship, between addition and subtraction and between multiplication and division in early stages of education.

FACT TRIANGLE: A triangular flash card labelled with the numbers of a fact family used to practice addition/subtraction and multiplication/division facts. The two 1-digit numbers and their sum or product (marked with a dot) appear in the corners of each triangle.



Addition/Subtraction Fact Triangle **Multiplication/Division Fact Triangle**

FACTOR: 1. Also known as **divisor** or **submultiple**, any integer or polynomial that divides into another number or polynomial exactly. For example, the factors of 24 are: 1, 2, 3, 4, 6, 8, 12 and 24; factors of the polynomial $f(x) = 2x^3 + 3x^2 - 5x - 6$ are $(x + 1)$, $(x + 2)$ and $(2x - 3)$. 2. Any analogous quantity or entity whose product with another is a given quantity or entity, such as any of a set of cycles of which the composition is a given permutation. 3. In statistics, a postulated causal influence derived from and used to explain a cluster of responses. 4. Anything that affects an organism living in a habitat; for example, abiotic and biotic factors. 5. Any substance that functions in a specific biochemical process such as blood coagulation or physiological process such as growth factors.

FACTOR ANALYSIS: In statistics, it is a mathematical method for reducing a large number of measurements or tests to a small number of factors that can completely account for the results obtained on all the tests, as well as for the correlations between them.

FACTOR FORMULAE: Trigonometric identities given by the following:

i. $\sin A+\sin B=2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)$,

ii. $\sin A-\sin B=2\sin\left(\frac{A-B}{2}\right)\cos\left(\frac{A+B}{2}\right)$,

iii. $\cos A+\cos B=2\cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)$, and

iv. $\cos A-\cos B=-2\sin\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right)$.

FACTOR GROUP (QUOTIENT GROUP): The group, denoted G/H, whose members are the cosets of a normal subgroup H in a group G, with the product of two cosets defined as the coset containing the product in the same order of arbitrary representatives of the given cosets. This is an Abelian group when the original group is.

FACTOR IX (CHRISTMAS FACTOR): A protein in blood that plays an important part in the clotting mechanism. A deficiency of factor IX causes a rare genetic bleeding disorder known as Christmas disease, also called haemophilia B.

FACTOR OF A COUNTING NUMBER: A counting number whose product with some other counting number equals n. For example, 2 and 3 are factors of 6 because 2 × 3 = 6. But 4 is not a factor of 6 because 4 × 1.5 = 6, and 1.5 is not a counting number.

FACTOR PAIR: Two factors of a counting number n whose product is n. A number may have more than one factor pair. For example, the factor pairs for 18 are 1 and 18, 2 and 9, and 3 and 6.

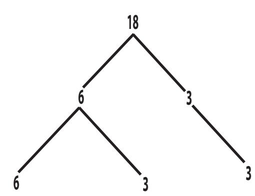
FACTOR RING (QUOTIENT RING; RESIDUE CLASS RING): The ring, denoted R/K, whose members are the cosets of an ideal K in a ring R; these cosets are called residue classes and are equivalence classes whose members differ by a member of K. Since K is an ideal, the sums and products of elements of the factor ring are unique and the cosets of a sum or product of ring elements equals the sum or product (in the same order) of their individual cosets. K is the zero of the factor ring and it has a unit element if R has.

FACTOR SPACE (QUOTIENT SPACE): A space obtained from another by identification of points that are equivalent to one another in some equivalence relation. For example, the factor group G/H of a group G by a normal subgroup H is the set of cosets of H in G and the factor ring R/K of a ring R by an ideal K is the set of cosets of K in R.

FACTOR STRING: A counting number written as a product of two or more of its counting-number factors other than 1. The length of a factor string is the number of factors in the string. For example, 2 × 3 × 4 is a factor string for 24 with length 3. By convention, 1 × 2 × 3 × 4 is not a factor string for 24 because it contains the number 1.

FACTOR THEOREM: A theorem in algebra, which is a special case of the polynomial remainder theorem and shows that a number or expression is the outcome of two or more factors multiplied together. It states that a polynomial $f(x)$ has a factor $(x-k)$ if and only if $f(k)=0$. For example, if $f(x)=x^3+2x^2+3x-6$. then $(x-1)$ is a factor, since $f(1)=(1)^3+2(1)^2+3(1)-6=0$, thus $f(1)=0$.

FACTOR TREE: A special diagram with branches used to find the prime factors of a number, until the factors are all prime numbers of the original number. The factor tree of the integer 18 is shown in the diagram.



Factor Tree of 18

FACTOR V: One of the blood proteins that maintains the balance between the blood clotting too easily or too slowly after an injury. People, who have an inherited mutation in the gene controlling factor V production, known as factor V Leiden, are at increased risk of deep-vein thrombosis, particularly if they take the oral contraceptive pill or undertake long aircraft journeys.

FACTOR VIII (ANTI-HAEMOPHILIC FACTOR): An essential blood-clotting factor that is deficient or absent in people with haemophilia. It is a soluble protein that brings about the conversion of prothrombin to thrombin, which in turn creates fibrin fibres round which a blood clot forms. In humans, Factor VIII is encoded by the *F8* gene. Defects in this gene result in haemophilia A, a recessive X-linked coagulation disorder. It is extracted from blood plasma or genetically engineered and used for treating haemophilia.

FACTORIAL: The function, symbolised ! that computes the product of the first n natural numbers, written $n! = n(n-1)!$. In other words, it is the product of a given number and all the whole numbers below it. For example, the factorial of 0 is written as 0! = 1; 1! = 1; 2! = 2 x 1 = 2; 3! = 3 x 2 x 1 = 6; 4! = 4 × 3 × 2 × 1 = 24.

FACTORIAL EXPERIMENT: An experiment in which all the different factors are combined in all possible ways, but it does not specify the experiment design used to perform such an experiment.

FACTORIAL SERIES: A series of the form, $\sum_{n=1}^{\infty}\frac{n!a_1}{z(z+1)(z+2)...(z+n)}$ $z \neq 0, -1, -2, ...$ It converges or diverges together with the ordinary Dirichlet series $\sum\frac{a_n}{n^z}$ except at $z = 0, -1, -2, ...$.

FACTORION: A natural number that equals the sums of the factorials of its digits in a given base. The only known decimal factorions are 1 = 1!, 2 = 2!, 145 = 1! + 4! + 5! and 40585 = 4! + 0! + 5! + 8! + 5!.

FACTORISATION (FACTORIZATION): Simplification or reduction of an object such as a number or algebraic expression into the parts (or factors) that form the original number or expression when multiplied together. For example, 24 may be factorised as: 2 x 2 x 2 x 3 and $x^2 - 1$ as $(x + 1)(x - 1)$.

FACULA: 1. A bright area on the face of the Sun, commonly seen near an active region such as a sunspot or where such a region is about to form. Faculae, which last on average about 15 days, are best seen in blue light and are not visible at all in H-alpha. They were named by Johannes Hevelius and are thought to be caused by luminous hydrogen clouds close to the photosphere. 2. A bright spot on the surface of a planet or moon.

FACULTATIVE: Possessing the ability to utilise certain circumstances or environmental conditions but not being dependent upon them. For example, a facultative parasite can grow either parasitically or saprobically.

FACULTATIVE ANAEROBES: Microorganisms such as certain bacteria, fungi and some internal parasites of animals that are able to alter their metabolism to grow well under aerobic and anaerobic conditions or in either the presence or absence of oxygen. The best known facultative anaerobe is *Saccharomyces cerevisiae*, the yeast used in brewing.

FAECES (FECES): Waste food materials left over from digestion or solid waste matter discharged from the alimentary canal composed mainly of undigested food, mainly dietary fibre, bile pigments, bacteria, mucus and water. The average amount discharged is about 100 g/day, but varies widely depending on the intake of dietary fibre. Nearly 450 grams of faeces can be produced per day on a high fibre diet. Faeces are formed in the colon, stored in the rectum and eliminated through the anus.

FAGACEAE (BEECH FAMILY): A family of hardwood trees containing about 1000 species but only eight genera. These include Quercus (oaks), Fagus (beeches), Castanea (chestnuts) and Nothofagus (southern beeches), which together make up a large proportion of

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the broad-leaved forests of the world. Perhaps in terms of biomass, this is the most abundant family of flowering plants. Representatives are not, however, found in the equatorial rain forests of Africa and South America. The flowers are unisexual and usually borne in catkins or spikes. The fruit is a nut, which is partly or completely enclosed within a cupule. Certain species of chestnut, especially sweet chestnut (*C. sativa*), are grown for their fruits. However, timber, obviously, is the most economically important product of this family and, on a far smaller scale, certain species of oak yield cork (Q. suber) and tannins. Many species are grown as ornamentals.

FAHRENHEIT SCALE: A scale for measuring temperature, devised by the German physicist Gabriel Daniel Fahrenheit. It was used widely in the English-speaking world until recently. It is now replaced by the SI Celsius temperature scale, except in the USA, where it remains in common use. In the Fahrenheit scale, at standard atmospheric pressure, water boils at 212 degrees Fahrenheit (100 °C or 373 K) and freezes at 32 degrees Fahrenheit (0 °C or 273 K).

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FAHRENHEIT, GABRIEL DANEIL (1686-1736): German physicist, who became an instrument maker in Amsterdam. In 1714 he developed the first thermometer, employing mercury instead of alcohol and devised the Fahrenheit scale. He discovered that other liquids besides water have a fixed boiling point and that these boiling points vary with changes in atmospheric pressure. To convert Fahrenheit temperatures to Celsius, subtract 32 from the Fahrenheit reading, multiply by 5 and divide by 9, i.e., °C = (°F – 32) × $^5/_9$. To convert Celcius into Fahrenheit, multiply the Celsius reading by 9/5 and add 32, i.e., °F = (°C × $^9/_5$) + 32.

FAILURE RATE (FORCE OF MORTALITY; HAZARD RATE FUNCTION): The probability of failure of a component or the death of an organism, within the next unit of time, Δt, given survival to time t. It is defined mathematically as $\Delta t \cdot h(t) = \mathrm{P}[\mathrm{X} \le (t + \Delta t) : \mathrm{X} > t]$.

FAINT (SYNCOPE): A sudden, usually temporary, loss of consciousness, caused by a reduction in the flow of oxygen through the blood to the brain. It can be caused by dehydration, medications, diabetes, pregnancy, anaemia, heart conditions, age and others.

FAIR: An experiment in which any outcome is equally likely; there is no bias. Each side of a flipped coin will land up about equally often: either heads or tails is equally likely.

FAIR TEST: An experiment that is carried out with all the known variables held constant except two.

FAIRING: A section of aircraft surface or an attached structure, designed to reduce drag. An example is the contoured sheet metal where a wing meets the fuselage.

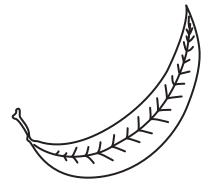
FAIRY RING: A complete circle or arc of fruiting bodies of fungi, particularly agarics, which commonly appear in fields and lawns. Each colony of fungi is derived from a single spore from which the hyphae grow out in all directions forming an invisible circular colony that only becomes visible when fruiting bodies are formed at its periphery.

FAITHFUL: In mathematics, a faithful representation ρ of a group G on a vector space V is a linear representation in which different elements g of G are represented by distinct linear mappings $\rho(g)$.

FAITHFUL MODULE: A module over a unit ring for which the annihilator A = $\{r \in \mathrm{R} : rx = 0$ for all $x \in \mathrm{M}\}$ is zero, where M is the module and R the ring.

FAJANS' RULES: Rules introduced by the Polish-American chemist Kasimir Fajans (1887-1975) indicating the extent to which an ionic bond has covalent characteristics caused by polarisation of the ions depending on the charge on the cation and the relative sizes of the cation and anion. Covalent character is more likely if: (i) the charge of the ions is high; (ii) the positive ion is small or the negative ion is large; (iii) the positive ion has an outer electron configuration that is not a noble-gas configuration.

FALCATE: Curved like a sickle or the beak of a falcon, as seen in the diagram.



Falcate leaf

FALCIFORM: Sickle shaped.

FALL-OUT: 1. Also known as **radioactive fall-out**, it is harmful radioactive material released into the atmosphere in the debris of a nuclear explosion or from a nuclear release accident that descends and subsequently deposits on the Earth. 2. In migratory birds, **fallout** is a term used to refer to birds that are forced by circumstances such as harsh weather to land in areas they would not normally inhabit.

FALLACY: An argument that is invalid or a false conclusion drawn from a set of statements called premises.

FALLOPIAN TUBES (OVIDUCT; UTERINE TUBES): The two tubes lined with ciliated epithelia, leading from the ovaries of female mammals into the uterus that carry egg cells from the ovary to the womb or uterus in mammals.

FALLOW: Agricultural land which is not used for the production of crops but is left unused for some time in order to restore its natural fertility.

FALLS: 1. Also known as **waterfall**, it is a place on the long profile of a river, where water makes a sudden vertical drop, making a stream of water. It occurs where a river or stream falls over the edge of a steep place. An example is the Niagara Falls in the U.S., with a drop of 55 m. 2. Sepals of an iris flower.

FALSE: One of the distinct truth-values of a two-valued logic; a unique anti-designated truth value.

FALSE NUMBER SENTENCE: In mathematics, a number sentence that is not true, because both numerical expressions do not evaluate to the same number. Examples are 12 = 6 + 5, since 6 and 5 do not add up to 12;and 5 < 12, since 12 is not less that 5.

FALSE POSITION (RULE OF FALSE POSITION; REGULA FALSI): A method of solving equations by using trial and error values for the variables and then adjusting the values accordingly. For example, in solving $x^2 + y^2 = 50$, test values like $x = 5$, $y = 6$ might be attempted, which gives $x^2 + y^2 = 25 + 36 = 61$, which is more than 50; therefore smaller values for the variables would have to be guessed until the equation is solved. There are two types of false position methods: (i) **simple false position** for solving problems involving direct proportion that can be written algebraically in the form $ax = b$, for the value of x to be found if the values of a and b are known. (ii) **Double false position** is used for solving more difficult problems that can be written algebraically in the form: $f(x) = b$ for the value of x to be determined, if it is known that $f(x_1) = b_1$ and $f(x_2) = b_2$.

FALSE VACUUM: A state in quantum field theory that is a local minimum but not the lowest energy state of the system overall, even though it may remain stable for some time. Tunnelling occurs from the false vacuum to the true vacuum.

FALSE VEINS: Small vein-like areas of thick-walled cells in the leaves of some lower vascular plants.

FALSE-COLOUR IMAGERY: A graphic technique that displays images in false colours that differ from those a full-colour photograph would show so as to enhance certain features.

FALSE-MEMORY SYNDROME (FMS): A psychological condition occurring in some people, in which the individual has a false memory and strongly believes in certain events that actually never occurred. For example, the patient may recall having been sexually abused as a child. This can be destructive to the individual as he/she avoids discussing issues that challenges his/her belief. FMS is controversial as it is very difficult to determine whether memories are false or genuine.

FAMILY: 1. A collection of things or entities grouped by their common attributes, for example, protein family and gene family. 2. A taxonomic rank fitting between order and genus. It may be divided into one or more subfamilies, ranking below a family and above a genus. A group of living organisms belonging to the same family consisting of one or several of closely related genes would have evolved from the same ancestors and share relatively common characteristics. 3. In mathematics, a term used for a collection of objects, especially a set of subsets of a topological space.

FAMILY OF CONFOCAL CENTRAL CONICS: The family of curves given by $\frac{x^2}{a^2+\lambda} + \frac{y^2}{b^2+\lambda} = 1$. The differential equation of this family of plane curves is $(xy' - y)(yy' + x) = (a^2 - b^2)y'$.

FAMILY OF CURVES: A set of similar curves that display a particular geometric pattern and whose equations can be obtained by varying a finite number of parameters in a particular general equation. For example, $y = 2x^2 + c$, where c is a constant representing family of curves.

FAMILY PLANNING: Regulating and spacing of children within a family and also deciding when to have children. It is achieved through use of contraceptive methods and the treatment of involuntary infertility. The benefits include: (i) to secure the well-being and autonomy of women, while supporting the health and development of communities; (ii) preventing pregnancy-related health risks in women; (iii) reducing infant mortality; (iv) reducing adolescent pregnancies; (v) slowing population growth; (vi) helping to prevent HIV/AIDS, when condoms are used.

FAMINE: A severe scarcity of food, usually accompanied by regional malnutrition, starvation, epidemic and increased mortality that may apply to any faunal species.

FANG: Long, pointed tooth located in the front of the mouth. In mammals, these fangs are called canine teeth and are used for tearing flesh. In snakes, fangs have hollow groove through which venom is injected into the victim.

FANJET (TURBOFAN): A type of aircraft gas turbine engine used by most airlines. It provides thrust using a combination of a ducted fan and a jet exhaust nozzle.

FANNING: In blast furnace operation, the idling period between the blowing periods when the blast pressure is reduced to a minimum. This also applies to plume behaviour under inversion conditions.

FAQ (FREQUENTLY ASKED QUESTIONS): A group of commonly asked questions about a subject along with the answers a user may have about a certain software program or that a newcomer to a Web site might have regarding the site. Web sites will often refer visitors to an FAQ before asking them to e-mail their questions, which helps cut down on technical support.

FAR POINT: The farthest point from the eye at which an image is clearly focused on the retina when accommodation of the eye is completely relaxed; the eye can clearly see an image at that point. The eye is unable to focus a sharp image on the retina of an object beyond this point. The far point for a normal eye should be at infinity; with respect to the normal eye, this is equivalent to any distance that is more than 6.1 metres. An eye that has a far point nearer than this is short-sighted.

FAR-RED LIGHT: Electromagnetic radiation of wavelength of between 710 and 850 nm. It has been shown that far-red light inhibits many physiological processes in plants, such as the germination of lettuce seeds. It also counteracts the effect of red light. For example, if an etiolated plant is exposed to short periods of red light this results in more normal growth. However, if the periods of red light are followed by periods of far-red light the plant remains etiolated. It is believed that far-red light is absorbed by a pigment, phytochrome, which mediates such responses.

FARAD: The SI unit of capacitance, named after the English physicist, Michael Faraday. A capacitance of 1 farad is defined as a capacitor that carries a charge of 1 coulomb when a potential difference of 1 volt is applied across it. Its symbol is F. $1F = 1CV^{-1}$.

FARADAIC CURRENT: Electric current flowing through an electrode that corresponds to the electrolytic oxidation or reduction of one or more chemical species. The net faradaic current is the algebraic sum of all the faradaic currents flowing through an indicator or working electrode.

FARADAIC CURRENT DENSITY: Measures of the rate of the interfacial reaction, the proportionality constant being the charge number of the reaction multiplied by the Faraday constant.

FARADAIC DEMODULATION CURRENT: Component of the current that is due to the demodulation associated with an electrode reaction and that appears if an indicator or working electrode is subjected to the action of two intermodulated applied potentials of different frequency.

FARADAIC RECTIFICATION CURRENT: A component of the current that is due to the rectifying properties of an electrode reaction and that appears if an indicator or working electrode is subjected to any periodically varying applied potential while the mean value of the applied potential is controlled.

FARADAY: A unit of electrical charge which is equal to the charge on 1 mole of an electron or that is capable of depositing or liberating 1 gram equivalent weight of a substance in electrolysis. Its value is 9.468×10^4 coulombs.

FARADAY CAGE (FARADAY SHIELD; HOFFMAN BOX): An earthed screen made of metal wire that surrounds an electric device in order to shield it from external static electric fields. It was named after the English scientist Michael Faraday, who invented it in 1836.

FARADAY CONSTANT: The magnitude of electric charge carried by one mole of electrons or singly ionised ions that is the product of the Avogadro constant and the charge on an electron. Its symbol is F and it has the value 9.648670×10^4 coulombs per mole.

FARADAY CUP: A hollow collector that is opened at one end and closed at the other used to collect beams of ions. In its simplest form it consists of a conducting metallic chamber or cup, which intercepts a particle beam. An electrical lead is attached which conducts the current to a measuring instrument. Detection can be as simple as an ammeter in the conducting lead to ground or a voltmeter or oscilloscope displaying the voltage developed across a resistor from the conducting lead to ground. A bias voltage applied either to the cup itself or a repelling grid preceding the cup or a magnetic field, is usually used to prevent secondary emission from distorting the reading.

FARADAY EFFECT (FARADAY ROTATION): The rotation of the plane of polarisation of electromagnetic radiation on passing through an isotropic medium exposed to a magnetic field or a plane-polarised microwave passing through a magnetic field along the lines of that field.

FARADAY, MICHAEL (1791-1867): British chemist and physicist, who received little formal education and best known for his discoveries of electromagnetic induction and of the laws of electrolysis. He was apprenticed to a bookbinder in London and read books on scientific subjects and experimented with electricity. In 1812, he attended lectures by Sir Humphrey Davy at the Royal Institution and a year later he became his assistant. He remained at the Institution until 1861. Faraday's chemical discoveries include the liquefaction of chlorine (1823) and benzene (1825) as well as the laws of electrolysis. In 1821, he demonstrated electromagnetic rotation (the principle of the electric motor) and in 1832 discovered electromagnetic induction (the principle of the dynamo). Faraday was elected to the Royal Society in 1824 and the following year was appointed director of the laboratory of the Royal Institution. In 1833, he succeeded Davy as professor of chemistry at the institution. In 1845, he discovered the Faraday Effect, the rotation of the polarisation plane of plane-polarised light when passing through particular substances, such as quartz, in a strong magnetic field.

FARADAY'S LAWS OF ELECTROLYSIS: Faraday's 1st Law of Electrolysis states that for a given quantity of electricity (electric charge), the amount of chemical change during electrolysis is proportional to the charge passed or the mass of an element discharged at the electrodes during electrolysis is directly proportional to the quantity of electricity passing through the electrolyte. Faraday's 2nd Law of Electrolysis states that for a given quantity of electricity (electric charge), the charge required to deposit or liberate a mass, m is given by Q = Fz/m, where F is the faraday constant, z the charge of the ion and M the relative ionic mass.

FARADAY'S LAW OF ELECTROMAGNETIC INDUCTION: A law which states that whenever the magnetic flux linking a coil is charged, electromotive force (emf) is induced in the coil whose magnitude is proportional to the rate of change of the magnetic flux.

FAREY SEQUENCE: A sequence of fractions in the form a/b arranged in their lowest terms in order of increasing size and either a or b is not less than 1; i.e. $1 \le a \le n$ *and* $1 \le b \le n$. It was named after John Farey (1766-1826), Englsh mathematician and engineer.

FARINALEOUS: Containing starch or having mealy appearance.

FARINOSE: Covered with a powdery substance or mealy dust.

FARKAS' LEMMA: The result that a linear inequality $(f_0, x) \le 0$ is a consequence of a system $(f_1, x) \le 0, \dots, (f_n, x) \le 0$, if and only if there are non-negative real numbers $\lambda_1, \dots, \lambda_n$ with $\sum_{k=1}^{n} \lambda_k f_k = f_0$.

The result holds with certain of the inequalities replaced by equalities if the corresponding multipliers are now arbitrary real numbers. This is the key result underpinning the Kuhn-Tucker theorem or linear programming duality.

FARM GATE VALUE: Value of an agricultural or aquaculture product when it leaves the farm excluding the costs of transporting from the farm gate to the nearest market or first point of sale and market charges.

FARROW: To give birth to a litter of young pigs.

FARROWING CRATE (GESTATION CRATE; SOW STALL): A special metal cage or pen used in intensive pig farming, in which a sow may be confined during pregnancy and most of her adult life and where it gives birth.

FARSIGHTEDNESS (HYPERMETROPIA; HYPEROPIA; LONG-SIGHTEDNESS): An abnormal condition of the eye in which vision is better for distant objects than for near objects. It results from the eyeball being too short from front to back, causing images to be focused behind the retina. As an object moves toward the eye, the eye must increase its optical power to keep the image in focus on the retina. If the power of the cornea and lens is insufficient, the image will appear blurred.

FARTHEST POINT: In mathematics, it is a point that is not in a given subset of a given metric space with a maximum distance from any point in the subset.

FAS SIGNAL PATHWAY: An intracellular pathway that initiates the process of programmed cell death (apoptosis) in most multicellular organisms. FAS is a cell surface receptor, when it binds its specific extracellular signal molecule, FAS ligand. The intracellular portion of the FAS protein changes shape, enabling it to bind adaptor proteins and recruit components of apoptosis, notably caspase enzymes, which act as executioners inside the cell.

FASCIA: A sheet of fibrous uninterrupted, three-dimensional web of tissue that extends from head to toe, from front to back, from interior to exterior. It occurs beneath the skin and also enveloping glands, vessels, nerves and forming tendon sheaths.

FASCIATED: Abnormally coalesced into a bundle, bearing flattened and elongated flower heads with numerous flowers. This may be a result of random genetic mutation, a bacterial or viral infection or damage to plants by frost. An example is *Euphorbias*.

FASCICLE (VASCULAR BUNDLE): A long continuous strand prominent in annual and young plants, which is a part of the transport system in vascular plants containing the xylem and phloem vessels, responsible for conducting sap through the stems and branches of a plant. In perennial and woody plants, they become part of an inner cylinder of vascular tissue.

FASCICULAR CAMBIUM (INTRAFASCICULAR CAMBIUM): The part of the vascular cambium that originates within the vascular bundles (fascicles) between the xylem and phloem.

FASCICULATION: 1. A brief involuntary contraction and relaxation of a group of muscle fibres visible under the skin arising from the spontaneous discharge of a bundle of skeletal muscle fibres. 2. The process of becoming organised into a bundle, as in the bundling of axons to form a nerve.

FASCIOLIASIS: A disease caused by an infestation of the liver and bile ducts by parasitic liver flukes of the family Fasciolidae.

FAST ATOM BOMBARDMENT IONISATION: This term is used to describe ionisation of any species by causing interaction of the sample (which may be dissolved in a solvent matrix) and a beam of neutral atoms having high translational energy.

FAST BREEDER (FAST BREEDER REACTOR, FBR): A type of nuclear reactor, in which the chain reaction is maintained mainly by fast neutrons having energy in excess of 0.1 Mev (1.6 x 10^{-4} J). This is sufficient to produce fission in uranium 238 (U-238) and breed fuel by producing more fissile material than it consumes. The reactors are used in nuclear power plants to produce nuclear power and nuclear fuel. FBRs use a mixture of plutonium oxide (PuO_2) and uranium oxide (UO_2) as fuel. In order to reach self sustaining nuclear chain reaction, the fuel used for FBR must have a higher enrichment in U-235. The breeder reaction happens, when U-238 in the fuel captures a fast neutron to form fissible Pu-239, which can be reprocessed for use as nuclear fuel.

FAST FOURIER TRANSFORM (FFI): An efficient algorithm, developed by James Cooley and John Tukey in 1965 for computing the discrete Fourier transform (DFT) of a set of data values (it breaks the DFT into smaller DFTs). Its uses include in digital signal processing, communication and high-performance computing; and applied for tasks such as spectral analysis, image compression, interpolation, solving partial differential equations, etc.

FAST GREEN (FAST GREEN FCF; FOOD GREEN 3; SOLID GREEN FCF): A green dye, with the chemical formula $C_{37}H_{37}N_2O_{10}S_{3+}$ and molar mass 765.89 g/mol. It is used in optical microscopy to reveal plaque deposits on teeth, cytoplasm, collagen and mucus green.

FAST NEUTRON: A neutron resulting from a nuclear fission that has lost little of its energy by collision and has energy in excess of 0.1 Mev (1.6 x 10^{-4} J), sufficient to produce fission in uranium 238.

FAST REACTOR: A nuclear reactor in which little or no moderator is used, the nuclear fissions being caused by fast neutrons.

FAST-ATOM BOMBARDMENT MASS SPECTROSCOPY (FAB MASS SPECTROSCOPY): A method in which ions are produced in a mass spectrometer from non-volatile or thermally fragile organic molecules by bombarding the compound in the condensed phase with energy-rich neutral particles.

FAST-ION CONDUCTOR (SUPERIONIC CONDUCTOR; SOLID ELECTROLYTES): A solid conductor of electricity in which the moving particles are ions, which transport electric charge by moving rapidly between vacancies or empty crystallographic positions in the lattice of a crystal. It is used in batteries and fuel cells.

FASTIGIATE: Having an erect and parallel branches tapering upwards, thereby giving it a broom like shape.

FAT: A wide group of compounds, either solid or liquid at room temperature that are generally soluble in organic solvents and largely insoluble in water. Fats are usually esters made from glycerol and fatty acids. They have about twice the energy content of carbohydrates. It is stored in the adipose and used as a source of food energy, insulate animals against heat loss and form a protective cushion around organs.

FAT 32 (32-BIT FILE ALLOCATION TABLE): A disk file allocation system from Microsoft, used on many computer systems and most memory cards that uses 32-bit values for FAT entries instead of 16-bit values used by the original FAT system, enabling partition sizes up to 2TB (terabytes). FAT 32 first appeared in Windows 95b and is also found in Windows 98 and Windows NT 5.0.

FAT BODY: 1. An abdominal fatty tissue in amphibians attached to the anterior of each kidney. It contains a reserve of fat that nourishes the gonad during the winter hibernation in readiness for the spring breeding season. 2. A food reserve of fatty tissue spreading throughout the body cavity of insects in which fats, proteins and glycogen are stored as a reserve for hibernation or pupation.

FAT CELL (ADIPOCYTES; LIPOCYTES): Any of the cells of adipose tissue, comprising white adipose tissue (WAT or white fat) and brown adipose tissue (BAT or brown fat) specialised in storing energy as fat. Fat cells contain enzymes that can break down fat into glycerol and fatty acids, which can be transported in the blood to the liver, where they are used in fatty-acid oxidation.

FATAL EXCEPTION ERRORS (FATAL OE): A software error that is returned by a program in the following cases: when a software program attempts to access an illegal instruction; when invalid data or other code has been accessed; or in the case of operational invalid privilege level. Depending upon the severity of the error, the exception is not recoverable and the system must either be restarted or shut down.

FATE MAP: A diagram or chart that traces the developmental path of adult structures from the different parts of the embryo and what it will become during normal development. It was developed by Walter Vogt as a means of tracing the development of specific regions of the early embryo.

FATHOM: A unit of length, used to express depth of water. It is equal to 1.83 m.

FATIGUE: 1. In metallurgy, it is a cumulative and localised structural damage that occurs in a metal after repeated applications of stress, none of which exceeds the ultimate tensile strength. Fatigue occurs when a material is subjected to repeated loading and unloading and above a certain threshold, microscopic cracks form at the surface. In a photochromic system, it is a loss of performance over time, due to chemical degradation. The major cause of fatigue is oxidation. 2. Also called **tiredness**, **exhaustion**, **lethargy** or **listlessness**, it is an extreme physical and or mental tiredness.

FATOU'S LEMMA: The theorem that establishes an inequality relating the integral (in the sense of Lebesgue) of the limit inferior of a sequence of functions to the limit inferior of integrals of these functions. If $\{f_n\}$ is a sequence of non-negative measurable functions, then $\int \liminf_{n\to\infty} f_n \, d\mu \le \liminf_{n\to\infty} \int f_n \, d\mu$. The theorem can be used to prove the Fatou–Lebesgue theorem and Lebesgue's dominated convergence theorem.

FATTY ACID: A substance, with the general molecular formula C_nH_{n+1}.COOH. In other words, it is a carboxylic acid having a long hydrocarbon chain derived from plants or animals. It is found especially in animal and vegetable fats and oils, made up of a long straight hydrocarbon chain compound, consisting of a hydrocarbon and a terminal carboxyl group, one of the end products of fat digestion. Most naturally occurring types have a chain of an even number of carbon atoms, from 4 to 28. Fatty acids are usually derived from triglycerides or phospholipids. Fatty acid is an important source of energy for many organisms. The lowest members are formic acid (HCOOH) and acetic acid (CH_3COOH).

FATTY ACID METABOLISM: Breakdown or synthesis of fatty acids. In plants, fatty acids can be broken down to yield carbon dioxide, water and energy but in many lipid-metabolising tissues, especially germinating oil-rich seeds, fatty acids are metabolised to sugars. The major route for fatty acid breakdown is the β-oxidation cycle, which degrades fatty acids to acetyl CoA. The acetyl CoA can then either undergo further oxidation or can be converted into sugar via the glyoxylate cycle. The enzymes responsible for fatty acid synthesis aggregate in synthetase complexes, which are found, associated with the various membrane-bound organelles of the cell. Three different types of synthetase complex have been identified, each with a different specificity.

FATTY LAYER (ADIPOSE TISSUE; ADIPOSE LAYER; SUBCUTANEOUS TISSUE): A layer of cells found chiefly below the skin and around major organs such as the kidneys, heart and the gut made up of fat and oils which act as energy reserve and also providing insulation against heat loss.

FATTY-ACID OXIDATION (β-OXIDATION): The metabolic pathway in which fats are metabolised to release energy, occurring mainly in the mitochondria in animal cells and in peroxisomes in plant cells. It consists of catabolic processes that generate energy and primary metabolites from fatty acids and anabolic processes that create biologically important molecules from fatty acids and other dietary carbon sources. Fatty-acid oxidation occurs continually but does not become a major source of energy until the animal's carbohydrate resources are exhausted, for example during starvation.

FAUCES: The arched opening at the back of the mouth leading to the pharynx, surrounded by the glossopalatine arch, which forms the anterior pillars of the fauces and the pharyngopalatine arch (the posterior pillars).

FAULT: 1. Also known as shift, is compressional or tensional forces causing a fracture or break in the Earth's crust, where a very large mass of rock splits with the sides moving or sliding past each other. Most earthquakes are caused by rapid movement along faults. Usually, faults occur together in large numbers, parallel to each other or crossing each other at different angles. 2. In electricity, any defect in an electric circuit, resulting in an interruption of current flow. The fault may be caused by a defective component in the circuit or poor insulation between conductors causing short-circuit or partial short-circuit, between an energized conductor and ground.

FAULT GOUGE: In geology, it is soft, unconsolidated, pulverized clay-like material found between the walls of a fault produced by the movement of the rocks on each side of a fault.

FAULT SCARP: A steep slope or cliff formed directly by movement along a fault and representing the exposed surface of the fault before modification by erosion and weathering.

FAULT TOLERANCE: The ability of a computer system to continue operation despite minor hardware faults to satisfactory level.

FAULT-BLOCK MOUNTAINS: Mountain range formed when sections of sedimentary rocks are tilted upward in sections and others collapse down caused by forces in the Earth's crust.

FAUNA: The animal life of a region, habitat or geological period.

FAUNAL REGION: A region of the Earth with distinct and characteristic geographic and environmental barriers, to which certain animal communities abound.

FAVEOLATE: Fitted with cell-like cavities; shaped like a honey comb.

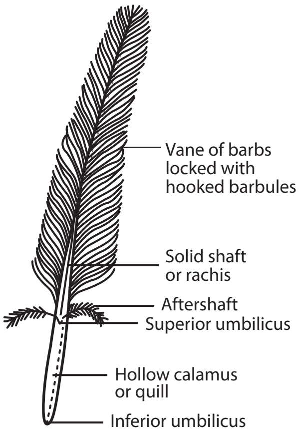
FAVOURABLE OUTCOME: An outcome that satisfies the conditions of an event of interest. For example, if a 6-sided die is rolled and the event of interest is "roll an even number"; there are six possible outcomes: roll 1, 2, 3, 4, 5, or 6. Of these, 3 are favourable: roll 2, 4, or 6.

FAVORITES: A term used by Microsoft for bookmarked Web pages, which is a collection of direct links to predefined web pages which is stored in the web browser. Favorites are used in browsers such as Internet Explorer and Opera and can be created and managed by the user. Most Web browsers store saved webpage locations as bookmarks.

FDDI (FIBRE-OPTIC DIGITAL DEVICE INTERFACE): In computing, it is a series of American National Standards Institute (ANSI) network protocols for transmitting digital data using fibre optic cable. It is capable of supporting data rates of up to 100 Mbps (100 million bits) per second. FDDI is usually used as the backbone for a wide area network (WAN).

FEASIBLE SET: 1. Of a constrained optimisation problem, possessing consistent constraints. In other words, having a non-empty feasible set. 2. Of a point lying in a prescribed feasible set. **Feasible set** is the set of all possible points that satisfy the restrictions or constraints, of an optimisation problem including inequalities, equalities and integer constraints.

FEATHERS: Exoskeletal structures of birds which are outgrowths of the epidermis or outer layer or plumage, made of the protein keratin. They are useful in flight, thermal insulation, waterproofing and colouration that help in communication and protection. Feathers consist of a shaft, called the **rachis** and the vanes on either side. Vanes are made of barbs that are arranged side by side up the shaft of the feather. Barbules grow from the barbs, which have tiny hooks that interlock. The short bare part at the base of the shaft is called the **calamus**. Some types of feathers include the following: (i) **Contour feathers**, which are small feathers that cover the body, wings and tail; (ii) **Flight feathers**, which are used in flight; (iii) **Down feathers**, which are soft under-feathers that provide insulation for warmth; and (iv) **Filoplumes**, which are specialised kind of bristle feathers with many nerve endings that act as sensory organs, registering pressure and vibration. They help keep feathers in place and adjust them for flight, insulation and bathing.



Flight feather (quill)

FECUNDITY: The ability or rate at which an organism reproduces or the fertility of an organism or the actual reproductive rate of an organism or population, measured by the number of gametes or eggs, seed set or asexual propagules. In higher animals, it refers generally to the female of the species.

FEDERAL COMMUNICATIONS COMMISSION IDENTIFICATION NUMBER (FCC ID): A unique code found on almost every piece of computer equipment, usually printed on the circuit board that identifies the manufacturer or other useful information such as the product's age and the part number. The Federal Communications Commission (FCC) is the regulatory authority in the United States for the wireless industry. FCC is recognised in many other countries. Almost every piece of computer equipment has a FCC ID number. The FCC identification database has a section, where FCC ID can be accessed at http://www.fcc.gov/oet/ea/. This is sometimes necessary to obtain the proper driver for the reuse of old but functioning hardware such as unmarked modems, graphics cards, drive interface cards and other devices.

FEED: 1. To give food, especially to animals or babies or a meal taken by an animal or baby. 2. The part of a machine through which it is supplied with power or fuel.

FEED ADDITIVE: A non-nutritive substance that is not obtained from regular meals added to feed to improve growth, increase feed efficiency or to maintain health.

FEEDBACK: 1. A procedure used in self-regulating control systems in which the effect of the output of a system is routed back to the input. It is important in nature in living and non-living things, in technology and in several endeavours of human life. There are two types of feedback: positive feedback and negative feedback. **Negative feedback** is a process that slows down or completely stops the output of a system. In biology, examples are the regulation of hormone synthesis, blood glucose levels, body temperature and blood pressure regulation. In mechanics, an example is found in temperature-controlled devices such as a heater in which a thermostat controls the temperature by either turning the heater on or off. **Positive feedback** is a process in which part of the output is used to increase or amplify the input. In human anatomy, the body senses a change and activates mechanisms that accelerate or increase that change. For example, when contraction occurs in labour, oxytocin (a hormone that is made in the hypothalamus) is released into the body making the uterus contract more frequently and with greater strength. 2. An unwanted noise such as whistling caused by an amplified or broadcast signal or a loudspeaker to the microphone that has been returned as input and retransmitted.

FEEDBACK INITIATION: A cellular control mechanism in which an enzyme's activity is inhibited by the end product of the biosynthetic pathway involved in its synthesis in order to control output.

FEEDBACK INHIBITION: A metabolic control mechanism in which the end product of a biochemical sequence is able to inhibit the activity of an early enzyme in the sequence, thereby controlling the metabolic flux through this pathway. An example is the inhibition by isoleucine of the enzyme threonine dehydratase, the first enzyme in the biosynthetic pathway from threonine to isoleucine. The opposite of feedback inhibition, **feedforward stimulation**, occurs when the first substrate of a reaction sequence stimulates subsequent reactions in the sequence. Variations of simple feedback inhibition are found in branching pathways.

FEEDBACK SYSTEMS: In biology, it is the control mechanisms, in which an increase or decrease in the level of a particular factor inhibits or stimulates the production, utilisation or release of that factor. It is important in the regulation of enzyme and hormone levels, ion concentrations, temperature and many other factors.

FEEDER: 1. A stream or river that joins the flow of a larger one. 2. An animal that is fattened for sale or slaughter. 3. A transmission line between an antenna and a transmitter.

FEEDLOTS: An area in which large numbers of animals are kept and fed and fattened for food.

FEEDSTUFF: Any edible material used for animal feed.

FEHLING'S TEST: A chemical test devised by the German chemist, H. C. von Fehling (1812-85) using Fehling's solution to determine whether an organic substance is aldehydes and sugars. Fehling's solution consists of Fehlings A (copper(II) sulfate solution) and Fehling's B (alkaline 2,3-dihydroxybutanedioate (sodium tartrate) solution), with equal amounts being added to the test solution. After boiling, a positive result is indicated by the formation of a brick-red precipitate of copper(I) oxide. Methanal, being a strong reducing agent, also produces copper metal; ketones do not react. Its uses include testing for glucose in urine to detect diabetes; in conversion or breakdown of starch to glucose syrup and maltodextrins, to measure the amount of reducing sugars and calculating the dextrose equivalent (DE) of the starch sugar. The test has been replaced by Benedict's test.

FEIT-THOMPSON THEOREM: The theorem of group theory that asserts that every finite simple non-abelian group has even order. In other words, there is no non-abelian finite simple groups of odd order; or every finite group of odd order is solvable. It was named after mathematicians Walter Feit and John G. Thompson.

FEJER'S THEOREM: It states that the arithmetic means of the partial sums of the Fourier series of any periodic function continuous of the interval $[-\pi, \pi]$ are uniformly convergent to the functions.

FELDSPAR: One of a large group of rock-forming minerals composed of aluminosilicates of potassium, sodium, calcium or occasionally barium occurring principally in igneous, plutonic and some metamorphic rocks. About 60 percent of the Earth's outer crust is composed of feldspar. All the feldspars weather readily to form a type of clay known as **kaolin**.

FELDSPATHOID: Any of a group of alkali aluminosilicate minerals such as nepheline and leucite resembling feldspars but containing less silica; or containing chloride, sulfide, sulfate or carbonate. Feldspathoids are found mainly in igneous and metamorphic rocks and are used as raw materials in the production of alum, glass and ceramics. Feldspathoid also includes kalsilite, sodalite and nosean.

FELSIC: Referring to silicate minerals, magma and rocks or any light coloured silicate material enriched in the lighter elements such as silicon, oxygen, aluminium, sodium and potassium. Examples of felsic minerals include quartz, muscovite orthoclase and the sodium-rich plagioclase feldspars. Granite is the most common felsic rock.

FEMALE: 1. The sex that gives birth to young or belonging to the sex that produces sex cells gametes that fuse with male sex cells during sexual reproduction. 2. Describing the part of a plant such as a carpel that produces the female sex cells. 3. Describing flowers that have carpels but no stamens. 4. Describing a component or part of a component such as an electric socket that has a recess designed to receive a corresponding projecting part.

FEMALE CIRCUMCISION (CLITORIDECTOMIES): The excision of the labia minora and/or clitoris, carried out in some societies in Africa and the Middle East that is usually performed as part of female initiation rites. It is also sometimes done to curb sexual desire in females.

FEMALE REPRODUCTIVE SYSTEM (FEMALE GENITAL TRACT): The organs of the female reproductive system, which include the ovaries, Fallopian tubes (oviducts), uterus, vagina, cervix, accessory glands and external genital organs. They produce and sustain the female sex cells (egg cells or ova), transport these cells to a site where they may be fertilised by sperm, provide a favourable environment for the developing foetus, move the foetus to the outside at the end of the development period and produce the female sex hormones.

FEMORAL ARTERY: A major blood vessel that supplies oxygenated blood to the leg. The femoral artery is the direct continuation of the external iliac artery of the abdomen and is the great arterial trunk of the leg. It begins behind the inguinal ligament, at the mid-inguinal point and it descends through the upper two-thirds of the thigh to the opening in the adductor magnus, through which it passes into the popliteal artery.

FEMORAL NERVE: One of the main nerves of the leg. It is made up of fibres from nerves in the second, third and fourth segments of the lumbar spinal cord. The nerves emerge from the lower back region of the spine and run down into the thigh, where they branch to supply the skin and muscles of the front of the thigh. The nerve branches that supply the skin supply sensation from the front and inner side of the thigh. The branches that supply the muscles stimulate contraction of the quadriceps muscle at the front of the thigh, causing the knee to straighten.

FEMTO: A prefix used in the metric system for physical units to denote a factor of 10^{-15}, for example, femtosecond. Its symbol is *f*.

FEMTOCELL: A low-powered wireless access point connected to a mobile operator's network via a standard broadband DSL or cable service that improves cellular reception inside a home or office building.

FEMTOCHEMISTRY: The science that studies chemical reactions using lasers that produce very short-duration pulses of light approximately 10^{-15} seconds (one femtosecond). The pulses are produced using dye lasers or using solid-state lasers made from titanium-doped sapphire. Femtosecond lasers can be used to study the breaking and formation of individual chemical bonds in compounds.

FEMTOMETRE (FERMI): A unit of length, with the symbol fm. It is equal to 10^{-15} metre and used in measuring nuclear distances.

FEMUR (THIGH-BONE): 1. The thigh bone or the bone closest to the body of terrestrial vertebrates. It articulates at one end with the pelvic girdle at the hip joint and at the other (via two condyles) with the tibia. In tetrapod (four-limbed) vertebrates such as dogs and horses, the femur is found only in the rear legs. Anything that relates to the thigh or femur (thigh bone) is described as **femoral**. 2. The third segment of an insect's leg, attached to the trochanter.

FENCHEL CONJUGATE: Of a convex function on a normed space, X, the convex function defined on the dual Banach space X* by the formula $f^*(y) = \sup\{y(x) - f(x) : x \in X\}$, where $f(x)$ is the given convex function.

FENCHEL'S DUALITY THEOREM: One of the central duality results in convex optimisation theory that if $f : X \to]-\infty,\infty]$ and $g: X \to]-\infty, \infty]$ are respectively a convex and a concave functions, then, if there is a point at which one function is continuous and the other is finite, $\inf_x f(x) = \max_{x^*}\{g^*(x^*) - f^*(x^*)\}$, where f^* is the Fenchel conjugate of f and $g^*(g^*) = -(-g)^*(-x^*)$.

FENESTRA: 1. Either of the two oval and round membranous windows between the middle ear and the inner ear. 2. Small pores in endothelial cells that allow for rapid exchange of molecules between sinusoid blood vessels and surrounding tissue. They may enlarge and contract at the action of various stimuli such as noradrenaline. 3. A transparent spot or marking, as on the wing of a moth or butterfly.

FENESTRA OVALIS (OVAL WINDOW; VESTIBULAR WINDOW): A membrane-covered opening between the middle ear and the vestibule of the inner ear to which the base of the stapes is connected and through which the ossicles of the ear transmit sound vibrations to the cochlea. Vibrations coming into contact with the tympanum are transmitted through the middle ear by the three ear ossicles and transmitted to the inner ear by the oval window which is connected to the third ear ossicle. Vibrations from the tympanum reaching the oval window are amplified over twenty times.

FENTON'S REACTION: A free radical reaction employed in organic chemistry, where hydroxyl radicals are generated using iron(II) sulfate and hydrogen peroxide. It was developed in the 1890s by Henry John Horstman Fenton. Fenton's reagent is used to oxidise contaminants or waste waters and used to destroy organic compounds such as trichloroethylene (TCE) and tetrachloroethylene (PCE).

FERAL ANIMAL: Any domesticated animal, such as a dog or a cat that has returned partly or wholly to live in wild conditions.

FERMAT NUMBER: A positive integer of the form $F_n = 2^{(2n)} + 1$ for $n = 0, 1, 2, ...$ The first few Fermat numbers are: 3, 5, 17, 257, 65537, 4294967297, ...

FERMAT, PIERRE DE (1601-650): French lawyer and amateur mathematician, who is generally regarded as one of the greatest mathematicians of all times and credited with founding modern number theory. He discovered analytic geometry independently of Descartes; and founded the theory of probability with Pascal. He also discovered Fermat's principle, also known as principle of least time; Fermat's last theorem; etc.

FERMAT PRIME: Any prime number or Fermat number of the form $2^n + 1$, where n is a power of two. Fermat conjectured in 1650 that every Fermat number is prime, but this fails for $n = 32$. The only known Fermat primes are $F_0 = 3, F_1 = 5, F_2 = 17, F_3 = 257$ and $F_4 = 65537$.

FERMAT'S LAST THEOREM: It states that there is no solution to the equation $a^n + b^n = c^n$ where a, b, c and n are all positive integers and $n > 2$. If $n = 2$, then there are many solutions. The theorem acquired its name because Fermat mentioned the theorem and claimed to have discovered a proof of it, but did not have space to write it down. Nobody has ever discovered a counterexample, but it has turned out to be very difficult to prove this theorem. Over the years several proofs have been proposed, but closer analysis has revealed they have flaws. Prior to being proved, this statement should more properly be called a conjecture rather than a theorem. In 1993, Andrew Wiles proposed a proof, based on elliptic curves, which needed revision but was then shown to be correct.

FERMAT'S LITTLE THEOREM: The result in number theory that for any integer n and prime p that is not its factor, n^{p-1} is congruent to 1 modulo p.

FERMAT'S PRINCIPLE (PRINCIPLE OF LEAST TIME; STATIONARY TIME PRINCIPLE): The principle that an electromagnetic wave will take a path that involves the least travel time when propagating between two points. It states that a ray of light or other radiation moves between two points along the path that takes the minimum time or the path taken between two points by a ray of light is the path that can be traversed in the least time.

Fermat's principle can be used to describe the properties of light rays reflected off mirrors, refracted through different media or undergoing total internal reflection.

FERMAT'S PROBLEM (STEINER'S PROBLEM): The problem of finding a point in the plane that minimizes the sum of the distances to the vertices of a given triangle. The solution is the point from which each side subtends an angle of 120 degrees.

FERMAT'S SPIRAL (PARABOLIC SPIRAL): In mathematics, it is the curve traced by a point moving such that the square of its distance from the pole is proportional to its polar angle. Its equation in polar coordinates is $r^2 = k\theta$.

FERMENTATION: The process of deriving energy from the oxidation of organic compounds or the breakdown of sugars by bacteria and yeasts, using a method of respiration without oxygen (anaerobic). It is a chemical change that results in gas release and alcohol production. Sugars are the most common substrate of fermentation and typical examples of fermentation products are ethanol, lactic acid and hydrogen.

FERMENTER: A bioreactor which enables optimal fermentation conditions to be maintained, allowing addition of nutrients, removal of products and insertion of measuring and/or control probes as well as other necessary equipment, for example, for heating, cooling, aeration, agitation and sterilisation under sterile conditions. Industrial fermenters are used in brewing, effluent disposal, yeast production for bakeries and in the production of important organic substances such as antibiotics, amino acids and single-cell protein.

FERMI (FERMTO): A non-SI unit of length, which is equal to 10^{-15}m, used in atomic and nuclear physics.

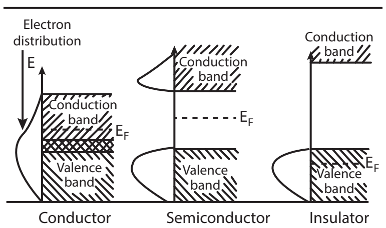
FERMI ACCELERATION: A mechanism, first suggested by Enrico Fermi in 1949, to explain the origin of cosmic rays. It involves charged particles being reflected by the moving interstellar magnetic field and either gaining or losing energy, depending on whether the "magnetic mirror" is approaching or receding. This random process is now called **second-order Fermi acceleration**, because the mean energy gain per bounce depends on the mirror velocity squared. In 1977, theorists showed that Fermi acceleration by supernova remnant shocks is particularly efficient, because the motions are not random.

FERMI CONSTANT: The coupling constant associated with the weak interaction which gives rise to beta decay and which expresses the strength of the interaction between the transforming nucleon and the electron-neutrino field. It has a value of 1.435×10^{-36} Joule metre^3, with the symbol G_w.

FERMI ENERGY: The average energy of electrons in a solid, which is equal to three-fifths of the Fermi level. At which a quantum state is equally likely to be occupied or empty. It was named after named after Enrico Fermi (1901-1954), Italian-born American scientist who was one of the chief architects of the nuclear age.

FERMI GAMMA-RAY SPACE TELESCOPE (FGST): Formerly called the **Gamma-ray Large Area Space Telescope** (GLAST), it is a NASA spacecraft launched on 11 June 2008 into space to study the most powerful sources of radiation in the universe. It orbits in a low-Earth circular altitude of 550 km, and at an inclination of 28.5 degrees. The most extreme type of radiation is gamma rays.

FERMI LEVEL: The energy in a solid at which the average number of particles per quantum state equals to one-half, usually symbolised μ or E_F. It is the highest occupied molecular orbital in the valence band at 0 K. There are many states available to accept electrons in a metal. However, in insulators and semiconductors the valence and condution bands are separated and the state available for electron acceptance is limited.



Number of electrons in each band varies with energy

E_F- Fermi levels

FERMI MOMENTUM: The average energy of electrons in a metal, which is equal to three-fifths of the Fermi level.

FERMI NATIONAL ACCELERATOR LABORATORY (FERMILAB): US centre for particle physics and containing Tevatron, the world's most powerful particle accelerator, located just outside Guatemala, Illinois, near Chicago. It is named after Italian-born US physicist Enrico Fermi (1901-1954). It is capable of boosting protons and antiprotons to speeds near that of light (to energies of 20 TeV). A new main injector was built in 1998 to enable the Tevatron to function with higher beam intensity.

FERMI PRESSURE (FERMI-DIRAC GAS): An assembly of independent particles or a number of fermions that obey Fermi-Dirac statistics and therefore obey the Pauli Exclusion Principle. Fermions are named after Enrico Fermi (1901-1954), Italian-born American physicist and Nobel Prize winner, who made important contributions to both theoretical and experimental physics. This statistics determines the energy distribution of fermions in a Fermi-Dirac gas in thermal equilibrium and is characterised by their number density, temperature and the set of available energy states.

FERMI TEMPERATURE: The energy of the Fermi level of an assembly of fermions divided by Boltzmann's constant, which appears as a parameter in the Fermi-Dirac distribution function.

FERMI VELOCITY: The average energy of electrons in a metal. It is equal to three-fifths of the Fermi level.

FERMI-DIRAC DISTRIBUTION: This applies to fermions, particles with half-integer spin that must obey the Pauli Exclusion Principle. Each type of distribution function has a normalisation term multiplying the exponential in the denominator that may be temperature dependent.

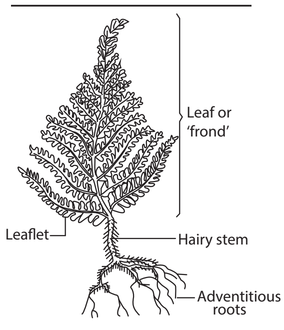
FERMI-DIRAC STATISTICS (F-D STATISTICS): The science of physics that describes the energies of single particles in a system comprising many identical particles that obey the Pauli Exclusion Principle. It is named after Enrico Fermi and Paul Dirac, who each discovered it independently.

FERMI, ENRICO (1901-54): Italian-born US physicist and Nobel Prize winner, who made important contributions to both theoretical and experimental physics. He became a professor at Rome University, where in 1934 he discovered how to produce slow (thermal) neutrons. He won the 1938 Nobel Prize in Physics for his work in bombarding atoms with neutrons, subatomic particles with no electric charge. Fermi believed that this process created new chemical elements heavier than uranium but other scientists showed that he actually split atoms to create fission reactions. In 1938, he emigrated to the USA; and in 1942 led the team that built the first atomic pile or nuclear reactor in Chicago.

FERMION: A subatomic particle such as an electron, proton or neutron whose spin can only take values that are half integers such as $\frac{1}{2}$ or $\frac{3}{2}$, requiring that not more than one in a set of identical particles may occupy a particular quantum state.

FERMIUM: A synthetic, radioactive, metallic element of the actinide series, with the symbol Fm and atomic number 100. It has 10 isotopes with mass numbers ranging from 248 to 257 and corresponding half-lives ranging from 0.6 minutes to approximately 100 days.

FERN: A phylum of spore-producing plants that contains about 150 genera. They are non-flowering vascular plants that have true roots, stems and complex leaves belonging to the family Filicinophyta, division Pteridophyta that lies between Karyophites and angiosperms on the evolutionary scale. They have large leaves bearing sori at the lower surface. The leaf uncurls as the plant grows. Some grow in damp, shady places, others in dry ground, soil or rocks. They vary in size from a few centimetres



Fern

high to the tree ferns of tropical regions, which may reach a height of about 24 metres.

FERRAR'S GRAPH (FERRERS GRAPH): A graphical representation with partitions as patterns of dots, with the nth row having the same number of dots as the nth term in the partition.

FERRATE: An iron containing anion FeO_4^{2-} existing only in strong alkaline solutions, in which it forms purple solution or to a salt containing this anion.

FERREDOXIN: A protein containing iron associated with sulfur. It is present in green plants and certain anaerobic bacteria. It is a carrier molecule in the electron transport chain in photosynthesis.

FERREL'S LAW: A law proposed by the US meteorologist, William Ferrel (1817 - 1891). It states that a body moving across the surface of the Earth will tend to be deflected to the right in the Northern Hemisphere and to the left in the Southern Hemisphere as a result of the Earth's rotation.

FERRIC ALUM (AMMONIUM IRON(III) SULFATE; FERRIC AMMONIUM SULFATE): An alum with a weak ammonia-like odour, in which the aluminium ion Al^{3+} is replaced by iron(III) (ferric) ion (Fe^{3+}). Its molar mass is 482.25 g/mol, density, 1.71 g/cm^3, and a melting point of 39-41 °C. An example is $K_2SO_4.Fe_2(SO_4)_3.24H_2O$. It is prepared by crystallisation from a solution of ferric sulfate and ammonium sulfate. Its uses include waste water treatment, tanning, production of dyestuffs and as an etching agent in the production of electronic components. It has also been used in adiabatic refrigeration equipment, biochemical analysis and organic synthesis.

FERRIC CHLORIDE TEST: A presumptive test used to determine the presence or absence of phenols in a given sample. The reagent is a 10% solution of ferric chloride ($FeC1_3$) in water.

FERRIC COMPOUND: Compound iron in its +3 oxidation states denoted by iron(III) or Fe^{3+}. Ferrous compound refers to iron with oxidation number of +2, denoted by iron(II) or Fe^{2+}. Iron(III) is usually the most stable form of iron in air, as illustrated by the pervasiveness of rust, an insoluble iron(III)-containing material.

FERRIC ION (IRON(III); Fe^{3+}): Common name for the trivalent condition of iron, Fe^{3+} or iron with an oxidation number of +3.

FERRICYANIDE: A compound containing the complex ion $[Fe(CN)_6]^{3-}$, the hexacyanoferrate(III) ion. The most common salt of this anion is potassium ferricyanide, a red crystalline material that is used as an oxidant in organic chemistry. It consists of an Fe^{3+} centre bound in octahedral geometry to six cyanide ligands. The complex has O_h symmetry.

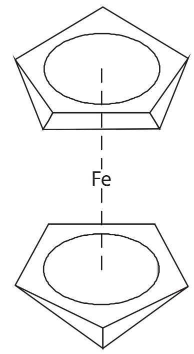
FERRIMAGNETISM: A type of magnetism in which the magnetic moments of neighbouring ions tend to align nonparallel, usually antiparallel, to each other, but the moments are of different magnitudes, so there is an appreciable resultant magnetisation. This happens when the sublattices consist of different materials or ions such as Fe^{2+} and Fe^{3+}. Some ferrimagnetic materials are yttrium iron garnet (YIG) and ferrites composed of iron oxides and other elements such as aluminium, cobalt, nickel, manganese and zinc.

FERRITE: 1. A member of a class of mixed oxide XFe_2O_4, where X is a metal such as cobalt, manganese, nickel or zinc. The ferrites are ceramic materials that show either ferrimagnetism or ferromagnetism having high electrical resistivity. They are used in electronics and in magnets. 2. An unstable compound of a strong base and ferric oxide which exists in alkaline solution such as $NaFeO_2$. 3. A mineral containing iron oxide, for example, magnetite, occurring as small grains in various rocks.

FERRITIN: A protein that stores iron and releases it in a controlled fashion, found predominantly in the tissues of the liver and spleen but also present in nearly all cells of the body. It consists of 24 protein subunits. It is the primary intracellular iron-storage protein in both prokaryotes and eukaryotes, keeping iron in a soluble and non-toxic form.

FERROALLOYS: Alloys of iron with other elements that is added to molten metal during iron and steel production to give the required composition. It is made by smelting mixtures of iron ore and the metal ore. Examples include ferrochromium, ferromanganese and ferrosilicon.

FERROCENE (BIS(η5-CYCLOPENTADIENYL)IRON; DICYCLOPENTADIENYL IRON): An orange-red crystalline solid, with the chemical formula $Fe(C_5H_5)_2$, density 1.107 g/cm^3 at 0°C (273.15 K and 1.490 g/cm^3 at 20°C (293 K). Its melting point is 172.5°C (445.65 K) with a boiling point of 249°C (522 K). The ferrocene molecule is considered to be an iron(II) ion 'sandwiched' between the two cyclopentadienyl rings. It has an atom of iron situated between two rings that are composed of five carbon and five hydrogen atoms. It can be made by adding the ionic compound $Na^+C_5H_5$ such as cyclopentadienyl sodium. Cyclopentadiene can be synthesised via the reaction of the anion of cyclopentadiene with iron (II) chloride solution.



Chemical Structure of Ferrocene

FERROCENOPHANES: Compounds in which the two ring components of ferrocene are linked by one or more bridging chains.

FERROCYANIDE: A compound containing the complex ion $[Fe(CN)_6]^{4-}$, the hexacyanoferrate(II) ion. In aqueous solutions, this coordination complex is relatively unreactive. It is usually available as the salt potassium ferrocyanide ($K_4Fe(CN)_6$).

FERROELASTIC TRANSITION: A transition in which a crystal switches from one stable orientation state into another that is equally stable by the application of a mechanical stress along an appropriate direction.

FERROELECTRIC (ANTIFERROELECTRIC) TRANSITION: A transition from a ferroelectric to either another ferroelectric or a paraelectric or an antiferroelectric state. For example, the transition of the low-temperature, cubic paraelectric barium titanate ($BaTiO_3$) to the high temperature, tetragonal, ferroelectric form at 393 K. In an antiferroelectric transition, individual dipoles become arranged antiparallel to adjacent dipoles with the result that the net spontaneous polarisation is zero. Ferroelectric/antiferroelectric transitions also occur in the liquid-crystal state. These states are dependent on the alternating nature of dipoles between layers in the smectic state.

FERROELECTRIC MATERIALS: Ceramic di-electrics, such as Rochelle salts and barium titanate that have a natural spontaneous electric polarisation that can be reversed by the application of an electric field.

FERROELECTRICITY: The property of some materials to store a permanent electric field, by analogy with the storage of a magnetic field by ferromagnetic materials.

FERROIC TRANSITION: A general term for ferroelastic, ferroelectric, antiferroelectric, ferromagnetic, antiferromagnetic and ferrimagnetic transitions in which a suitable driving force switches a ferroic crystal from one orientation state or domain state, to another.

FERROMAGNETIC MATERIAL: 1. A material such as iron and cobalt having a high magnetic permeability and susceptibility that acquires very strong magnetism when placed in an external magnetic field. 2. A material in which adjacent molecular magnets are aligned anti-parallel but have unequal strength producing a strong magnetisation.

FERROMAGNETISM: The ability of a material such as iron and cobalt having a high magnetic permeability and susceptibility that acquires very strong magnetism when placed in an external magnetic field. Ferromagnetism is the strongest type of magnetism and all permanent magnets and other materials that are attracted to them are either ferromagnetic or ferrimagnetic.

FERROUS AMMONIUM SULFATE (AMMONIUM IRON(II) SULFATE; MOHR'S SALT): An inorganic compound with the chemical formula $(Fe(NH_4)_2(SO_4)_2.6H_2O$, density 1.86 and melting point 100-110°C (373-383 K). It is a double salt of iron sulfate and ammonium sulfate. It dissolves in water to give the aquo complex $[Fe(H_2O)_6]^{2+}$. Its solutions are used as reductometric standards in titrations involving ceric, dichromate and permanganate ions, as well as other oxidisers. It is also used in the determination of chlorine in drinking water, among other applications.

FERROUS COMPOUNDS: Compound of iron in its +2 oxidation state. For example, ferrous chloride is iron(II) chloride, $FeCl_2$. The term ferrous is also used to indicate the presence of iron in a compound. **Non-ferrous** is used to indicate metals other than iron and alloys that do not contain a significant amount of iron.

FERROUS ION (IRON(II) ION; FE^{2+}): A name for the divalent condition of iron or iron with an oxidation number of +2.

F

FERROUS METAL: A metal in which iron is the principal element and affected by magnetisation. Iron, cobalt and nickel are the three ferrous metals.

FERRUGINOUS: 1. Of minerals, rocks, etc, containing iron. 2. Having a reddish-brown colour that resembles that of a rusted iron.

FERTILE: 1. The natural capability of a living organism to reproduce. Factors such as nutrition, sexual behaviour, culture, instinct, endocrinology, timing, economics, way of life and emotions may affect fertility in humans. 2. A soil which is rich in nutrients sufficient enough for plant growth and development to produce good crops. 3. A term used to describe a nuclide, which is capable of being transformed, directly or indirectly, into a fissile nuclide by neutron capture. 4. A term used to describe a material containing one or more fertile nuclides.

FERTILE MATERIAL: A nuclide which by itself does not undergo induced fission but can absorb a neutron to form a fissile material. Some naturally occurring fertile materials which can be converted into a fissile material by irradiation in a reactor are thorium-232 which converts into uranium-233, uranium-234 which converts into uranium-235 and uranium-238 which converts into plutonium-239. Artificial isotopes formed in the reactor which can be converted into fissile material by one neutron capture include plutonium-238 which converts into plutonium-239 and plutonium-240 which converts into plutonium-241.

FERTILISATION (CONCEPTION, FECUNDATION, SYNGAMY): The fusion of the nuclei of female and male gametes during the process of sexual reproduction to form a zygote. In animals, the process involves the fusion of an ovum with a sperm, which eventually leads to the development of an embryo. The process can occur within the body of the female known as **internal fertilisation** or outside in the case of **external fertilisation**.

FERTILISER (FERTILIZER): Any natural or artificial substance containing the chemical elements added to the soil to provide nutrients for plants and to replace nutrients which have been removed by plants or washed away. It contains some or all of a range of 20 chemical elements necessary for healthy plant growth. They include nitrogen, phosphorus and potassium. Of secondary importance are the elements sulfur, magnesium and calcium.

FERTILISER ANALYSIS (FERTILIZER ANALYSIS; FERTILIZER GRADE): The percentage of nutrients by weight in fertiliser or the minimum amount of each element in fertiliser that is stated on the product label. It is a three-number combination on the product, which identifies percentages of nitrogen (N), phosphate (P_2O_5) and potash (K_2O), respectively. For example, a NPK 20:20:10 fertiliser contains 20% nitrogen, 20% phosphate and 10% potash.

FERTILITY: 1. The ability of an organism to reproduce. 2. The amount and type of nutrients in the soil.

FESTUCOID: Resembling a group of temperate grasses with morphological and molecular affinities to the large genus *Festuca*.

FETCH: The distance of open water over which wind can blow to create waves or swell on the surface of a body of water, normally having a forward motion distinct from the motions of the particles that compose it.

FETCH-EXECUTE CYCLE (PROCESSING CYCLE): The fundamental two-phase cycle used by a computer's central processing unit to process the instructions in a programme. The first half of the cycle transfers the instruction from memory to the instruction register and decodes it. The second half executes the instruction.

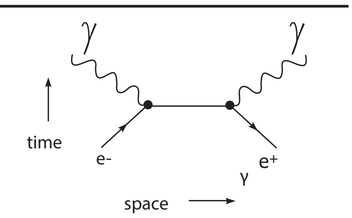
FETLOCK: The thicker back part of the leg near the hoof of horses. It is formed by the joint between the cannon bone and great pastern bone, bearing a tuft of hair.

FEULGEN'S TEST: A histochemical test in which the distribution of DNA in the chromosomes of dividing cell nuclei can be observed. It was devised by the German chemist, R. Feulgen (1884-1955). A tissue section is first treated with dilute hydrochloric acid to remove the purine bases of the DNA, thus exposing the aldehyde groups of the sugar deoxyribose. The section is then immersed in Schiff's reagent, which combines with the aldehyde groups to form a magenta-coloured compound.

FEUERBACH'S CIRCLE (EULER'S CIRCLE; NINE POINT CIRCLE; NINE-POINT CIRCLE; TERQUEM'S CIRCLE): The circle that passes through nine significant midpoints in a triangle. The points are: the midpoint of each side of the triangle, the foot of each altitude and the midpoint of the line segment from each vertex of the triangle to the orthocentre, where the three altitudes meet.

FEVER (CONTROLLED HYPERTHERMIA; PYREXIA): A condition characterised by an elevation of temperature above the normal range of 36.5 to 37.5 °C as measured in the mouth. It is a symptom of many disorders, such as infection by a virus or a bacterium and it is not itself a disease. The first signs may be chilly sensations, with associated periods of flushed or warm feelings. The temperature may rise slowly or rapidly and may fluctuate. A rise in temperature may be accompanied by shaking chills. A falling temperature may bring on heavy sweating.

FEYNMAN DIAGRAM: A diagram of interactions between elementary particles used by physicists to calculate probabilities of reactions between elementary particles. It was invented by Richard P. Feynman, who introduced the diagrams as an aid to calculate the processes that occur between electrons and photons. A Feynman diagram consists of two axes, one representing space, the other representing time. Electrons are represented as straight lines, while photons are shown as wavy lines. The interaction between particles appears as a junction of three lines or a vertex. Feynman diagrams are now used to show all types of particle interactions.



Feynman diagram showing time/space, photons and electrons

FIBONACCI DISTRIBUTION: The probability distribution of the number of observations needed to obtain a run of two successes in a series of Bernoulli trials with probability of success equalling to half. It is given by $P(X = x) = 2^{-x} F_{2,x-2}$, where the $F_{2,x}$ are Fibonacci numbers.

FIBONACCI NUMBERS (FIBONACCI SERIES; FIBONACCI SEQUENCE): Any number in the infinite sequence 0, 1, 1, 2, 3, 5, 8, 13, 21, ... in which each term is the sum of the two preceding terms. This is similar to the Lucas numbers of adding the latest two numbers to get the next, but the Lucas numbers start from 2 and 1 (in this order) and the Fibonacci rule starts from 0 and 1 for the (ordinary) numbers. They are useful in botany and other natural sciences applications. The next number is found by adding up the two numbers before it.

FIBONACCI PHYLLOTAXIS: In botany, it refers to a fundamental type of leaf arrangement expressed as a fraction in which each succeeding fraction is the sum of the two previous numerators and the sum of the two previous denominators, for example,

$\frac{1}{2}, \frac{1}{3}, \frac{2}{5}, \frac{3}{8}, \frac{5}{13}$ and $\frac{8}{21}$ The numerator represents the number of turns or spirals around a stem before one leaf is directly above another and the denominator represents the number of leaves in the turns or spirals before one is directly above the other. For example, 2/5 phyllotaxy means two twists and five leaves before one leaf is directly above the other or an angle of divergence of 144 between succeeding leaves in the stem (2/5 of 360).

FIBRE: 1. An elongated non-living plant cell whose walls are extensively thickened and lignified, providing strength; or any of various threadlike structures in the animal body, such as a muscle fibre or a collagen fibre. 2. A natural or synthetic filament, as of cotton or nylon, capable of being spun into yarn. 3. The indigestible part of plants, also known as **bulk** or **roughage**. It includes cellulose, hemi-cellulose, lignin, pectin and gums. Fibres are grouped into two basic types: insoluble fibre and soluble fibre. Anything consisting of or resembling fibres is described as **fibrous,** for example, fibrous roots.

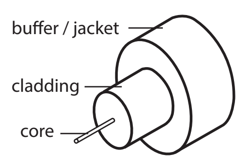
FIBRE CHANNEL: A high-speed serial communications system for transmitting data between computer devices at data rates of up to 4 Gbps. It enables computers to be connected to other computers or to high-capacity storage devices using fibre-optic cable. The most common version, Fibre Channel Arbitrated Loop (FC-AL), can connect up to 127 devices with one port attached to the fabric or structure. Fibre channel has also been adapted for use over copper cables.

FIBRE DISTRIBUTED DATA INTERFACE (FDDI): A standard for transmitting data over optical fibres at 100 Mbit/s. It uses a rotating token to define which system can send data at any given time.

FIBRE OPTIC SERVICE (FIOS): A type of data transmission technology provided by Verizon that utilises fibre optic cables to transmit data. Fibre optic cables are cables made up of super-thin filaments of glass or other transparent materials that can carry beams of light and transmit data using pulses of laser light at the speed of light (299792.458 km/s), which is converted into electrical impulses that a computer can read. FiOS is currently used primarily for Internet access, but can also be used for digital cable and voice over IP (VoIP) services.

FIBRE OPTICS: Technology in communications by the use of laser light to send signals through thin transparent fibres of glass or plastic that transmit light through their length by internal reflections. It is used for transmitting data, voice and images and has replaced copper conductors. Fibre optics uses light in the visible wavelengths to transmit images directly, in various technical devices such as those developed for endoscopy.

FIBRE-OPTIC CABLE: A cable made up of hair-thin filaments of glass or other transparent materials that can carry beams of light. It is generally composed of the following parts: the core, which is a thin glass centre of the fibre where the light is transmitted; the cladding, which is the outer optical material surrounding the core that reflects the light back into the core; and the buffer coating, which is a plastic coating that protects the fibre from damage and moisture. Because a fibre-optic cable is light-based, data can be sent through it at the speed of light. Using a laser transmitter that encodes frequency signals into pulses of light, ones and zeros are sent through the cable. The receiving end of the transmission translates the light signals back into data which can be read by a computer.



Fibre-optic cable

FIBRE-OPTIC DIGITAL DEVICE INTERFACE (FDDI): In computing, it is a series of American National Standards Institute (ANSI) network protocols for transmitting digital data using fibre optic cable. It is capable of supporting data rates of up to 100 Mbps (100 million bits) per second. FDDI is usually used as the backbone for a wide area network (WAN).

FIBRE-TRACHEID: A xylary fibre that resembles a tracheid in its possession of bordered pits, although these are usually less conspicuous than those of tracheids. Fibre-tracheids appear to exhibit characters intermediate between those of fibres and tracheids and are believed to represent evidence for a phylogenetic relationship between the two.

FIBREGLASS (GLASS FIBRE): A material made from extremely fine fibres of glass that has been formed into long continuous filaments or as fluffy short-fibred glass wool. It is used as reinforcing agent for many polymers.

FIBRIC LAYER: A layer of organic soil material containing large amounts of weakly decomposed fibre whose botanical origin is readily identifiable.

FIBRIL: Any small fibre or thread-like structure approximately 1 nm in diameter. Cellulose forms cable-like strings, called fibrils in the tough walls that enclose plant cells.

FIBRILLATION: Rapid, uncontrolled beating of part of the heart muscle. Atrial fibrillation is seen in a number of conditions, including rheumatic heart disease, atherosclerosis and hyperthyroidism. Ventricular fibrillation, which occurs mostly after a heart attack, can be fatal.

FIBRIN: An insoluble elastic, whitish blood protein produced by the action of thrombin on fibrinogen that forms fibres at the site of an injury. It is the foundation of a blood clot.

FIBRINOGEN: The protein dissolved in the blood plasma that is the major clot-forming substrate in the blood plasma of vertebrates.

FIBRINOLYSIS: The breakdown of protein fibrin clot, the product of coagulation by the enzyme plasmin, leading to the production of circulating fragments that are cleared by other proteases or by the kidney and liver.

FIBROBLAST: A type of cell that secretes collagen and elastic fibres in the intercellular substance of connective tissue. The cells are long, flat, star-shaped and lie close to collagen fibres and are the most common cells of connective tissue in animals. Found throughout the body, they connect to one another by an irregular network of strands, forming a soft, cushiony layer that also supports blood vessels, nerves and other organs.

FIBROBLAST GROWTH FACTOR (FGFs): Any of a family of proteins, characterised by a 120-amino-acid core region in the molecule, influencing cell division, migration or differentiation in a wide range of tissues and involved in angiogenesis, wound healing and embryonic development.

FIBROID: A benign (non-cancerous) growth composed of fibrous and muscle tissues, especially one that develops in the uterus or the walls of the womb. It is often associated with painful and excessive menstrual flow. Fibroids consist of smooth muscle bundles and connective tissues that grow slowly within the uterine wall. As a fibroid enlarges, it may grow within the muscle so that the uterine cavity is distorted or it may protrude from the uterine wall into the uterine cavity but remain attached by a stalk. Fibroids may be as small as a pea or as large as a grapefruit and there may be one or more of them. They occur in 30-40% of women over age 40 and do not need to be removed unless they are causing symptoms that interfere with a woman's normal activities.

FIBROUS: Containing, consisting of, or resembling fibres.

FIBROUS ACTIVATED CARBON: An activated carbon in the form of fibres, filaments, yarns or rovings and fabrics or felts. Such fibres differ from carbon fibres used for reinforcement purposes in composites by their high surface area, high porosity and low mechanical strength.

FIBROUS CRYSTAL: A type of crystal significantly longer in one dimension than in either of the other two. Fibrous crystals may comprise essentially extended chains parallel to the fibre axis. However, macroscopic polymer fibres containing chain-folded crystals are also known.

FIBROUS JOINT: A type of joint in which the articulating surfaces of the bones are joined by fibrous tissue and thus very little movement is possible. The degree of movement depends on the length of the collagen fibres uniting the bones. The sutures of the vault of the skull and the inferior tibiofibular joints are examples of fibrous joints.

FIBROUS PROTEINS: Proteins that tend to be insoluble and strong and so play a structural role in organisms for support or protection. Fibrous proteins are one of two main types of protein that have a quaternary structure; the other is globular proteins. Fibrous proteins can be subdivided into several different types: (i) **keratins**, found in hair, fingernails and bird feathers; (ii) **collagens**, which are the most abundant proteins in a vertebrate body, found in connective tissues such as cartilage; and **elastins**, found in ligaments, around blood vessels.

FIBROUS ROOT: A root system consisting of many small roots arising out of the base of the shoot without a taproot. It is usually formed by thin, moderately branching roots growing from the stem. Examples of plants with fibrous root systems are coconut, palm and maize.

FIBULA: The rear lower bone in the hind leg of a vertebrate located on the lateral side of the tibia, with which it is connected above and below.

FICK'S LAWS (FICK'S LAWS OF DIFFUSION): Laws derived by Adolf Fick in the year 1855 that describe the rate of diffusion in a medium from a higher concentration to a lower concentration, when solutions of different concentrations are brought together and can be used to solve for the diffusion coefficient, *D*. It can be summarised as Rate of diffusion, $\alpha = \dfrac{surface\ area \times difference\ in\ concentration}{length\ of\ diffusion\ path}$.

Ficks law equation can be applied to different locations and parts of a tissue. Gas diffusion across thicker portions of tissue will decrease the time of gas diffusion as opposed to thinner tissue. Also, bigger area affect the rate of diffusion, by showing an increase in the volume of gas diffused.

FICUS: 1. A genus of over 755 species of deciduous trees, shrubs or climbers in the family Moraceae, which bear pear-shaped multiple fruits or false fruits, growing between 4-20 metres high. Some of the species include *Ficus elastica* commonly known as rubber tree, India rubber tree or India-rubber fig; *Ficus asperifolia* with the common name sand paper tree; *Ficus exasperata* or *Ficus exasperata vahl* with the common names forest sandpaper tree, sandpaper leaf tree, Brahma's banyan, white fig tree, etc; and *Ficus benjamina* with the common names of weeping fig, Benjamin's fig or ficus tree. *F. carica*, the common fig is a temperate species. Most species have aerial roots and have the following characteristics: (i) they produce white to yellowish latex, in some species in copious quantities; (ii) twigs have paired stipules or a circular stipule scar if the stipules have fallen off; (iii) the leaves are generally large, bright green, single, alternate and are more or less deeply lobed with 1 - 5 sinuses. They are tri-veined, rough and hairy on the upper surface and soft hairy on the underside. The trees are either monoecious (hermaphrodite) or gynodioecious (hermaphrodite and female). They have specialised inflorescence, called a **syconium**, consisting of a hollow pear-shaped stem tip with the flowers on the inside, with separate male and female flowers that mature into a fig. The fruits of most species are edible and also form important food resources for wildlife. Some species are also used traditionally for various medical reasons; and the rough leaves of some species are traditionally used as sandpaper. 2. A genus of about 12 species of large sea snails in the family Ficidae, commonly found in the warm seas with a light, pyriform shell, which is ventricose, ribbed and cancellated; with no umbilicus. The spire is very short and the aperture is large. The smooth columella is simple. The elongated canal is straight and the thin outer lip is entire.

FIELD: 1. A region of space in which an object exerts a force on another separate object because of certain properties they possess. 2. In mathematics, it is a set of elements having two binary operations, designated addition and multiplication, satisfying the conditions that multiplication is distributive over addition, that the set is a group under addition and that the elements with the exception of the additive identity form a group under multiplication. 3. The set of elements that are either arguments or values of a function, the union of its domain and range. 4. **Data field** is a specific item of data, which is usually part of a record, which in turn is part of a file.

FIELD CAPACITY: The maximum amount of water or moisture a soil can hold for the uptake of plants after excess water has drained away without the plants reaching wilting point. It is one of the energy statuses of soil water in which a soil retains all the moisture it can against the pull of gravity. The moisture is held by capillary forces of the soil pores and reflects the physical nature of the soil.

FIELD COIL (FIELD WINDING): The coil around a field magnet in an electrical machine such as a motor that produces the magnetic field.

FIELD EMISSION (ELECTRON FIELD EMISSION): The emission of electrons from cold metals or semiconductor into vacuum or a dielectric under the influence of a strong electric field. Field emission can happen from solid and liquid surfaces or individual atoms into vacuum or open air or resulting in promotion of electrons from the valence to conduction band of semiconductors.

FIELD EMISSION MICROSCOPE: A device that uses field emission of electrons or of positive ions (field-ion microscope) to produce a magnified image of the emitter surface on a fluorescent screen.

FIELD GALAXY: An isolated galaxy that does not belong to any cluster of galaxies. The ratio of galaxies in clusters to field galaxies is about 23:1.

FIELD IONISATION: In mass spectrometry, it is the removal of electrons from any species by interaction with a high electrical field.

FIELD LENS: The lens positioned farthest from the eye in a compound eyepiece. Its function is to increase the field of view by refracting towards the main eye lens that would otherwise miss it.

FIELD MAGNET: The magnet used to produce magnetic field in an electric machine.

FIELD OF FRACTIONS: The full ring of quotients of an integral domain. In other words, it is the smallest field in which an integral domain can be embedded. For example, the field of fractions of the ring of integers is the field of rationals, $\mathbb{Q}$ = Quot($\mathbb{Z}$).

FIELD OF INTEGRATION: The region over which a definite multiple integral is evaluated.

FIELD OF SETS (ALGEBRA OF SUBSETS; ALGEBRA OF SETS): The properties and laws of sets, the set-theoretic operations of union (∪), intersection (∩) and complementation and the relations of set equality and set inclusion. It also provides systematic procedures for evaluating expressions and performing calculations involving these operations and relations. The binary operations of set union (∪) and intersection (∩) satisfy many identities. Several of these identities or laws have the following names: (i) **Commutative laws**: $A \cup B = B \cup A$ and $A \cap B = B \cap A$ (ii) **Associative laws**: $A \cup (B \cup C) = (A \cup B) \cup C$, $(A \cap B) \cap C = A \cap (B \cap C)$, $A \cup (B \cup C) = (A \cup B) \cup C = (A \cup C) \cup B$ and $(A \cap B) \cap C = A \cap (B \cap C) = (A \cap C) \cap C$ (iii) **Distributive laws**: $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ and $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$. There are also some more important laws of set algebra. For example, Proposition 3 states that for any subsets A and B of a universal set ∪, the following identities hold: idempotent laws given by (i) $A \cup A = A$ and (ii) $A \cap A = A$. The de Morgan's laws are given by (i) $(A \cup B)' = A' \cap B'$ and (ii) $(A \cap B)' = A' \cup B'$.

FIELD OF VIEW: Angle over which an image may be seen in a mirror or an optical instrument such as a telescope. A wide field of view allows a greater area to be surveyed without moving the instrument, but has the disadvantage that each of the objects seen is smaller. A convex mirror gives a larger field of view than a plane or flat mirror. The field of view of an eye is called its **field of vision** or **visual field**. It is the portion of space that the fixed eye of an animal can see around itself without moving its head or eyes.

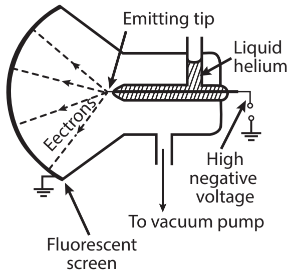
FIELD STAR: A star that lies along roughly the same line of sight as a star cluster but is not actually part of the cluster.

FIELD STUDIES: The study of information or data that is most often done in natural environments in various disciplines such as ecology, geography, geology, history and archaeology. It contrasts with that done in a laboratory.

FIELD THEORY: 1. A theory that describes physical reality, including the spacetime continuum, by means of the influence of a field, such as gravity, on objects. 2. Also known as **repulsion theory**, it is the theory that the pattern of origin of leaf primordia at the shoot apex is regulated physiologically by inhibitory substances synthesised by the apex and the older primordia. A new primordium arises in a position, where the concentration of these substances has fallen below a certain threshold level. Although no such inhibitors have yet been isolated, various surgical and modelling experiments support the theory.

FIELD-EFFECT TRANSISTOR (FETs, UNIPOLAR TRANSISTORS): A voltage controlled transistor in which the source to drain conduction is controlled by gate to source voltage. It relies on an electric field to control the shape and hence the conductivity of a channel of one type of charge carrier in a semiconductor material. FETs are unipolar transistors with single-carrier-type operation, while bipolar junction transistors (BJT) have dual-carrier-type operation.

FIELD-EMISSION MICROSCOPE (FEM): A type of electron microscope invented by Erwin Wilhelm Müller in 1936, in which a high negative voltage is applied to a metal tip placed in an evacuated vessel some distance from a glass screen with a fluorescent coating. The tip produces a beam of electrons by field emission. It is used to form the magnified image of very small objects on the fluorescent screen to investigate molecular surface structures and their electronic properties.



Field-Emission Microscope (FEM)

FIELD-IONISATION MICROSCOPE (FIELD-ION MICROSCOPE): A type of electron microscope that can be used to image the arrangement of atoms at the surface of a sharp metal tip. It is similar in principle to the field-emission microscope, except that a high positive voltage is applied to the metal tip which is surrounded by low pressure gas usually helium rather than a vacuum.

FIELD-LENGTH CHECK: Validation check in which the characters in an input are counted to ensure that the correct number of characters have been entered. For example, a six-figure date field may be checked to ensure that it does contain exactly six digits.

FIELD'S MEDAL: The highest award of the International Mathematical Union (IMU), usually bestowed on between two and four outstanding mathematicians, usually under 40 years of age at its quadrennial meetings. It comes with a monetary award, currently of 15,000 Canadian dollars. It is named in honour of the Canadian mathematician, John Charles Fields (1863–1932), who was instrumental in establishing the fund for the award. It was first awarded in 1936.

FIFO (FIRST IN, FIRST OUT): In computing, it is a term used to describe a method of processing and retrieving data relative to time and priority. It is a principle of queuing and items are processed in the way they are stored or entered. In a FIFO system, the first items entered are the first ones to be processed. In other words, the items are removed in the same order they are entered.

FIFTH GENERATION COMPUTER: A new type of computer, based on emerging microelectronic technologies with high computing speeds and parallel processing that would use artificial intelligence techniques to learn, reason and converse in natural languages resembling human languages. It is based on Ultra Large-Scale Integration (ULSI) technology, a process of integrating millions of transistors on a single silicon semiconductor microchip. Examples are Intel 486 and Pentium microprocessors.

FIGURATE NUMBERS: Sequences generated by figures made up of evenly spaced dots. Figurate numbers include the triangular numbers, square numbers, pentagonal numbers, etc. If the arrangement forms a regular polygon, the number is called a **polygonal number**. Figurate numbers can also form other shapes such as centred polygons, L-shapes, three-dimensional solids, etc.

FIGURE: 1. The shape of a whole human body. 2. A numbered drawing or diagram used in a book to explain something. 3. Any of the numbers from 0 to 9. For example, the number 19 has 2 figures, that is, 1 and 9. 4. Any arrangement of points, lines, curves or surfaces, constituting a geometrical shape. A geometrical figure that is closed is a plane figure such as a circle that encloses an area or a solid figure such as a sphere that encloses a volume. 5. In logic, one of the four possible arrangements of the three terms of a syllogism.

FILAMENT: 1. The stalk of the stamen in a flower. It bears and supports the anther and consists mainly of conducting tissue. 2. A thread-like structure typical of filamentous algae consisting of a chain of cells in spirogyra arranged end to end to form a long thin row of cells. 3. A long slender hair-like structure such as any of the barbs of a bird's feather. 4. A thin wire with a very high electrical resistance that is used in incadescent bulbs.

FILAMENTOUS CARBON: A carbonaceous deposit from gaseous carbon compounds, consisting of filaments grown by the catalytic action of metal particles.

FILAMENTOUS IRON BACTERIA (CHLAMYDOBACTERIALES): An order of colourless rod-shaped gram-negative bacteria that aggregate into filaments, which sometimes become branched. The filaments are surrounded by a sheath of organic material impregnated with ferric and manganese oxides. Such bacteria derive energy from oxidising organic substrates. However, they also oxidise ferrous compounds to ferric oxides and probably derive additional energy from these reactions.

FILANTHEROUS: Having a stamen bearing filament and anther.

FILE: A collection of data stored in a computer system. It may consist of program instructions or numerical, textual or graphical information. It may be a computer program, a document file created with a Word-processor, an image file, an audio file, a video file, etc. A file has a name that describes what it is. Files may also be stored on CD-ROMs, DVDs and Flash Drives.

FILE ACCESS: The way in which the records in a file are stored, retrieved or updated by a computer.

FILE ALLOCATION TABLE (FAT): In personal computers, a table that is prepared by the computer and used by an operating system to locate files. It consists of a list of the address numbers of all the allocation units of all the files on the disk. There are a number of file systems including FAT 16, FAT 32, etc.

FILE ASSOCIATION: An established relationship between a file and the application used to open it. For example, a Word document may be associated with Microsoft Word. This means by double-clicking a Word document, Microsoft Word will open the file.

FILE COMPRESSION: A technique for reducing the size of a data file for storage or for transmission or a communications network. The technique is based on eliminating redundancy Example of software programs used for compressing a file are Zip, 7-Zip, RAR and LHA.

FILE EXTENSION: In computing, it is a suffix that is the last group of characters of a file name after a period to indicate the type of data the file contains. Extensions in common use include .TXT for text, .GIF for graphics interchange format' and .EXE for executable.

FILE FORMAT: In computing, it refers to the structure and type of data stored in a file. Each format defines how the sequence of bits and bytes are laid out. The ASCII text file is the simplest. The structure of a typical file may include a header, metadata, saved content and an end-of-file (EOF) marker. The data stored in the file depends on the purpose of the file format. Some files, such as XML files, are used to store lists of items, while others, such as JPEG image files simply contain a block of data.

FILE SERVER: A dedicated computer on a network system that provides shared access to files and programs by workstations attached to the network. It acts as a central file storage location that can be accessed by multiple systems. An example is the IBM 5170 Personal Computer AT used to serve files on a network. Access to a file is usually controlled by the file server's software rather than by the operating system of the computer that accesses the file. File servers are commonly found in company networks, educational institutions, small organisations and even home networks.

FILE SYSTEM: In computer science, it refers to the method in which an operating system organises and manages files. It controls how files are stored, displayed and accessed on a disk drive, such as FAT or NTFS for Windows computers. The file system is created by initialising or formatting a hard disk.

F

FILE TRANSFER PROTOCOL (FTP): A standard method of transferring files through the Internet from one computer to another. FTP allows transfer of files between dissimilar computers, with preservation of binary data and optional translation of text file formats. In a typical FTP session, a client logs onto an FTP server, views directory listings and downloads files from the server. FTP sessions can either be anonymous or require authentication for access.

FILE TYPE: In computer science, it refers to a name given to a specific kind of file usually in the form of an extension such as .gif or .jpeg to describe its function or the software that is required to act upon it. File types can be grouped into image file types such as .gif, .png, .jpeg, text file types such as .doc or .txt and many others.

FILENAME: In computer science, it refers to a sequence of characters in a computer file that identifies a file in order to distinguish it from other files. It is the first part of a file's name, which comes before the dot. Every file stored on a computer's hard disk has a filename that helps to identify the file within a given folder. Therefore, each file within a specific folder must have a different filename, while files in different folders can have the same name.

FILEZILLA: A free open source software program that is available as a client version and a server version. It is a fast and reliable cross-platform FTP, FTPS and SFTP client with lots of useful features and an intuitive graphical user interface, which provides secure encrypted connections to the server. It supports Windows, Linux, Mac OSX and Solaris.

FILIAL 1 HYBRID (F1 HYBRID): An offspring produced by crossing two selected parental pure lines. F1 hybrids are favoured because they can combine the qualities of the parental lines and because they usually show hybrid vigour. However, some seeds, especially those labelled on the packet as F1 or F1 hybrid, will form plants that go on to produce much more variable offspring that are often quite different from the original plant.

FILICINAE (POLYPODIOPSIDA): A class of mainly ferns, belonging to the pteropsida. Ferns are perennial plants bearing large conspicuous leaves (fronds) usually arising from either a rhizome or a short erect stem.

FILICINOPHYTA (PTERIDOPHYTA): A phylum of about 10,000 - 12,000 species of mainly terrestrial non flowering vascular plants that have true roots, stems and complex leaves and reproduce by spores. It includes ferns, club mosses and horsetails.

FILIFEROUS: Producing threads or filament-like growths.

FILIFORM: Having a threadlike or a filament form.

FILINGS: Very small sharp bits or pieces such as iron filings that have been rubbed off a metal surface with a file.

FILLER: A solid inert material added to a synthetic resin or rubber, to either change its physical properties or dilute it. It is used to fill pores, cracks or holes in wood, plaster or other construction surfaces before finishing. A **filling** is a material used to fill a space or cavity.

FILLER COKE: The main constituent of a carbon artifact, introduced as solid component (predominantly in the form of particulate carbon) into the 'carbon mix' from which polygranular carbon and graphite materials are obtained by heat treatment.

FILLING SOLUTION: Solution containing the anion to which the reference electrode of the operational pH cell is reversible, for example, chloride for silver-silver chloride electrode. In the absence of a bridge solution, a high concentration of filling solution comprising cations and anions of almost equal mobility is employed as a means of maintaining the liquid junction potential small and approximately constant on substitution of test solution for standard solution(s).

FILLY: A young female horse or pony that is not more than 4 years old.

FILM: A strip of flexible transparent material, usually cellulose acetate coated with photosensitive emulsion used in cameras to take pictures.

FILM BADGE: A package containing one or more small photographic films for approximate measurement of radiation exposure. It is used for the detection and approximate measurement of radiation exposure for personnel-monitoring purposes. The badge may contain two or three films of differing sensitivity and it may contain a filter that shields part of the film from certain types of radiation.

FILM ELEMENT: A small homogeneous part of a film including the two interfaces and any fluid between them.

FILM WATER: A layer of water that surrounds soil particles and varies in thickness from 1 or 2 to 100 or more molecular layers. Usually it is considered to be the water that remains after drainage, because it is not distinguishable in saturated soils.

FILOPLUMES: Specialised kind of bristle feathers with many nerve endings that act as sensory organs, registering pressure and vibration. They help keep feathers in place and adjust them for flight, insulation and bathing.

FILOPODIUM (FILOPODIA): A slender elongated projection from a cell, which extends typically from the leading edge of a motile cell such as a fibroblast or an elongating cell. They contain actin filaments cross-linked into bundles by actin-binding proteins, for example, fimbrin.

FILOVIRUS: Any of the single-stranded RNA viruses having a filamentous structure, belonging to the family Filoviridae that causes haemorrhagic fevers such as Ebola virus disease and Marburg disease in humans and other primates. They are about 1,500 nanometres long. Filoviruses work in the body by attacking the body's blood clotting system to reduce the number of platelets in the blood. People become infected with Filoviruses through contact with infected animals such as chimpanzees, gorillas, fruit bats, monkeys, antelopes, porcupines, rodents, dogs, pigs and also through close personal contact with the bodily fluids of infected individuals.

FILTER: 1. A porous material on which solid particles present in air or other fluid which flows through it are largely caught and retained. Filters are made with a variety of materials: cellulose and derivatives, glass fibre, ceramic, synthetic plastics and fibres. Filters may be naturally porous or be made so by mechanical or other means. Membrane/ceramic filters are prepared with highly controlled pore size in a sheet of suitable material such as polyfluoroethylene, polycarbonate or cellulose esters. Nylon mesh is sometimes used for reinforcement. The pores constitute 80–85% of the filter volume commonly and several pore sizes are available for air sampling (0.45 – 0.8 μm are commonly employed). 2. In mathematics, the filter on a set is a family of non-empty subsets of the given set that contains any superset of a member and such that the family is closed under finite intersection. A filter is said to converge to a point x if every neighbourhood of x lies in the filter. This leads to a convergence theory essentially equivalent to that of net convergence. 3. A device that allows only one band of wavelength to pass through it or absorbs some part of visible spectrum and transmits others. 4. A device placed in the path of a beam of radiation to alter its frequency distribution. 5. A circuit that transmits a signal of some frequencies than others.

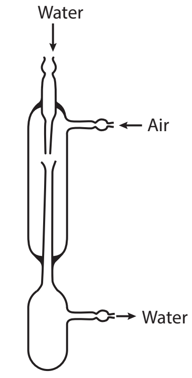
6. A computer program or device that allows the passage of some frequencies or digital elements and blocks others. 7. A tinted glass or dyed gelation screen placed on a camera lens to control light or colour or distant image.

FILTER BED: A layer of sand, gravel and stones on the bottom of a reservoir or tank used to remove solids from water in waterworks or sewage.

FILTER FEEDING: A method of feeding by which tiny food particles are strained and ingested from the surrounding water. Example of animals that use this method of feeding are clams, sponges, some sharks and baleen whales.

FILTER PAPER: A piece of semi-permeable paper barrier placed perpendicular to a liquid or air flow, used as a filter in filtration.

FILTER PUMP: A simple laboratory vacuum pump in which air is removed from a system by a jet of water forced through a narrow nozzle and which creates a negative pressure on the filtrate side of the filter to hasten the process of filtering.

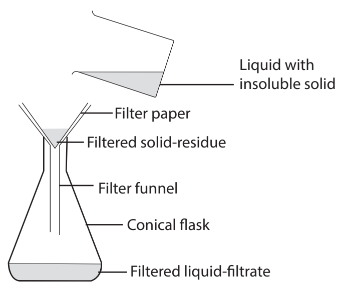


Filter pump

FILTER SPECTROMETER: A spectrometer which has one or more spectral filters for isolating one or more spectral bands.

FILTRATE: 1. The clear liquid or solution that passes through a filter during filtration. 2. Fluid extracted by the excretory system from the blood in the malpighian tubules. The excretory system produces urine from the filtrate after extracting valuable solutes from it by selective reabsorption and concentrating it.

FILTRATION: Mechanical or physical process by which solid particles suspended in a fluid are removed by passing a mixture through medium which only the fluid can pass.



Filtration

FIMBRIATE: Fringed, mostly with hairs or hair-like structures along the margin as in some petals in flowers and antennae in some insects.

FIMBRIA: Appendages that can be found on many gram-negative and some gram-positive bacteria, used in attachment to other surfaces it comes in contact with.

FIMBRILLA: A single unit of marginal fringe.

FIN: A surface used to produce lift and thrust or to steer while travelling in water, air or other fluid media. For example, the locomotory organs of aquatic vertebrates such as fishes, are typically one or more dorsal and ventral fins whose function is balance, a caudal fin around the tail which is the main propulsive organ, two paired fins and the pectoral fins attached to the pelvic girdle which are used in steering. Fins may be used in missiles, rockets, self-propelled torpedoes and kinetic energy penetrators.

FINAL ENERGY CONSUMPTION: Energy supplied that is available to the consumer to be converted into usable energy such as electricity at the wall outlet.

FINDER: 1. A small low-powered, astronomical telescope, with a wide field of view that is fixed to a large astronomical telescope so that the large telescope is pointed in the same direction to observe a particular celestial body. 2. In computer science, it is the part of the operating system of Macintosh computers, which allows users to move, copy, delete and open files, navigate through folders and move windows around the desktop. The Finder loads automatically when the computer starts up and is always running in the background.

FINE CHEMICALS: Chemicals such as a flavour or vitamin produced in small quantities usually aiming at a certain degree of purity for specific purposes; or chemical products that are made in relatively small quantities and typically at high cost. Examples include dye, petrochemicals and pharmaceuticals.

FINE EARTH: Soil particles having a high proportion of finer particles such as silt and clay with sizes of less than 2 mm in diameter.

FINE STRUCTURE: 1. Closely spaced optical spectral lines arising from transitions between energy levels that are split by the vibrational or rotational motion of a molecule or by electron spin. 2. In botany, it is also known as **ultrastructure**, which is structural details of cells below the limit of resolution of the light microscope and only revealed by the electron microscope.

FINE STRUCTURE CONSTANT (SOMMERFELD FINE-STRUCTURE CONSTANT): The dimensionless constant, with the symbol α. It was introduced by Arnold Sommerfeld in 1916 with a value of about 1/137, that characterises the strength of the electromagnetic interaction. It is given by $\alpha = \frac{e^2}{hc}$, where e is the charge on an electron, h is the Dirac constant and c is the speed of light in free space.

FINENESS OF GOLD: A measure of the purity of gold alloy, defined as the parts of gold in 1000 parts of the alloy by mass.

FINGER: 1. A digit of the hand, sometimes excluding the thumb, which is an organ of manipulation and sensation, found in the hands of humans and other primates. Humans have five digits, termed **phalanges**, on each hand. The first digit is the thumb, followed by index finger, middle finger, ring finger and little finger or pinky. 2. An approximate unit of measurement, which is equal to the length or width of a finger.

FINGER DOMAIN (RING FINGER DOMAIN): A finger-shaped structure produced in a protein when a series of the constituent amino acids bind two zinc cations. Finger domains are often found repeated in transcription factors. This protein domain contains from 40 to 60 amino acids. The acronym **RING** stands for **R**eally **I**nteresting **N**ew **G**ene.

FINGERLING: A small fish less than one year old, especially a salmon or trout.

FINGERPRINT: 1. An impression of the friction ridges of all or any part of the finger. 2. The evidence for the presence or identity of a substance that is obtained by techniques such as electrophoresis, chromatography or spectroscopy.

FINGERPRINTING: The characteristic two–dimensional paper chromatogram obtained from the partial hydrolysis of a protein or a nucleic acid.

FINISHED SEQUENCE: In bioinformatics, a high-quality determination of the base sequence of a chromosome or genome that conforms to an agreed level of accuracy. It corrects and refines the draft sequence using various experimental approaches and data analyses. It allows for only an average of one error in every 10 000 bases.

FINITARY: Of a proof, not involving infinite sets either explicitly or implicitly; for example, a proof, in a denumerable theory, involving an assertion of existence that is not supported by a construction of the entity in question essentially involves quantification over an infinite domain and so is not finitary.

FINITE: Limited in extent or scope. In mathematics, a finite set is such that the number of elements it contains can be described by a natural number. For example, the set of integers between -18 and 5 is finite, because it has a natural number (17) of elements. The set of all prime numbers, on the other hand, is not finite.

FINITE CHARACTER: The property of a collection of sets or a property that a set is in the collection or has the property if and only if this is also true of every non-empty finite subset of the given set.

FINITE ELEMENT METHOD: A numerical technique for finding approximate solutions to boundary value problems for partial differential equations. It uses subdivision of a whole problem domain into simpler parts, called finite elements and variational methods from the calculus of variations to solve the problem by minimising an associated error function. It has been used in solving structural, mechanical, heat transfer and fluid dynamics problems as well as problems of other disciplines.

FINITE ENERGY SOURCE: Energy source that cannot replenish itself after it has been exhausted. They are primary sources of energy such as fossil fuel (coal, oil, natural gas).

FINITE EXTENSION: A field L that contains a given field K and is a finite dimensional vector space over K.

FINITE FOURIER TRANSFORM (DISCRETE FOURIER TRANSFORM, DFT): In mathematics, the problem of determining the coefficients the unique polynomial p of degree n that interpolates given values a_i; at w^i for $i = 0, 1, ..., n$ where w is a primitive $(n + 1)^{th}$ root of unity either in the complex field or in a finite field. This finds application in fields such as digital signal processing, where it is widely used in signal processing and analysis. It converts a finite list of equally spaced samples of a function into the list of coefficients of a finite combination of complex sinusoids ordered by their frequencies, that has those same sample values. It can be said to convert the sampled function from its original domain (often time or position along a line) to the frequency domain. They are extremely useful because they reveal periodicities in input data as well as the relative strengths of any periodic components.

FINITE GEOMETRY: Any geometric system that has only a finite number of points. An example is geometry based on the graphics displayed on a computer screen, where the pixels are considered to be the points. The Euclidean geometry is not finite, because a Euclidean line contains infinitely many points.

FINITE GROUP: A group with cardinality of its underlying set (its order) that is finite. The cardinality of a set is the number of elements in the set.

FINITE INDUCTION (MATHEMATICAL INDUCTION): A two-part method of proving a theorem involving an integral parameter that all integers have a certain property. First the theorem is verified for the smallest admissible value of the integer. Then it is proven that if the theorem is true for any value of the integer, it is true for the next greater value. The final proof contains the two parts.

FINITE INTERSECTION PROPERTY: In mathematics, the property of a family of closed subsets so that if the intersection of any finite number of sets is non-empty, then the intersection of all the members of the family is non-empty.

FINITE MEASURE: A measure that assigns a finite value to each measurable set in its measure ring.

FINITE POPULATION: A population of finite size.

FINITE PROJECTIVE PLANE: A square block design or configuration that consists of $n^2 + n + 1$ points and $n^2 + n + 1$ lines with $n + 1$ points on each line, $n + 1$ lines through each point, each pair of lines meeting at one point and dually each pair of points lying on one line, where n is the order of the finite projective plane. Such planes exist for all prime power orders and no other order is known to be possible; no plane of order 6 exists and the case of order 10 is not settled.

FINITE SERIES: A series that has a limited number of terms in an algebraic function or a series with a finite fixed number of terms. For example, 1 + 2 + 3 +...+ 100.

FINITE-DIMENSIONAL: Of a vector space, possessing a finite maximal set of linearly independent vectors. For example, the set of ordered pairs of real numbers is finite-dimensional, but the set of real continuous functions on [0, 1] is not.

FINITE-STATE MACHINE (STATE MACHINE): A mathematical abstraction of computation used in the design of programs of sequential logic circuits. In diagrammatic representation, it is depicted as states (nodes) and the transitions between these states.

FINITELY GENERATED: Of an algebraic structure, generated by a finite number of elements.

FINITISM: The philosophy of mathematics that the only entities that may be admitted to mathematics are those that are constructible and only those propositions may be entertained whose truth can be proved in a finite number of steps.

FINSEN UNIT: A unit for measuring the intensity of ultraviolet (UV) radiation, which is equal to the intensity of ultraviolet radiation at a specified wavelength whose energy flux is 100,000 watts per square metre. Its symbol is Fu.

FIORD (FJORD): A narrow sea inlet enclosed by high cliffs often extending well inland. It results from marine inundation of a glaciated valley.

FIRE: The manifestation of rapid but persistent chemical reaction accompanied by the emission of light and heat. The reaction is self-sustaining, unless extinguished, to the extent that it continues until the fuel concentration falls below a minimum value. Most commonly, it results from a rapid exothermic combination with oxygen by some combustible material. Flame, which is the visible manifestation of fire, results from a heating to incandescence of minute particulate matter composed principally of incompletely burned fuel.

FIRE EXTINGUISHER: An active fire protection device used to extinguish or control small fires, often in emergency situations. Many domestic fire extinguishers contain liquid carbon dioxide under pressure.

FIRE POINT: The lowest temperature at which a volatile combustible substance such as fuel vapourises rapidly enough to form above its surface or an air vapour which burns continuously for at least 5 seconds when ignited by a flame.

FIRE TRIANGLE (COMBUSTION TRIANGLE): A diagram for understanding the three conditions needed for fires or combustion to occur. They are air/oxygen, fuel and heat. Removing any one of them will put out the fire.

FIRE-FLY (LIGHTNING BUGS): Any of the several winged nocturnal beetles of the family Lampyridae, consisting of about 1,900 species emitting light by the oxidation of luciferin. Adult fireflies are 525 mm long and have light-producing organs on the underside of the abdomen, which plays a role in mating.

FIREBALL: A meteor that is brighter than any planet or star, with an apparent magnitude of -5 or greater. Fireballs are often followed by meteorite falls.

FIREBLIGHT: A serious disease of pears and other trees of the family Rosaceae caused by the bacterium *Erwinia amylovora*. The disease attacks blossoms and leaves in the spring and, as the disease progresses, individual branches or whole trees look as if they have been scorched by fire. The host forms cankers in response to infection. The bacteria overwinter in the cankers and are spread by insects and rainsplash to infect flowers in the spring. Hawthorn (Crataegus), an alternate host, can also be a source of inoculum.

FIREBRICK: A brick made in a variety of shapes for use in structures, such as furnace linings, that are exposed to high temperatures. Firebrick is composed of fire clay and other non-metallic, high melting point (refractory) minerals, particularly those rich in silica, aluminium oxide, magnesium and carbon.

FIRECLAY: Clay with refractory characteristics or clay that can resist high temperatures without becoming glassy. It must withstand a minimum temperature of 1515°C (1788 K), with fusion points higher than 1600°C (1873 K). Fire clays consist of natural argillaceous materials, mostly kaolinite group clays, along with fine-grained micas and quartz and may also contain organic matter and sulfur compounds. It is used for lining furnaces, as fire brick and in the manufacture of utensils used in the metalworking industries, such as crucibles, saggars, retorts and glassware. Because of its stability during firing in the kiln, it can be used to make complex items of pottery such as pipes and sanitary ware.

FIREDAMP: A flammable gas that occurs in coal and is explosive when mixed with air in certain proportions. The gas accumulates in pockets in the coal and adjacent strata and when they are penetrated, the release can trigger explosions.

FIREWALL: A security device placed on a private network to protect it from Internet intruders, usually consisting of either a special kind of hardware router, a piece of software or both. It limits the data that can pass through it and protects a networked server or client machine from damage by unauthorized users.

FIREWIRE (IEEE 1394): In information technology, it is a high-speed Input/Output (I/O) bus mostly developed and promoted by Apple for video transmission. It has a bandwidth of 400-800 Mbps and can handle up to 63 units on the same bus. Introduced in 2000, FireWire is used for connecting peripheral devices that require fast data transfer speeds. Examples include external hard drives, video cameras and audio interfaces. On Macintosh computers, FireWire can be used to boot a computer in target disk mode, which allows the hard drive to show up as an external drive on another computer. Mac OS X also supports networking two computers via a FireWire cable.

FIREWORKS: Combustible or explosive preparations used for entertainment. Compounds of carbon, potassium and sulfur are the prime constituents in fireworks, colours being produced by metallic salts (for example, blue, copper, yellow, sodium, red, lithium or strontium, green, barium), sparks and crackles by powdered iron, carbon or aluminium or by certain lead salts.

FIRING: Heating of ceramics until they are chemically changed, the process prevents a return to the plastic state. Many ceramic clays contain matter that must vitrify (fuse) to render the ware non-porous. This may require temperatures of up to 1,450 °C. Firing is done in kilns or ovens.

FIRMWARE: A combination of hardware and software, which is usually rather small, fixed and held permanently in a computer's ROM (Read Only Memory) chip as opposed to a program that is read from external memory, as it is needed. Examples of devices containing firmware are remote controls or calculators, computer parts and devices like hard disks, keyboards, TFT screens or memory cards, mobile phones, digital cameras and synthesizers. A firmware cannot be changed or deleted by an end-user without the aid of special programs.

FIRST AID: The provision of initial care for an illness or injury until more professional medical treatment is available. It generally consists of a series of simple and in some cases, potentially life-saving techniques that an individual can be trained to perform with minimal equipment.

FIRST CLASS LEVER: A type of lever in which the pivot or fulcrum or point of support or axis of rotation is between the load and the effort. An example is a pair of scissors. In the human body, a first class lever is used when the head is raised off the chest.

FIRST COUNTABLE SPACE: A topological space having a countable base for the topology at each point of the space, as in any metric space.

FIRST DERIVATIVE: A term used for first differential, i.e., $\frac{dy}{dx}$ or $f'(x)$. In other words, the derivative of a given function, the derivative of first rather than the derivative of any derivative.

FIRST DERIVATIVE TEST: A test for optimality of a critical point of a given function using only the first derivative. If the first derivative of a function $f(x)$ is zero at a point x_0, then the point has horizontal tangent at that point. The point may be a local maximum, local minimum or neither. The critical point c is a local minimum if in some neighbourhood of c the derivative $f(x)$ is strictly positive to the left of c and strictly negative to its right. It is a local maximum if the derivative is strictly negative to the left and strictly positive to the right of c.

FIRST DIVIDED DIFFERENCE SEQUENCE (DIVIDED DIFFERENCE): A sequence of difference quotient of the form $\frac{f(x_k+1)-f(x_k)}{x_k+1-x_k}$. The second and higher divided differences are defined recursively: the $(n + 1)^{th}$ divided difference sequence is the first divided difference sequence of the n^{th} divided difference sequence with respect to the same points.

FIRST ELECTRON AFFINITY: The energy given off when 1 mole of electron is added onto the outermost shell of an isolated gaseous atom to form 1 mole of a singly charged anion. First electron affinities have negative values. For example, the first electron affinity of chlorine is -349 kJ mol^{-1}. By convention, the negative sign shows a release of energy.

FIRST FILIAL GENERATION (F1): The first offspring resulting from a cross between selected parents.

FIRST GENERATION BIOFUEL: A biofuel produced primarily from food crops such as cereals, sugar cane and vegetable oils. Second generation biofuels, also known as advanced biofuels, are fuels that can be manufactured from various types of biomass, thereby reserving the food crops from which second generation biofuels are derived for consumption.

FIRST GENERATION CELLULAR NETWORKS (1G): The first wireless cellular technology that was developed for mobile telecommunications and was introduced in the 1980s. 1G network used analogue signals, and was superseded by 2G which uses digital signals.

FIRST IN, FIRST OUT (FIFO): In computing, it is a term used to describe a method of processing and retrieving data relative to time and priority. It is a principle of queuing and items are processed in the way they are stored or entered. In a FIFO system, the first items entered are the first ones to be processed. In other words, the items are removed in the same order they are entered.

FIRST IONISATION ENERGY: The minimal amount of energy required to remove 1 mole of electron from each atom or molecule isolated in free space and in its ground electronic state.

FIRST ISOMORPHISM THEOREM (HOMOMORPHISM THEOREM): Any theorem stating that some specific algebraic structure, G, has the property that if θ is a homomorphism, then G/kerθ, where kerθ is the kernel of the homomorphism, is isomorphic to the image of G under the homomorphism.

FIRST LAW OF THERMODYNAMICS: It says that any increase in internal energy ΔU of the system, is the sum of the heat ΔQ flowing into the system and the work done ΔW on the system. It can be expressed as: $\Delta U = \Delta Q + \Delta W$. For work done by the system it is expressed as $\Delta U = \Delta Q - \Delta W$. The first law makes use of the key concepts of internal energy, heat and system work. It is used extensively in the discussion of heat engines. The standard unit for all these quantities would be the joule.

FIRST ORDER PREDICATE CALCULUS (LOWER PREDICATE CALCULUS, LPC): A formalisation of predicate calculus in which quantification is only over individuals and not over classes or predicates.

FIRST PRINCIPLES: A differentiation method that does not depend on any theorem. For example, if $y = x^3$, then $f'(x) = 3x^3$. To find this using the first principles method requires the following process:

$$f^{'}(x) = \lim_{h\to 0}\left[\frac{f(x+h)-f(x)}{h}\right] = \lim_{h\to 0}\left[\frac{f(x+h)^3-x^3}{h}\right]$$

$$= \lim_{h\to 0}\left[\frac{x^3+3x^2h+3xh^2+h^3-x^3}{h}\right]$$

$$= \lim_{h\to 0}\left[\frac{x^3+3x^3h+3xh^2+h^3-x^3}{h}\right] = \lim_{h\to 0}\left[\frac{3x^2h+3xh^2+h^3}{h}\right]$$

$$= \lim_{h\to 0}\left[\frac{h(3x^2)+3xh+h^2}{h}\right] = 3x^2+3x(0)+(0)^2 = 3x^2.$$

Also, the following method can be used:

$$y = x^3 ------(i)$$

$$y+\Delta y = (x+\Delta x)^3 ------(ii)$$

$$y+\Delta y - y = (x+\Delta x)^3 - x^3$$

$$\Delta y = x^3+3x^2\Delta x+3x(\Delta x)^2+(\Delta x)^3-x^3$$

$$\Delta y = 3x^2\Delta x+3x(\Delta x)^2+(\Delta x)^2$$

$$\Delta y = \Delta x(3x^2+3x\Delta x+(\Delta x)^2)$$

$$\lim\frac{\Delta y}{\Delta x} = \frac{dy}{dx} = 3x^2+3x(0)+(0)^2 \text{ as } \Delta x \to 0$$

$$\frac{dy}{dx} = 3x^2$$

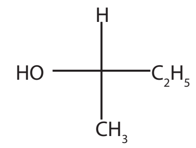
FIRST-KIND INDUCTION (INCOMPLETE INDUCTION; SPECIAL INDUCTION): An induction in which the inductive step is from the integer n to $n + 1$.

FIRST-ORDER: 1. Being or relating to the first derivative of a function. 2. Of an ordinary differential equation, involving the first derivative, but no higher order differential coefficients, of the dependent variable with respect to the independent variable. 3. Of a partial differential equation, involving no partial differential coefficient of order greater than 1. 4. In logic, quantifying only over individuals and not over predicates or classes.

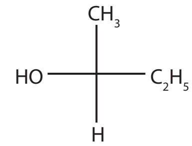
FIRST-ORDER DIFFERENTIAL EQUATIONS: A differential equation in which the highest derivative of the dependent variable is a first derivative. For example, $3x\frac{dy}{dx}+4=0.$

FIRST-ORDER REACTION: A chemical reaction whose rate depends on reactant concentration raised to the first power; or a reaction in which the rate of decrease of concentration of component A with time is proportional to the concentration of A.

FIRST-PASS EFFECT: Biotransformation and, in some cases, elimination of a substance in the liver after absorption from the intestine and before it reaches the systemic circulation.



Fischer projection of (2R)-2-butanol



Fischer projection of (2S)-2-butanol

FISCHER PROJECTION (FISCHER–TOLLENS PROJECTION): A way of representing molecules in a two dimensional configuration, the four bonds to the chiral carbon are represented by a cross with horizontal bonds representing bonds coming out of a page and vertical bonds depicting bonds lying in the plane of the page or behind the page.

FISCHER–TROPSCH PROCESS: An industrial method of making hydrocarbon fuels from carbon monoxide and hydrogen. Hydrogen and carbon monoxide are mixed in the ratio 2:1 and passed at 200 °C (473 K) over a nickel or cobalt catalyst. The resulting hydrocarbon mixture can be separated into a higher boiling fraction for diesel engines and a lower-boiling gasoline fraction. The gasoline fraction contains a high proportion of straight-chain hydrocarbons and has to be reformed for use in motor fuel. Alcohols, aldehydes and ketones are also present. The process is also used in the manufacture of SNG from coal. It is named after the German chemist Franz Fischer (1852-1932) and the Czech Hans Tropsch (1839-1935).

FISH: A limbless aquatic vertebrate belonging to three groups, the Agnathans (jawless vertebrates), cartilaginous fishes (the Chondrichthyes) and bony fishes (the Osteichthyes). They typically have an elongated somewhat spindle-shaped body (usually covered with scales) terminating in a broad caudal fin, limbs in the form of fins when present at all and a two-chambered heart by which blood is sent through thoracic gills to be oxygenated. It uses its gills for obtaining oxygen from the water and its fins and tail to swim. Fish can be bred in a specially enclosed area called a fish farm. An example of fish is tilapia, a freshwater fish.

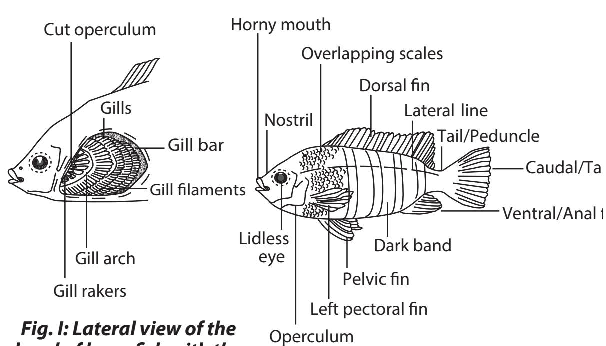


Fig. I: Lateral view of the head of bony fish with the operculum removed

Fig. II: Lateral view of a bony fish (Tilapia)

FISH-HOOK: In logic, the symbol ⥽ used to represent the relation of entailment.

FISHER INDEX: A composite number index used to study inflation or the increase in the cost of living. It is the geometric mean of two index numbers, the Laspeyres index and Paasche index. Mathematically, it is given by $\sqrt{\left(\frac{\sum P_n Q_0}{\sum P_0 Q_0}\right)\cdot\left(\frac{\sum P_n Q_n}{\sum P_0 Q_n}\right)}$, where P_n and Q_n represent the price of goods and quantity of them sold in the current period n. P_0 and Q_0 are the price and quantity sold of the same goods during reference period 0.

FISHER-BEHRENS PROBLEM: In statistics, it is the problem of determining a test for the equality of the means of two normally distributed populations with different variances, given a sample from each. It is named after Walter Ulrich Behrens and Ronald Fisher.

FISHER'S INEQUALITY: Named after Ronald Fisher, it is the result for a block design, stating that the number of blocks is always greater than or equal to the number of points. In other words, if a balanced incomplete block design is specified with parameters (v, b, r, k, λ) then $v \leq b$.

FISHERY: 1. A region of water in which fish are reared or where industrial fishing is practised. 2. A business that harvests, processes or sells fish. In other words, it is the catching, processing or selling of fish, including the industries and occupations involved in these activities.

FISHMEAL: A nutritive mealy substance prepared from fish or fish parts used as animal feed, fertiliser, etc.

FISSILE MATERIAL: A nuclide of an element that undergoes nuclear fission, either spontaneously or when irradiated by neutrons. Fissile materials can be used to fuel thermal reactor, with a neutron moderator; fast reactor, with no moderator; and nuclear explosive.

FISSION: 1. The splitting or breaking of a heavy atomic nucleus such as uranium into two or more major fragments or a form of radioactivity in which an unstable nucleus splits into two roughly equal parts. The process proceeds with the emission of neutrons and gamma rays. The neutrons are available to initiate further fissions and thus a nuclear chain reaction. Some nuclides such as Californium-252 (Cf-252) can undergo spontaneous fission, although fission is induced in other nuclides such as Uranium-235 (U-235) by incoming neutrons. The fission fragments can undergo further radioactive decay, producing a host of fission products. The following are examples of fission reaction: $(i)\,{}^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{236}_{92}U \rightarrow {}^{140}_{54}Xe + {}^{94}_{38}Sr + 2\,{}^{1}_{0}n + energy$; $(ii)\,{}^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{90}_{36}Kr + {}^{143}_{56}Ba + 3\,{}^{1}_{0}n + energy$. 2. An asexual production in unicellular organisms in which the main or parent cell divides to form two or more daughter cells.

FISSION FRAGMENTS: Nuclei species produced, which possess kinetic energy acquired from the fission process when an atom such as uranium-238 or plutonium-239 undergoes fission.

FISSION NEUTRONS: Neutrons originating in the fission process which have retained their original energy.

FISSION PRODUCTS: Atomic fragments produced by fission and the daughter products of these nuclides.

FISSION YIELD (FISSION PRODUCT YIELD): The ratio of the nuclide of the fission product to the original nuclide expressed as a percentage.

FISSION-TRACK DATING: A method of dating geological and archaeological specimens and other mineral objects by observing the tracks made in them by the fission fragments of the uranium nuclei that they contain by using a chemical reagent to dissolve the altered material; the tracks can then be seen in an ordinary microscope.

FISSURE: A long, narrow opening or line of breakage made by cracking or splitting in a rock along, which there is a distinct separation. Fissures are often filled with mineral-bearing materials.

FISTULA: In medicine, it is an abnormal passage between a hollow or tubular organ and the body surface or between two hollow or tubular organs. Its causes include infection, Crohn's disease, tumours, trauma and surgery. The gastrointestinal tract, particularly the duodenum, pancreas, bladder and female genital tract are particularly susceptible.

FIT: In statistics, it is the degree of correspondence between observed and predicted characteristics of a distribution or model. 2. A sudden violent attack of an acute disease or the sudden appearance of its symptoms. 3. The suitability of an organism to prevailing conditions of its environment. This is an indication of its ability to live, survive and reproduce

FITNESS: 1. The state of being in a healthy physical condition or the condition of an organism that is well adapted to its environment, as measured by its ability to reproduce itself. Fitness involves not only physical factors, but also has intellectual, emotional, social and spiritual components. Fitness is also specific to a particular physical activity. 2. In evolutionary theory, it is the relative ability of an organism to survive and reproduce and is equal to the average contribution to the gene pool of the next generation that is made by an average individual of the specified genotype or phenotype. It can be defined either with respect to a genotype or to a phenotype in a given environment.

FITZGERALD CONTRACTION: The contraction of a moving body in the direction of its motion relative to an observer.

FIXATION: 1. The first stage in the preparation of a specimen for microscopic examination in which the tissue is killed and preserved in as natural a state as possible by immersion in a chemical fixative. 2. A theoretical strong libidinal attachment to a person or object formed during early childhood that results in neurotic or arrested psychosexual behaviour in adulthood. 3. The conversion by soil bacteria of atmospheric nitrogen to a stable biologically available form. 4. The process of stabilising a chemical or compound.

FIXED ACTION PATTERN: An innate, highly stereotyped behaviour that must be completed once initiated. Once initiated, fixed action patterns cannot be interrupted. Examples are yawning and some moths instantly fold their wings and drop to the ground if they encounter ultrasonic signals such as those produced by bats.

FIXED IONS: In an ion exchanger, the non-exchangeable ions which have a charge opposite to that of the counterions.

FIXED NEUTRAL LOSS (GAIN) SCAN: A scan which determines, in a single experiment, all the parent ion mass-to-charge ratios which react to the loss or gain of a selected neutral mass.

FIXED POINT: 1. A temperature that can be accurately reproduced to enable it to be used as the basis of a temperature scale. In the Celsius scale, for example, the fixed points are the melting ice temperature at 0 °C (273.15 K) and boiling water temperature at 100 °C (373.15 K). 2. Also known as **invariant point**, it is the point of a function that is mapped into itself by a given transformation; for example, 0 and 1 are fixed points of $f(x) = x^2$, since $f(0) = 0$ and $f(1) = 1$. 3. A point that lies in its own image under a given correspondence. 4. Of, relating to or being a mathematical notation, as in a decimal system, writing numerical quantities in full with the point separating the integral from the fraction part being fixed as contrasted with floating point notation in which variable multiples of the base are extracted as factors.

FIXED POINT ITERATION: An approximate method for finding the root of an equation $p(x) = 0$. The first step is to write the equation as $x = f(x)$. The next procedure is to choose a value x_0 to be an approximation to the true root. Subsequently, approximate values of the root are found by using the equation $x_{n+1} = f(x_n)$. If the values of x_n tend to a limiting value of L as n increases then $L = f(L)$. This means L is the root of the equation $p(x) = 0$.

FIXED POINT THEOREM: A theorem that gives conditions for a mapping to have a fixed point. Examples are Banach fixed point theorem and Brouwer's fixed point theorem. For Banach fixed point theorem, for example, let f be a contraction mapping from a closed subset F of a Banach space E into F; then there exists a unique $z \in F$ such that $f(z) = z$.

FIXED POPULATION: A group of individuals defined by a common fixed characteristic, such as all girls born in Accra in 1998. Membership of such a population does not change over time by immigration or emigration.

FIXED PRECURSOR ION SCAN: A scan which determines, in a single experiment, all the daughter ion mass-to-charge ratios that are produced by the reaction of a selected parent ion mass-to-charge ratio.

FIXED PRODUCT ION SCAN: A scan which determines, in a single experiment, all the parent ion mass-to-charge ratios that react to produce a selected daughter ion mass-to-charge ratio.

FIXED PRODUCT ION SPECTRUM: A spectrum obtained when data are acquired that determine all the parent ion mass-to-charge ratios that react to produce a selected daughter ion mass-to-charge ratio.

FIXED SET: A set S such that, for a given mapping T, that may be multivalued, $T(S) = S$.

FIXED STAR: One of many heavenly bodies so distant from Earth that it does not appear to alter its position on the celestial sphere. Its movements can be measured only by precise observations over long periods of time.

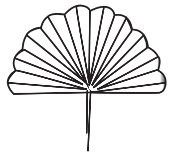
FIXED TEMPERATURE: The point of reference of temperature variation of matter.

FIXED-POINT NOTATION: A system in which numbers are represented, using a set of digits with the decimal point always in its correct position. The position of the point determines the absolute precision of the representation. If the point is fixed at the right-hand end of the number then all of the fixed-point numbers are integers.

FIXING NITROGEN: Any process which converts atmospheric nitrogen (N_2) into compounds which can be used by plants and thus become available to animals and humans.

FIZEAU'S METHOD (FIZEAU EXPERIMENT): A method of measuring the speed of light in moving water. It was invented by the French physicist Armand Fizeau. In his original experiment conducted in 1849, he sent light pulses through a rotating toothed wheel. A distant mirror on the other side reflected the pulses back through gaps in the wheel. By rotating the wheel at a certain speed, each light pulse that went through a gap on the way out was blocked by the next tooth as it came around. Knowing the distance to the mirror and the speed of rotation of the wheel enabled Fizeau to obtain one of the earliest measurements of the speed of light.

FLABELLATE: Shaped like a fan; having the upper portion prolonged into long branches.



Flabellate structure

FREE: In botany, it describes a plant that has become soft and less rigid due to the shrinking and contraction of the cytoplasm away from the cell walls as a result of the loss of water.

FLACCIDITY: Flaccidity refers to the condition of being flaccid or describes something that is weak, soft or lacking vigor as for example, a plant cell in an isotonic solution, such that the plasma membrane is not pressed tightly against the cell wall and, therefore, is neither turgid (swollen) nor plasmolyzed. The plant cell will lose its flaccidity when placed in a hypotonic or hypertonic solution.

FLAG: 1. An Boolean variable used as indicator in computing for the result of a test and so can be used as a condition to switch different parts of a program into action. For example, it can be used to indicate whether the end of a file has been reached or whether an overflow error has occurred. 2. In geometry, a triple consisting of a half plane, a boundary half line and its endpoint.

FLAG VALUE (DATA TERMINATOR; ROGUE VALUE; SENTINEL VALUE; SIGNAL VALUE; TRIP VALUE): In computing, it is a special value in the context of an algorithm that is used to mark the end of a list of input data items. The computer must be able to detect that the data terminator is different from the input data in some way, for example, a negative number such as -1 might be used as the sentinel value as the computation will never encounter that value as a legitimate processing output; otherwise the presence of such values would prematurely signal the end of the data. A lot of old programs used data terminators and many programmers used 00 as a year data terminator, leading to problems in the year 2000.

FLAGELLATE: Any single-celled organisms of the class Mastigophora. Flagellates possess, at some stage of their development, one or several whip-like structures called flagella for locomotion and sensation. They are divided into two major groups: the phytoflagellates resemble plants (in that they obtain their energy through photosynthesis) and the zooflagellates resemble animals because they obtain their energy through feeding. Reproduction may be asexual fission or sexual.

FLAGELLIFORM: Long, thin and tapering; whip-like as in the antennae of certain insects.

FLAGELLUM: A relatively long fine hair-like structure possessed by certain cells, particularly motile reproductive cells such as spermatozoa, algae, bacteria and certain protozoan. Flagellar motion causes water currents necessary for respiration and circulation in sponges and cnidarians.

FLAMBOYANT (FLAME TREE; ROYAL POINCIANA; *Delonix regia*; *Poinciana regia*): A member of the bean family, Leguminosae, that grows up to 15 m tall. Its numerous striking flame-like scarlet and yellow flowers are clustered above the leaves making it as one of the most beautiful tropical trees in the world. The leaves are bipinnately compound, fern-like, with small individual leaflets, which fold up at dusk. As the trees mature, they develop broad umbrella-shaped crowns that can be wider than its height, providing good shades. Its woody seedpods are brown to blackish, reaching up to 60 cm. It is native to Madagascar but widely cultivated in tropical and sub-tropical regions worldwide. It is propagated from seeds and cuttings.

FLAME: A hot luminous mixture of gases undergoing the process of combustion or light and heat energy given off when certain substances react and hot gases are produced. Flames generally consist of a mixture of oxygen or air and another gas, usually combustible substances such as hydrogen, carbon monoxide or hydrocarbon.

FLAME CELL: A hollow specialised excretory cell that has a tuft of projections or cilia resembling hairs, whose movement serves to force out waste products in some invertebrates such as flatworms.

FLAME LILY (CLIMBING LILY; CREEPING LILY; GLORY LILY; GLORIOSA LILY; FIRE LILY; *Gloriosa superba*): A perennial herbaceous climbing plant in the family Colchicaceae, native to much of Africa and Asia, growing in sparse savanna woodlands, grasslands, sand dunes, in abandoned fields or at the boundaries of cultivated ground and roadsides; in sandy-loam soil. It grows from a fleshy rhizome, is scandent, with a stem reaching 4 metres long and climbs using leaf-tipped tendrils. The leaves are lance-shaped and may be alternately or oppositely arranged with wavy margins. The flowers are bright-red to orange at maturity. It is propagated vegetatively by dividing the rhizome or sexually by seeds. It is grown as medicinal and ornamental plant. All parts of the plant, especially the tubers are extremely poisonous due to the presence of toxic alkaloids, including colchicine; and is toxic enough to cause fatalities to humans and animals if ingested. Medicinally, it has been used to treat various diseases such as gout, infertility, wounds, snakebite, ulcers, arthritis, cholera and colic.

FLAME NETTLE: Any of low growing, semi-woody, evergreen plants of the genus *Coleus* in the Lamiaceae family, having multicoloured decorative leaves, The plants grow quickly to heights of about 30-90 centimetres. The leaves are arranged oppositely and have serrated edges on branching plants, have varying shapes and sizes, with an infinite number of leaf colours such as yellow, black orange, red, green or brown or a combination of these colours. Some of the leaves are heart-shaped, others are slender or contorted pendulously. The flowers are inconspicuous and the plants are grown as houseplants for their variegated decorative and colourful foliage or leaves. Plants can be propagated by seeds or cuttings.

FLAME PHOTOMETRY: The use of emission spectroscopy in the ultraviolet and visible regions to identify and estimate the amounts of various elements which are excited in a flame, an arc or high voltage spark.

FLAME TEST: The use of a flame to detect metal cations present in a solid, based on each element's characteristic emission spectrum.

FLAME-IN-TUBE ATOMIZER: An atomizer in which hydrogen, used as carrier gas, is partially combusted in the inlet arm of a T shaped tube furnace by introducing a limited amount of oxygen or air.

FLAMING: In computer science, it is the act of posting or sending offensive messages known as **flames** over the Internet. The most common area where flaming takes place is online discussion forums, which are also called **bulletin boards**.

FLAMMABLE (INFLAMMABLE): A term describing a substance that is easily ignited and capable of burning rapidly, usually a gas or the vapour from a volatile liquid such as methane and petrol.

FLANGE: In mechanical engineering, it is a projecting collar, rib or rim on an object for fixing it to another object, holding it in place or strengthening it. Flanges are often found on shafts and pipes.

FLARE: A brilliant eruption on the Sun above a sunspot, thought to be caused by the release of magnetic energy. Flares develop in a few minutes and may last several hours, releasing intense x-rays and streams of energetic particles. Flares can disrupt radio communications and cause auroras and may pose a radiation hazard to astronauts.

FLASH: 1. A sudden quick very bright light. 2. The method or apparatus used for taking photographs in darkness.

FLASH BIOS: BIOS is the abbreviation for **Basic Input Output System** and Flash Bios is a chip that is capable of being written to by software or code rather than by a ROM chip. To update a ROM BIOS, you have to remove the chip and replace it with a newer version. However, with the flash BIOS you can boot with a special disk or execute a set of instructions and update the BIOS without having to open the case.

FLASH DRIVE (USB DRIVE; PEN DRIVES; USB KEYCHAIN DRIVES): In computer science, it is a plug-and-play portable data storage device that uses flash memory and has a built-in USB connection. Flash drives can be written and rewritten to almost unlimited number of times, much like a standard hard drive. Unlike most removable drives, a USB drive does not require rebooting after it is attached, does not require batteries or an external power supply and is not platform dependent. Storage capacity ranges from about 16MB to 1 TB.

FLASH FLOOD: A flood of water in a normally dry low-lying area brought about by a sudden downpour of rain or a dam break.

FLASH LAMPS: Lamps which contain an inert gas which can be rapidly pulsed.

FLASH MEMORY: A type of EEPROM (Electronically Erasable-Programmable Read-Only Memory) memory that retains its data when the power is turned off and can be electronically erased and reprogrammed without removal from the computer.

FLASH PAPER (CELLULOSE NITRATE; NITROCELLULOSE): A highly flammable material made by treating cellulose (wood pulp) with concentrated nitric acid. Despite the alternative name *nitrocellulose*, the compound is in fact an ester containing $CONO_2$ groups, not a nitro compound, which would contain $C–NO_2$. It is used in making explosives, plastics and varnishes.

FLASH PHOTOLYSIS: A technique for studying free-radical reactions in gases in which a sample is firstly excited by a strong pulse called pump pulse of light from a laser of nanosecond, picosecond or femtosecond pulse width or by a short-pulse light source such as a flash lamp.

FLASH POINT: The lowest temperature at which a liquid or volatile solid heated under standard conditions, gives off sufficient vapour to ignite on the application of a small flame or the temperature at which the vapour above a volatile liquid forms a combustible mixture with air.

FLASH SPECTRUM: An emission spectrum of the solar chromosphere, obtained by placing an objective prism in front of the telescopic lens the instant before or after totality in a solar eclipse.

FLASH VACUUM PYROLYSIS (FVP): Thermal reaction of a molecule by exposing it to a short thermal shock at high temperature, usually in the gas phase.

FLASHBACK: A reversal of flame propagation in a system, counter to the usual flow of the combustible mixture.

FLASK: 1. A narrow-necked bottle or a container with a round body and a narrow neck used for holding liquids in laboratories. Examples are the flat-bottom flask, round bottom flask, volumetric flask and the conical flask. 2. Also called **vacuum flask** or **thermos flask**, it is a container designed to keep warm drinks warm and refrigerated drinks cold. A **thermos flask** prevents heat loss from the fluid inside it, due to any of the three heat transfer mechanism: convection, conduction or radiation. Similarly it prevents the heat from outside, from getting into the material.

FLAT DECK PIGGERY: A piggery used for rearing weaned piglets from between two and eight weeks of age. It has a mesh floor, sell-feed hoppers and controlled heating and ventilation.

FLAT FIELD: A lens or optical system that corrects spherical aberration and gives uniform focus across the field of view.

FLAT FILE DATABASE: A database that stores data in a plain text file. Each line of the text file holds one record, with fields separated by delimiters, such as commas or tabs. While it uses a simple structure, a flat file database cannot contain multiple tables like a relational database can. Usually, there is no structural relationships between the records.

FLAT SURFACE: A two dimensional surface contained entirely in one plane without any thickness.

FLATBED SCANNER: A type of image scanner that uses a flat, glass surface for scanning documents or other objects. In a flatbed scanner, the original is placed face-down on a sheet of glass and an arm equipped with a light and an electronic sensor sweeps under the glass. Flatbed scanners are particularly effective for bound documents.

FLATWORM: A type of worm belonging to the phylum Platyhelminthes, which has a flat non-segmented body with a mouth at one end. There are three main types: turbellarians, trematodes and cestodes. Turbellarians are mostly free-swimming, but trematodes and cestodes are parasites. Flatworms reproduce asexually both by binary fission and by regeneration or producing an entire new worm from a piece that has been cut off.

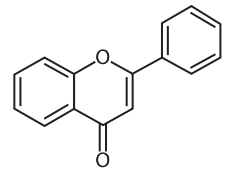
FLAVANONES: Any of a large group of water-soluble plant pigments, including the anthocyanins, with members lacking a double bond between carbons two and three of the C_3 group in the centre of the flavonoid nucleus. An example is naringenin, which is responsible for the bitterness of lemons.

FLAVIN: Any of various water-soluble yellow pigments, including riboflavin, found in plant and animal tissue as coenzymes. In plants, they have an absorption peak around 370 nm that are active in certain plant growth responses to light. Flavins have been implicated in phototropism and are also thought to transfer light energy from blue to red wavelengths hence influencing the function of phytochrome.

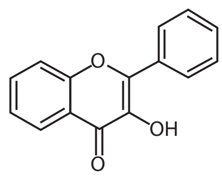
FLAVIN ADENINE DINUCLEOTIDE (FAD): A coenzyme, with the chemical formula $C_{27}H_{33}N_9O_{15}P_2$, which is important in various biochemical reactions. It is a derivative of phosphorylated vitamin B_2 (riboflavin) molecule linked to the nucleotide adenine monophosphate (AMP). The primary biochemical role of $FADH_2$ in eukaryotes is to carry high-energy electrons used for oxidative phosphorylation.

FLAVIN MONONUCLEOTIDE (FMN): A riboflavin-derived coenzyme, similar to flavin adenine dinucleotide (FAD), that acts as a prosthetic group to several dehydrogenases. NADH dehydrogenase, the enzyme that catalyses transfer of electrons from NADH to coenzyme Q in the respiratory chain, is an FMN-containing enzyme.

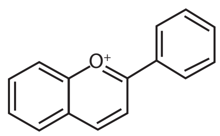
FLAVONE: 1. A colourless, water-insoluble compound crystalline compound with the molecular formula $C_{15}H_{10}O_2$ that gives the ivory to deep yellow colouring of plants. It occurs on the leaves or in the stems and seed capsules of many primroses. 2. Any of the derivatives of this compound, which occur as yellow plant pigments in the form of glycosides and used as yellow dyes.



Flavone backbone



Flavonol backbone



Anthocyanine backbone

FLAVONOID: One of the group of naturally occurring phenolic compounds containing no nitrogen, many of which are plant pigments. They include the anthocyanins, flavonols and flavones. Anthoxanthins give yellow colours, often to flower petals; and anthocyanins, are largely responsible for the red colouring of buds and young shoots and the purple and purple-red colours of autumn leaves.

FLAVONOLS: A group of flavonoid compounds, quercitin and kaempferol, with members having a hydroxyl group on the third carbon of the C_3 group in the centre of the flavonoid nucleus. They are formed by the addition of a hydroxyl ion to a flavanone.

FLAVOPROTEIN: Any of a group of enzymes with flavin bound to protein that acts as dehydrogenases. The general equation for a flavoprotein reaction is: flavoprotein + reduced substrate ↔ reduced flavoprotein + oxidised substrate. Like Nicotinamide adenine dinucleotide (NAD), reduced flavoproteins can be reoxidised in the respiratory chain. Flavoproteins are also components of the electron transport system of photosynthesis and the microsomal electron transport system.

FLAVOUR: In physics, it refers to any of six classifications of quark varieties distinguished by mass and electric charge. The six flavours are: **up** (**u**), **down** (**d**), **strange** (**s**), **charm** (**c**), **bottom** (**b**) and **top** (**t**), in increasing order of mass. These govern the weak interaction dynamics of the quarks, the flavours being essentially the 'weak charges' on the quarks. For each charged lepton flavour there is a corresponding neutrino flavour.

FLAX: An annual plant of the Linaceae family grown for its fibres of cellulose which are used to make linen, thread, twine and writing paper. Some varieties are grown for their seed, the **flaxseed** to produce linseed oil.

FLEA: A common name for insects of the order Siphonaptera, which are small jumping insect without wings, with mouthparts adapted for piercing skin and sucking blood and feeds on the blood of humans and animals.

FLEA BEETLE: Any of a subfamily (Alticinae, especially genera Alticia and Epitrix) of small chrysomelid beetles with legs adapted for leaping. It is tiny and less than 6 mm long and dark or metallic in colour. It causes damage to crops such as grapes, cucumbers, melons, tobacco, potatoes and tomatoes, especially during hot weather period between April and mid-May.

FLEDGE: 1. The stage in a young bird's life, when the feathers and wing muscles are developed and the bird is capable of flight. 2. The act of raising chicks to a fully-grown state by the parents.

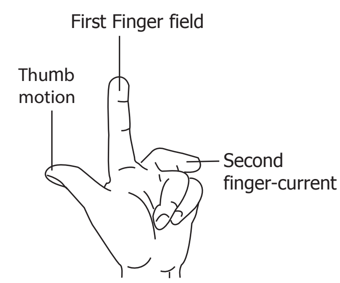
FLEDGLING: A young bird that has recently fledged, but is still being fed and cared for by the parents.

FLEECE: A coat of wool covering a sheep or goat consisting of cylindrical fibres of keratin covered by minute overlapping scales. It is valued as a textile fabric.

FLEMING, ALEXANDER (1881–1955): British biologist, pharmacologist and botanist, who is remembered for the discovery of lysozyme and penicillin, an antibiotic. Lysozyme is an enzyme present in many body tissues and lethal to certain bacteria. For the discovery of penicillin from the mould *Penicillium notatum* in 1928, he shared the Nobel Prize in Physiology or Medicine in 1945 with Howard Florey and Ernst Boris Chain.

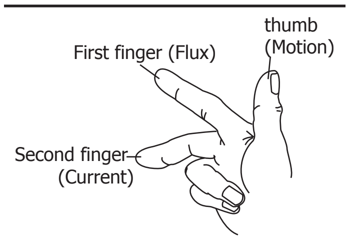
FLEMING, JOHN AMBROSE (1849-1945): English electrical engineer, who is often called the father of modern electronics and best known for developing the first successful thermionic valve (also called a vacuum tube, a diode or a Fleming valve) in 1904, which gave birth not only to radio communications, but to the entire electronics industry. Fleming's valve was a rectifier or diode. It consisted of two electrodes in an evacuated glass envelope resembling an electric light bulb. The diode permitted current to flow in one direction only. It could detect radio signals but could not amplify them. He is also remembered for Fleming's left hand rule.

FLEMING'S LEFT-HAND RULE: A rule showing the direction of motion of the force acting on a conductor when the electric current flowing in it is at right angles to a magnetic field. It can be used to find the direction of rotation of simple electric motors. If the forefinger, second finger and thumb of the left hand are extended at right angles to each other, the forefinger indicates the direction of the field, the seCond finger the direction of the Current and the thumb the direction of Motion of the force. It was invented by John Ambrose Fleming (1849-1945), English electrical engineer and physicist.



Fleming's Left-Hand Rule

FLEMING'S RIGHT-HAND RULE: A way of determining in which direction current flows invented by John Ambrose Fleming (1849-1945), English electrical engineer and physicist to remember the rules for the relative directions of the field, current and force in electrical generators. If the thumb and first two fingers of the right hand are held at right angles to each other with the First finger pointing in the direction of the magnetic Field or flux and the thumb in the direction of Motion of the conductor relative to the magnetic field, then the second finger points in the direction of the induced current. This means the finger points from north to south. When a conductor such as the wire on the armature of a generator moves between the poles of a magnet, the conductor crosses a magnetic field facing from the north (N) pole to the south (S) pole of the magnet; electromotive force is thus generated in the conductor and current flows. Fleming's right-hand rule is used to determine the relationship between the directions of the magnetic field, the movement of the conductor and the current in generators and dynamos.



Fleming's Right-Hand Rule

FLESHY: In biology, having a soft thick inner part. For example, most fleshy fruits such as mango and pawpaw are made of living cells and are often juicy and sweet. Also, the part of an animal that does not contain bones such as the abdomen can be described as fleshy.

FLEX-FUEL (FLEXIBLE-FUEL VEHICLE (FFV); DUAL-FUEL VEHICLE; FLEX-FUEL VEHICLE): A vehicle which is designed to run on petrol blended with either ethanol or methanol fuel.

FLEXOGRAPHY: A rotary letterpress printing using flexible plates and fast-drying inks that can print on any type of substrate, including plastic, metallic films, cellophane, and paper; commonly used for printing on non-porous substrates required for various types of food packaging.

FLEXOR: A muscle that when contracted acts to flex or bend a joint or limb in the body. For example, the elbow is flexed when the hand is brought closer to the shoulder. The trunk may be flexed toward the legs or the neck to the chest. Examples of flexor muscles are brachialis, brachioradialis and biceps brachii in the forearm at the elbow; pectoralis major, anterior deltoid and coracobrachialis in the humerus at the shoulder; and flexor carpi radialis and flexor carpi ulnaris of the carpus at the wrist.

FLEXUOSE: With curved branches, winding or zigzagged.

FLICKER: A visible change in brightness or unsteadiness in an image displayed on screens. Flicker occurs when the video rate is very low. When this occurs frequently, it makes the user of the experience eyestrain.

FLICKR: An online image and video hosting platform that is popular with users who share and embed photographs. It was created by Ludicorp on February 10, 2004, and acquired by Yahoo in 2005. The As of October last year, Flickr had more than 112 million users in more than 63 countries.

FLIGHT: 1. Any form of locomotion in air, which can be active or passive such as gliding. Mechanisms of flight have evolved mainly in birds, bats and insects. These animals are adapted for flight by the presence of wings, which increase the ratio of surface area to body weight. 2. Part of a survival mechanism in an animal that is generated in response to a threatening situation. A potentially dangerous situation can induce the release of adrenaline, which prepares the animal for 'fight or flight' by increasing the blood pressure and heart rate and diverting the blood flow to the muscles and heart.

FLIGHT FEATHERS (REMIGES): Stiff long feathers of a bird that is adapted for flight, consisting of primary, secondary, and tertial feathers used to provide lift and control direction during flight. **Primary flight feathers** are about 9-10 feathers responsible for thrust to propel the bird forward. **Secondary flight feathers** are, innermost, stiff long flight feathers close to the body of the bird, positioned just behind the primary flight feathers on the rear edge of the wings. Many birds have six secondary feathers. **Tertial flight feathers** are three flight feathers that are closest to the bird's body along the wing, located next to the secondary flight feathers.

FLIGHT MODE (AIRPLANE MODE): A setting for devices such as cell phones and other mobile communication devices that disables all wireless communications without having to turn off the device. This is to prevent any possible interference with the airplanes onboard communication systems and also with cellphones on the ground. During a flight, many airlines and governments will request electronic devices be turned off or switched to flight mode. Flight mode does not disable non-wireless functions such as listening to music and playing games.

FLIGHT PATH: 1. The line, course or track along which an aircraft is flying or is intended to be flown. 2. Also known as the **spacecraft's trajectory**, it is the path followed by the centre of gravity of a spacecraft during flight. The path is tracked with reference to any point on Earth or it may be tracked with reference to a star.

FLIGHT SIMULATOR: A computer controlled pilot training device, consisting of an artificial cockpit mounted on hydraulic legs that stimulates the experience of flying a real aircraft. Flight simulation is used for a variety of reasons; including flight training mainly of pilots, for the design and development of the aircraft itself and for research into aircraft characteristics, control handling qualities, etc.

FLIGHTLESS BIRD: A bird which has small wings and lacks the ability to fly, relying instead on its ability to run or swim. Examples are ostrich, emu, cassowary, rhea, kiwi and penguin.

FLINT (FLINTSTONE): A compact, hard, brittle mineral, brown, black or grey in colour, found in nodules and masses in sedimentary rocks, such as chalks and limestones. It produces a spark when struck with steel. It also refers to a form of chert which occurs in chalk or marly limestone.

FLINT GLASS: Flint glass is a highly-refractive lead-containing glass used to make lenses and prisms. Because it absorbs most ultraviolet light but comparatively little visible light, it is also used for telescope lenses. Flint glass typically has a much higher refractive index than does crown glass and exhibits much higher dispersion, however it is less resistant to damage. Flint glass is used in conjunction with crown glass to make achromatic lenses. The most common types of glasses used in optics are flint glasses and crown glasses, designations based on their dispersions.

FLIP FLOP (BISTABLE CIRCUIT): An electronic circuit that has two stable states (0 and 1) that can be used to store state information. It switches from one stable state to the other by means of a triggering pulse. types of flip flop include SR, JK, D and T.

FLIP-FLOP: The movement of a lipid molecule from one surface of a lipid bilayer membrane to the other, which occurs at a very slow rate. This contrasts with the much faster rate at which lipid molecules exchange places with neighbouring molecules on the same surface of the membrane.

FLIXWEED: A common annual weed *Descurainia sophia* of the mustard family, which reproduces by seeds.

FLOATING: A state of equilibrium in which a body rests on or is suspended in the surface of a fluid (liquid or gas). According to Archimedes' principle, a body wholly or partly immersed in a fluid will be subjected to an upward force or upthrust, equal in magnitude to the weight of the fluid it has displaced.

FLOATING POINT NUMBER (FLOATING-POINT NUMBER; FLOATING-POINT NOTATION): A binary number carried out to a specific number of digits and containing a radix (or decimal) point somewhere in the digit sequence. Such numbers are stored in three parts: the sign (either plus or minus), the mantissa (sequence of meaningful digits) and the exponent (power or order of magnitude), which determines the position in the mantissa. Examples are 5.5, 0.001 and -2,345.6789. Numbers that do not have decimal places are called **integers**. The main operations of the FPU consist of conventional arithmetic such as addition and multiplication. Some FPUs can also perform more sophisticated functions such as exponentials, logarithms and trigonometry.

FLOATING POINT UNIT (FPU; MATH COPROCESSOR; NUMERIC COPROCESSOR): A specialised coprocessor of a computer system specially designed to carry out operations on floating point numbers. The FPU is now a physical part of the microprocessor chip.

FLOATING RIB: Any of four atypical ribs attached only to the spine and not to the breastbone. In humans, the two lower ribs on each side are floating ribs.

FLOCCOSE: Bearing tufts of long, soft, tangled hairs. **Flocculent** means bearing tufts of very fine woolly hairs.

FLOCCULATION: 1. A process that causes particles suspended in water to aggregate into clumps or masses that then sink or can be removed by filtering or aggregate in this way. 2. To form fluffy masses or cause clouds to form fluffy masses.

FLOCCULE: A small loosely aggregated mass of material suspended in or precipitated from a liquid; a cluster of particles.

FLOCK: A large group of birds or some farm animals such as sheep and goats.

FLOCK MATING: A mating system in which several males are mated with the females of a flock.

FLOCKMASTER: A farm worker in charge of a flock of sheep or goat.

FLOOD: A large amount of water covering land that is usually dry. It is caused by phenomena such as melting snow, heavy rain, high tides or storms.

FLOOD IRRIGATION: A method of surface irrigation using water brought down by a river in flood, which is applied and distributed over the soil surface by gravity. The floodwaters are led off into specially prepared basins or ditches and water is released from them to flood over the land.

FLOOD PLAIN: An area of relatively flat land of periodic flooding along the course of river valleys. It is built up of sediments called alluvium, from overflow of the river and subject to inundation when the river is at flood stage. Such plains usually have extremely fertile soil and are often used, as near the River Nile, for intensive cultivation.

FLOOR FUNCTION (GREATEST INTEGER FUNCTION; INTEGER VALUE): The floor function of x is the greatest integer less than or equal to x or floor(x) = $\lfloor x \rfloor$ is the largest integer not greater than x.

FLOPPY DISK (DISKETTE): A round, flat flexible plastic disk with a magnetic coating encased in a stiff cover. It is used to store information in a small computer system. Data is stored on a floppy disk by the disk drive's read/write head, which alters the magnetic orientation of the particles. Orientation in one direction represents binary 1; orientation in the other direction is binary 0. It can store between a few hundred thousand to over one million bytes of data.

FLOPS (FLOATING POINT OPERATIONS PER SECOND): A measure of the performance of a computer's processor. It is usually noted as MFLOPS or Million FLOPS. While clock speed, which is measured in megahertz, is often seen as an indicator of a processor's speed, it does not define how many calculations a processor can perform per second. Therefore, FLOPS is a more raw method of measuring a processor's processing speed.

FLOQUET THEOREM: A second-order linear differential equation whose coefficients are periodic single-valued functions of an independent variable x has a solution of the form $e^{\mu x} P(x)$, where μ is a constant and $P(x)$ a periodic function.

FLORA: 1. All the plant life of a particular place or period. It is spelt with a lower case 'f'. 2. Flora spelt with a capital 'F' refers to a book or other work describing a flora including aids for the identification of the plants it contains such as botanical keys and line drawings that illustrate the characters that distinguish the different plants.

FLORAL DIAGRAM: A diagram showing a flower in cross section showing the number and arrangement of floral parts. An ovary is drawn in the centre, surrounded by representations of the other floral parts, in the correct sequence indicating the position of each at its base.

FLORAL ENVELOPE (PERIANTH): The part of a flower that surrounds the stamen and pistil. It is a collective term for corolla and calyx. **Floral tube** is a tube formed by the fusion of parts of a perianth, usually, with their stamens at their base.

FLORAL FORMULA: A symbolic representation of the structure of a flower using specific letters, numbers and symbols. Each kind of floral part is represented by the letter K for calyx, C for corolla, P for perianth, A for androecium, G for gynoecium and a number to indicate the quantity of the part present, for example, C_5 for a flower with five petals. It also indicates whether the ovary is inferior or superior.

FLORAL TUBE: An elongated tubular portion of a perianth.

FLORET: A small flower, especially one in a cluster or an inflorescence, as in the grass (Gramineae) and daisy (Compositae). Flower florets have no sepals or petals and the structures are represented by two tiny scales, called **lodicules**, at the base of the floret, outside the usually three stamens. Examples are Hebianthus (sunflower), Tithonia and *Tridax* species. Grass florets are grouped into units termed spikelets. A spikelet consists of an axis, called a **rachilla**, along which are borne the florets. Each floret is enclosed by a scale called a **lemma** on the outside and another called a **palea** on the inside. At the base of the rachilla are two scales called **glumes**, which are empty. The glumes and lemmas are usually hard in texture and often extend into a bristle or awn; the palea is delicate and membranous.

FLORICANE: The flowering and fruiting stem of a biennal plant, for example, raspberry.

FLORICULTURE: A discipline of horticulture concerned with production and distribution of cut flowers, potted plants, greenery and flowering herbaceous plants.

FLORIDEAN STARCH: The assimilation product of algae in the Rhodophyta. It has a structure similar to amylopectin.

FLORIFEROUS: Bearing flowers, especially blooming freely.

FLORIGEN (FLOWERING HORMONE): A hypothetical plant growth molecule produced in the leaves that acts in the shoot apical meristem of buds and growing tips. It is postulated to transmit the stimulus for flowering which is a response to photoperiod from the leaves to the apex of a plant. This stimulus would prompt the apical meristem to switch from producing leaves to making flowers.

FLORIST: A shop or somebody that sells flowers and other ornamental plants.

FLORISTICS: The branch of botany concerned with identifying and listing all the plant species present in a particular area. The check list so produced gives the floristic composition of the region. Once this has been established, further work may be undertaken to describe the abundance and distribution of the various species and to consider the affinities of the regional flora.

FLORY TEMPERATURE (THETA TEMPERATURE): The unique temperature at which the attractions and repulsions of a polymer in a solution cancel each other or the temperature at which the coiled polymer molecules expand to their full contour lengths and become rod-shaped.

FLOSS: 1. The mass of fine silken fibres that covers the seeds of a cotton plant or of the silk-cotton tree. It is also short or waste silk fibres, especially from the outer surface of the cocoon of a silkworm or soft, loosely twisted thread, as of silk or cotton, used in embroidery. 2. Also known as **dental floss**, it is a tuft of thin filaments used to remove food particles and dental plaque from inbetween the teeth.

FLOTATION: 1. The power or ability of a body or an object to float in a fluid. The law of flotation states that a floating body displaces its own weight of fluid in which it floats. 2. The separation of the particles of a mass of pulverized ore according to their relative capacity for floating on a given liquid.

FLOTATION PROCESS (FLOTATION SEPARATION): A common method for separating different materials, used in the wastewater treatment and mineral processing industries by agitating a pulverized mixture of the materials with water, oil and chemicals. The process of differential wetting of the suspended particles causes unwetted particles to be carried by air bubbles to the surface for collection. The various processes include the following: dissolved air flotation; induced gas flotation; and froth flotation, typical in the mineral processing industry.

FLOUR CORN: A variety of maize with large soft grains and friable endosperm, which is ideal for grinding into flour.

FLOURITE STRUCTURE: A type of ionic crystalline structure with the cations having an expanded face-centred cubic arrangement with the anions occupying both types of tetrahedral hole. The cations have a coordination number of eight and the anions have four. Examples are calcium fluoride (CaF_2), barium chloride ($BaCl_2$) and plumbic oxide (PbO_2).

FLOW ANALYSIS: The generic name for all analytical methods, which are based on the introduction and processing of test samples in flowing media.

FLOW CHART: 1. A drawing in which particular shapes and connecting lines are used for showing how each particular action in a system is connected with or depends on the next or another. An example is a flow chart showing the production process. 2. A diagram used to show the possible paths that data can take through a computer system or program.

FLOW RATE: The volume of a fluid, such as liquid, vapour or gas passing through a pipe, duct or open channel in unit time. The flow rate is usually measured at column outlet, at ambient pressure and temperature. Flow rate is usually measured in litres per minute (lpm).

FLOW REACTOR: A reactor through which the reactants pass while catalysis is in progress.

FLOW ROUTING: A network routing technology that takes variations in the flow of data into account to increase routing efficiently.

FLOW VELOCITY: The volume of water transferred per unit of time and per unit of area in the direction of the net flow of water in soil.

FLOWER: The reproductive structure in flowering plants that bears the organs of sexual reproduction or the reproductive unit of a flowering plant. The biological function of a flower is to mediate the union of male sperm with female ovum in order to produce seeds. The process begins with pollination, followed by fertilisation, leading to the formation and dispersal of the seeds. **Cut Flower** is a flower that is harvested for its brightly coloured and showy blossom.

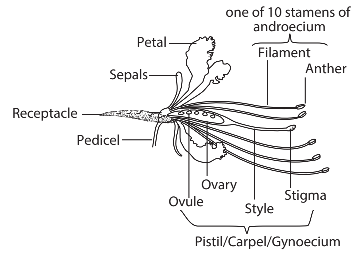


Fig I: Longitudinal section of a flower

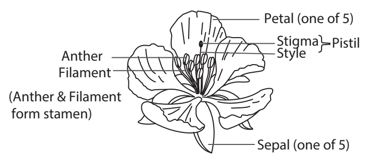


Fig II: Complete flower

FLOWERING GLUME (LEMMA): The lower of a pair of bracts immediately enclosing the floret in most Gramineae. The name of the second bract is **palea**, which is usually thinner and more delicate than the lemma, as found in the male flowers of maize.

FLOWERING PLANTS (ANGIOSPERMS; ANGIOSPERMAE; MAGNOLIOPHYTA): The most diverse group of land plants in which the seeds are enclosed in an ovary, which ripens to a fruit, forming angiosperms. They are divided into monocotyledons or single seed leaf in the embryo and dicotyledons or two seed leaves in the embryo.

FLOWERS OF SULFUR (SUBLIMED SULFUR): One of three forms of pharmaceutical sulfur, made by sublimation, which is used to dust on plants to prevent mildew. The other two forms are precipitated sulfur and washed sulfur.

FLUCTUATION-DISSIPATION THEOREM: It relates quantities in equilibrium and non-equilibrium statistical mechanics and microscopic and macroscopic quantities. The principle underlying this theorem is that a non-equilibrium state may have been reached either as a result of a random fluctuation or an external force such as an electric or magnetic field and that the evolution towards equilibrium is the same in both cases. The fluctuation-dissipation theorem enables transport coefficients to be calculated in terms of response to external fields.

FLUCTUATIONS: 1. Random deviations in the value of a quantity about some average value. Quantum fluctuations occur in quantum mechanics and thermal fluctuations occur in thermodynamics. 2. Wavelike motion of water, which is the variations of water-level height from mean sea level that is not due to tide-producing forces.

FLUE GAS: A mixture of waste gases resulting from combustion processes and other reactions in a furnace, passing off through the smoke flue. It is composed largely of nitrogen, carbon dioxide, carbon monoxide, water vapour and often sulfur dioxide. It sometimes serves as a source from which carbon dioxide or other compounds are recovered

FLUE GAS SCRUBBER: Equipment for removing fly ash and other objectionable materials from the products of combustion by means of sprays or wet baffles.

FLUE-GAS DESULFURIZATION (FGD): The process of removing harmful sulfur dioxide (SO_2) pollutant from the exhaust flue gases of plants. Tall flue disperses emissions by diluting the pollutants in ambient air and transporting them to other regions.

FLUENCE: A measure of the quantity of light or other radiation falling on a surface, expressed in terms either of particles or energy per unit area. It is defined as the quantity of radiant energy falling on a small sphere divided by the cross-sectional area of the sphere and can be expressed in two ways: particles or energy per unit area.

FLUID: Any substance, either liquid or gas in which the molecules are relatively mobile and can flow under an applied shear stress. Fluids are a subset of the phases of matter and include liquids, gases, plasmas and, to some extent, plastic solids and glass.

FLUID BALANCE: A state in which the volume of a body of water and its solutes is within normal limits and there is normal distribution of fluids within the intracellular and extracellular compartments.

FLUID BILAYER MODEL: A model, which depicts the organisation of the components of biological membranes such as the plasma membrane. This model is the widely accepted model to describe the structural organisation of biological membranes, such as the plasma membrane. In this model, the phospholipid component is arranged in bilayer whereas the proteins are seemingly floating on the bilayer, indicating fluidity. Although it is now recognised that some proteins are restrained by interactions with cytoskeletal elements and that the phospholipid annulus around a protein may contain only specific types of lipid, the model is still considered broadly correct.

FLUID CATALYTIC CRACKING (FCC): The most important conversion process used in petroleum refineries. It is widely used to convert the high-boiling, high-molecular weight hydrocarbon fractions of petroleum.The catalyst is a solid sand-like material that is made fluid by the hot vapour and liquid fed into the FCC, much as water makes sand into quicksand. Because the catalyst is fluid, it can circulate around the FCC, moving between reactor and regenerator vessels. The FCC uses the catalyst and heat to break apart the large molecules of gas oil into the smaller molecules that make up gasoline, distillate and other higher-value products like butane and propane. After the gas oil is cracked through contact with the catalyst, the resulting effluent is processed in fractionators, which separate the effluent based on various boiling points into several intermediate products, including butane and lighter hydrocarbons, gasoline, light gas oil, heavy gas oil and clarified slurry oil.

FLUID COKE: The carbonisation product of high-boiling hydrocarbon fractions (heavy residues of petroleum or coal processing) produced by the fluid coking process. Fluid coke consists of spherulitic grains with a spherical layer structure and is generally less graphitisable than delayed coke. Therefore, it is not suitable as filler coke for polygranular graphite products and is also less suitable for polycrystalline carbon products. All cokes contain a fraction of matter that can be released as volatiles during heat treatment. This mass fraction, the so-called volatile matter, is in the case of fluid coke about 6 wt.%.

FLUID DYNAMICS: A branch of fluid mechanics dealing with the properties and behaviour of fluids, such as liquids and gases at rest (fluid statics) or in motion (fluid dynamics). Because of their ability to flow, liquids and gases have many properties in common not shared by solids. The study of fluids in motion or fluid dynamics, makes up the larger part of fluid mechanics. Branches of fluid dynamics include hydrodynamics (study of liquids in motion) and aerodynamics (study of gases in motion) as well as vortex dynamics, gas dynamics, computational fluid dynamics (CFD), convection heat transfer, flows of turbo machinery, acoustics, bio-fluids, physical oceanography, atmospheric dynamics, wind engineering and the dynamics of two-phase flows. Modern designs of aircraft, spacecraft, automobiles, ships, land and marine structures, power and propulsion systems or heat exchangers rely on a clear understanding of the relevant fluid mechanics.

FLUID FEEDERS: Animals such as aphids, ticks and mosquitoes that pierce the body of a host plant or animal and obtain food from ingesting its fluids. Examples are haematophagy, feeding on blood; nectarivore, feeding on nectar; and plant sap feeders.

FLUID FRICTION (VISCOSITY): A measure of the resistance to flow that a fluid offers, as a result of its internal friction between the molecules. It applies to the motion of an object moving through a fluid as well as the motion of a fluid passing on an object. Viscosity is a major factor in determining the forces that must be overcome when fluids are used in lubrication or transported in pipelines. The viscosity of liquids decreases rapidly with an increase in temperature, while that of gases increases with an increase in temperature. The SI unit for viscosity is the newton-second per square metre ($N\text{-}s/m^2$). **Viscous** is a term used to describe a fluid that has a relatively high resistance to flow.

FLUID MECHANICS: The study of fluids and the forces on them both at rest and in motion. Fluid statics is concerned with the pressure and forces exerted on liquids and gases at rest and fluid dynamics is the study of fluids in motion. A modern discipline, called computational fluid dynamics (CFD), is devoted to this approach to solving fluid mechanics problems.

FLUID MOSAIC MODEL: A widely accepted model, proposed by S. J. Singer and G. L. Nicholson in the 1970s that describes the structure of cell membranes. It is based on a flexible layer made of lipid molecules interspersed with large protein molecules that act as channels through which other molecules enter and leave the cell.

FLUID OUNCE: 1. A unit of volume or capacity in the U.S. Customary System, which is equal to 29.57 millilitres and used in liquid measurement. 2. A unit of volume or capacity in the British Imperial System, used in liquid and dry measure. It is equal to 28.41 millilitres.

FLUID, SUPERCRITICAL (SUPERCRITICAL FLUID): A substance which can be either liquid or gas, used in a state above its critical temperature and critical pressure where gases and liquids can coexist. It shows properties that are different from those of either gas or liquid under standard conditions. A supercritical fluid has both the gaseous property of being able to effuse through solids and the liquid property of being able to dissolve materials into their components. Supercritical fluids are suitable as a substitute for organic solvents in a range of industrial and laboratory processes. Carbon dioxide and water are the most commonly used supercritical fluids. Carbon dioxide is used for decaffeination and water for power generation.

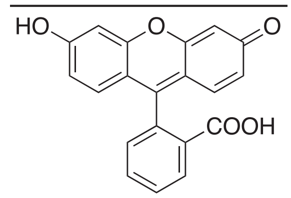
FLUIDICS (FLUIDIC LOGIC): The use of a jet of fluids or compressible medium in pipes to perform many of the control functions, usually performed by electronic devices. The physical basis of fluidics is pneumatics and hydraulics, based on the theoretical foundation of fluid dynamics. A jet of fluid can be deflected by a weaker jet striking it at the side. This provides non-linear amplification, similar to the transistor used in electronic digital logic. It is used mostly in environments such as systems exposed to high levels of electromagnetic interference or ionising radiation, where electronic digital logic would be unreliable.

FLUIDIZATION: A roasting process in which finely divided mass of solid particles act as a fluid by agitation or by gas passing through. It is used in the calcination of various minerals, in Fischer-Tropsch synthesis and in the coal industry.

FLUIDIZED BED: A catalyst bed in which the flow of gases is sufficient to cause the finely divided particles to behave like a fluid.

FLUKE: 1. A very small parasitic flatworm of the class Turbellaria related to the tapeworm with a body covered with a thick cuticle. It is a parasite of fish, frogs and turtles, also humans, domestic animals and invertebrates such as molluscs and crustaceans. 2. Any of various flatfishes, especially a flounder of the genus *Paralichthys*.

FLUNITRAZEPAM (DATE RAPE): A benzodiazepine, with the molecular formula $C_{16}H_{12}FN_3O_3$. It is the fluorinated methylamino derivative of nitrazepam. It is used medically in some countries as a hypnotic, sedative, anticonvulsant, anxiolytic, amnestic and muscle relaxant. It is quickly eliminated from the body, difficult to detect and causes amnesia.



Chemical Structure of Fluorescein

FLUORESCEIN: A dye that dissolves in water to make a yellowish-green fluorescent solution. Its uses include in microscopy, in a type of dye laser as the gain medium, in forensics and serology to detect latent blood stains and in dye tracing.

FLUORESCENCE: 1. The emission of light or other radiation from atoms that have absorbed light or other electromagnetic radiation of a different wavelength or the ability of some substances to absorb light of one frequency and emitting light of a different frequency. 2. Very short-lived luminescence (a glow not caused by high temperature). However, the term is used generally for any luminescence regardless of the persistence.

FLUORESCENCE IN SITU HYBRIDISATION (FISH): A technique developed by Christoph Lengauer, used to detect and localise the presence or absence of specific DNA sequences on chromosomes. It is used in genetic mapping for locating specific genes within a chromosome set.

FLUORESCENCE MICROSCOPY: A technique for examining samples under a microscope without slicing them into thin sections. Instead, fluorescent dyes are introduced into the tissue and used as a light source for imaging purposes.

FLUORESCENCE YIELD: For a given transition from an excited state of a specified atom, the ratio of the number of excited atoms which emit a photon to the total number of excited atoms.

FLUORESCENT ANTIBODY TECHNIQUE (IMMUNOFLUORESCENCE): A laboratory test used to detect the presence of specific proteins and enzymes in tissue samples by applying antibodies or disease-causing agents that have been labelled with fluorescent dye. The labelled antibody combines with the protein under investigation and the resulting complex can be identified under a microscope by its fluorescence.

FLUORESCENT LAMP (FLUORESCENT TUBE): A type of gas-discharge lamp that uses electricity to excite mercury vapour producing ultraviolet light when a current flows through a heated low pressure gas, for example, neon. The lamp employs a ballast to regulate the flow of current through the lamp.

FLUORIDATION: The process of adding very small amounts of fluorine salts to drinking water to help prevent tooth decay. The fluoride becomes incorporated into the fluoroapatite of the growing teeth and reduces the incidence of dental caries.

FLUORIDE: The negative ion (F^-), the reduced form of fluorine formed when hydrogen fluoride dissolves in water or the compound formed between fluorine and another element in which the fluorine is the electronegative element. Fluorine-containing compounds range from potent toxins such as sarin to life-saving pharmaceuticals such as efavirenz and from inert materials such as calcium fluoride to the highly reactive sulfur tetrafluoride. Compounds containing fluoride anions and in many cases those containing covalent bonds to fluorine are called **fluorides**.

FLUORIMETER: An instrument used to measure the intensity and the wavelength distribution of the light emitted as fluorescence from a molecule excited at a specific wavelength or wavelengths within the absorption of a particular compound. Characteristic fluorescence bands may be used to identify specific pollutants such as the polynuclear aromatic hydrocarbons. Excitation spectra of impurities can be observed by scanning the wavelength of the excitation light which is incident on the sample over a range of wavelengths and observing the relative intensity of the fluorescence emitted at a given wavelength. These spectra are also characteristic of the impurity.

FLUORINATION: A chemical reaction in which fluorine atom is introduced into a molecule.

FLUORINE: A chemically reactive, poisonous, gaseous element in Group 17 (VIIA) elements in the Periodic Table. Its chemical symbol is F, atomic number 9, atomic weight 18.9984, freezing point -219.62 °C (53.53 K), melting point -223 °C (50 K) and boiling point -188.14 °C (85 K), specific gravity of liquid 1.108 (at boiling point) and valence 1. It is the most chemically active nonmetallic element and the most electronegative of all the elements. Its chief source is fluorite. It also occurs in cryolite, fluorapatite, seawater, bones and teeth. Hydrogen fluoride (HF) is a raw material for many other fluorides. It can be produced only by electrolysis under special conditions. It is an essential trace element needed for the healthy development of bones and teeth and to prevent tooth decay. Fluorine compound, fluorite is used as a flux in refining iron; another compound, cryolite serves as the electrolyte in the production of aluminium. Compounds of fluorine are also used in the ceramic and glass industries. Hydrofluoric acid is used to etch glass and in the manufacture of light bulbs. Sodium fluoride (NaF) is added to water and tin fluoride (SnF_2) to dental-care products to reduce dental caries.

FLUORITE (FLUORSPAR): A glassy, brittle mineral, composed of calcium fluoride CaF_2, forming cubes. It is octahedral in shape. Fluorite is used in the manufacture of steel, aluminium fluoride, artificial cryolite and aluminium. It is also used in glassmaking, in iron and steel enamelware, in the production of hydrofluoric acid, in the refining of lead and antimony and as a catalyst in the manufacture of high-octane fuels.

FLUOROCARBON (PERFLUOROCARBONS): Compounds consisting entirely of fluorine and carbon bonded together in strong carbon–fluorine bonds, formed by replacing the hydrogen atoms of a hydrocarbon with fluorine.

FLUOROGENIC: Property of a material in which fluorescence is induced or enhanced by the addition of another molecular entity. An example is a non-fluorescent compound consisting of a pyrene connected to a maleimide group (which quenches the pyrene fluorescence). It is a reaction in which the maleimide moiety is converted to a succinimide derivative resulting in the recovery of the pyrene fluorescence.

FLUOROPHORE: Molecular entity, often organic that emits fluorescence.

FLUOROSCOPE: An instrument with a fluorescent screen used for viewing x-ray images by projecting them on to a film or screen containing a fluorescent material without taking and developing x-ray photographs. It is used in medical diagnosis and engineering quality control and allows the direct observation of an x-ray beam which is being passed through the object under examination. **Fluoroscopy** is the process of using a fluoroscope to view x-ray images by projecting them on to a film or screen containing a fluorescent material using a fluoroscope.

FLUOROSIS: An abnormal condition caused by excessive intake of fluorine in drinking water or food. It affects the bones, causes discolouration of the teeth and affects the milk yield of cattle.

FLUSHING: 1. The process by which soluble substances in the lower layers of the soil are washed upwards and deposited on or near the soil surface. It occurs, for example, in water-meadows, fenland and where there are springs. 2. In agriculture science, it is a high nutritional feed given to ewes before mating to increase increase the rate of ovulation and, hence, lambing rate.

FLUTED: Having grooves or furrows.

FLUVIAL DEPOSITS: Sediments that are transported and deposited by rivers. The various types include the following: (i) alluvial fans, which are fan-shaped sediment bodies that form at the bases of mountain slopes at the mouths of rivers; (ii) fan deltas, which also form at the bases of mountain slopes, but which are deposited very near a marine shoreline and in marine waters; (iii) braided-river deposits, which form at and beyond the bases of mountains, where the gradient of the ground surface is relatively steeply inclined; (iv) meandering-river deposits, which form on more gently inclined floodplains, and (v) incised-valley-fill deposits which fill preexisting continental valleys. Each of these types of deposits exhibits a unique set of properties that distinguish it from the others, including grain size, sand-body geometry, orientations, flow barriers, etc.

FLUVIO-GLACIAL: A process or landform, associated with glacial melt-water, flowing beneath or ahead of a glacier, capable of transporting rocky material and creating a variety of landscape features such as eskers, kames and outwash plains.

FLUX: 1. A field that represents the transfer of some quantity per unit area. Examples are heat flux, energy flux, mass flux and magnetic flux. The charge per unit area that would be displaced across a layer of conductor placed across an electric field. 2. A measure of the rate of flow of light or a measure of the quantity of magnetism, taking account of the strength and extent of a magnetic field or a measure of the number of particles flowing per unit area of cross-section in a beam of particles. 3. A compound that reduces the melting point of a refractory material. For example, cryolite reduces the melting point of aluminium for the electrolytic production of aluminium. 4. Any substance introduced in the smelting of ores to promote fluidity and to remove objectionable impurities in the form of slag. Limestone is commonly used for this purpose in smelting iron ores. Other materials used as fluxes are silica, dolomite, lime, borax and fluorite. In soldering, the flux removes oxide films, promotes wetting and prevents reoxidation of the surfaces during heating.

FLUX ADJUSTMENT: In climatology, in climate modelling, the practice of modifying the fluxes or exchanges of heat and water between the atmosphere and ocean in coupled atmosphere-ocean models. This modification is designed to minimise the climate drift that occurs during model integration. These flux adjustments are typically a function of location and season.

FLUX DENSITY: 1. Magnetic flux density is the amount of flux per unit area or the force that a magnet or electromagnetic source exerts on other magnets or charged particles, such as electrons. Its SI unit is tesla and the symbol is T. 2. The number of particles flowing per unit area of cross section in a beam of particles.

FLUX DEPRESSION: The lowering of the particle or photon flux density in the neighbourhood of an object due to absorption of particles or photons in the object.

FLUX QUANTUM: A fundamental unit of magnetic flux that occurs in a superconducting ring such as a hollow cylinder. The magnetic flux is quantized in multiples of $\frac{h}{2e}$, where h is the Planck constant and e is the charge of an electron, which is approximately 2.07×10^{-15} Weber. The magnetic flux quantum Φ0 is the quantum of magnetic flux passing through a superconductor.

FLUXIONAL MOLECULE: A molecule that undergoes very rapid rearrangement of its atoms and only has a specific structure for a very short period of time. For example, the molecule chlorine trifluoride (ClF_3) has a T-shape at low temperatures (-60 °C, 213 K). At room temperature the fluorine atoms change position very rapidly and appear to have identical positions.

FLUXMETER: An instrument for measuring magnetic flux.

FLY: The common name for any of a variety of winged insects, but properly restricted to members of phylum Arthropoda, class Insecta order Diptera, the true flies, which includes the housefly, gnat, midge, mosquito and tsetse fly. All have sucking or piercing-and-sucking mouthparts and, except for a few wingless species, bear one pair of wings. The larvae, which occupy a wide variety of ecological niches, typically require a moist environment such as rotting flesh, decaying fruit or the internal organs of other animals. Adults often feed on nectar and plant sap, but some, such as the female horsefly and female mosquito, feed on blood; the adults of some species do not feed at all.

FLYTHROUGH (FLY-THROUGH): 1. A computer-animated simulation that allows the viewing of a particular real or imaginary site as if one were inside and moving through it. 2. A flight by a spacecraft through a particular area for the purpose of observation.

FLYING SAUCER: A term, which refers to unknown disk-like aerial objects, often believed to be extraterrestrial spacecraft. The term has now been largely superseded by unidentified flying object (UFO).

FLYWHEEL: A heavy revolving wheel in a machine that that spins and stores energy. It is used to increase the machine's momentum or used for storing mechanical energy for providing greater stability or a reserve of available power during interruptions in the delivery of power to the machine. Since energy is stored as kinetic energy of rotation, it is advantageous for the wheel to have as large as possible a moment of inertia and thus for its mass to be concentrated near the rim.

FM (FREQUENCY MODULATION): A method by which radio waves are altered by varying the frequency of the carrier wave in accordance with the signal being transmitted, used for transmitting broadcasting signals. Compared to amplitude modulation (AM), FM offers important advantages such as extra bandwidth occupancy and negative feedback, to minimise noise problems and receiver distortion.

FM SYNTHESIZER (FREQUENCY MODULATION SYNTHESIZER): In audio and music, a method for generating sounds based on techniques used to transmit FM radio signals.

FOAL: 1. A newborn horse or pony of up to six months old. 2. To give birth to a pony.

FOAM: A dispersion in which a large proportion of gas by volume in the form of gas bubbles is dispersed in a liquid, solid or gel. Pure liquids do not foam. The diameter of the bubbles is usually larger than 1 µm, but the thickness of the lamellae between the bubbles is often of colloidal size range.

FOAM FRACTIONATION: A method of separation in which a component of the bulk liquid is preferentially adsorbed at the liquid/vapour interface and is removed by foaming.

FOAM RUBBER: A soft rubber full of air bubbles used widely including upholstery, insulation, cushion and mattresses. It is made by beating air into latex and then curing it.

FOAMING AGENT (BLOWING AGENT): A substance that generates inert gases on heating, causing the resin to assume a cellular structure. It is used to produce liquid or solid foam such as expanded plastic. It is a surfactant which when present in small amounts facilitates the formation of a foam or enhances its colloidal stability by inhibiting the coalescence of bubbles.

FOCAL: Of, relating to, situated at or measured from the focus.

FOCAL CHORD: A chord of a conic that passes through a focus.

FOCAL LENGTH (FOCAL DISTANCE): The distance between a lens or a spherical mirror and its focal point (principal focus). Diverging lenses and mirrors have virtual focal lengths, whereas converging lenses and mirrors have real focal lengths. The focal length of a given lens depends on both the refractive index of the material relative to that of the medium it is in and radii of curvature r_1 and r_2. The relationship that exists between the focal length f of a lens, its refractive index n and its radii of curvature r_1 and r_2 is expressed as: $\frac{1}{f}=(n-1)(\frac{1}{r_1}+\frac{1}{r_2})$. This formula is also called the lens' maker's formula.

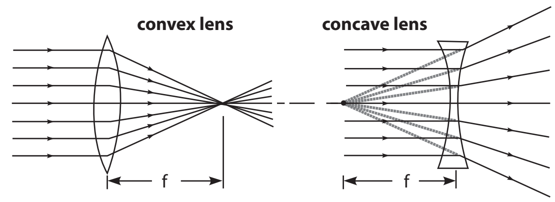


Fig1 A:
The focal length of a converging lens

Fig 1B:
The focal length of a diverging lens

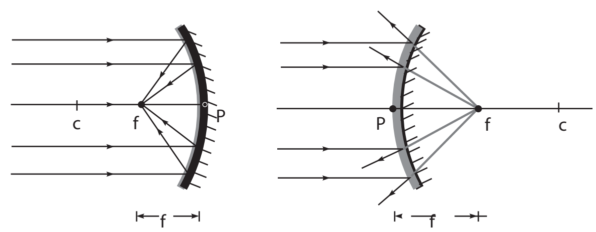


Fig 2A:
The focal length of a converging mirror

Fig 2B:
The focal length of a diverging mirror

FOCAL PLANE: The flat plane at right angles to the optical axis onto which a lens will focus an image. Curvature of field results in an extended image forming on a curved surface known as the focal surface.

FOCAL POINT (PRINCIPAL FOCUS): The convergence point in space where parallel light rays meet after passing through the lens or bouncing off the mirror or sharpest point of focus, for an image. For a positive (converging) lens or mirror, the focal points of the element are the points at which a ray, approaching the element parallel to the optical axis, crosses the optical axis on the opposite side. For a negative (diverging) lens, the focal points are the points at which the backward extension of the diverging ray intersects the optical axis.

FOCAL RADIUS: A line segment from the focus of an ellipse to a point on the perimeter of the ellipse.

FOCAL RATIO (F-NUMBER OR F-STOP): The ratio of the focal length (F) of a mirror or lens to its diameter (D) expressed as a number. It is a measurement that describes the relative opening or aperture. For example, a lens with a 50mm focal length with the aperture set at f/2 indicates an aperture diameter of 25mm, whereas f/4 indicates an aperture of 12.5mm. In photography, small focal ratios, below about f/6, are said to be fast and result in a brighter image for a given aperture. Large focal ratios, equal to or above about f/8, are said to be slow.

FOCAL REDUCER: An auxiliary lens, used mainly for photographic or CCD-imaging purposes. It reduces a telescope's effective focal length, allowing images to be taken much faster and also gives a wider field of view. Some focal reducers may be used visually to obtain a wider field of view, but others are intended for photographic/CCD use only.

FOCK DEGENERACY: A type of degeneracy that occurs in the spectrum of the hydrogen atom as a result of the rotational invariance in the four dimensions associated with the Coulomb interaction between the proton and electrons.

FOCUS: 1. Also called **principal focus** or **focal point**, it is any of two points on the axis of a mirror, lens or other optical system, one point being such that rays diverging from it are deviated parallel to the axis upon refraction or reflection by the system and the other point being such that rays parallel to the axis of the system converge to the point upon refraction or reflection by the system. 2. To adjust an optical device as for example, a microscope, to obtain a sharp image or picture. 3. The distance that a lens must be moved in order to focus a sharp image on the light sensitive film at the back of a camera. 4. A point at which rays of light, heat or other radiation, meet after being refracted or reflected. 5. In geometry, the focus of a conic section is a point having the property that the distances from any point on a curve to it and to a fixed line have a constant ratio for all points on the curve. 6. In geology, also known as **hypocentre**, it is a point beneath the Earth's surface, directly below the epicentre where the vibrations of an earthquake originate; the rocks start to fracture at this spot, then vibrating waves radiate outward in all directions.

FOCUS GROUPS: A research technique used to collect data through group interaction on a topic to be determined by the researcher. The goal in such groups is to learn the opinions and values of those involved with products and/or services from diverse points of view.

FOCUSING: In animal physiology, the process of directing and concentrating light from a source onto the retina of the eye, by means of the lens in order to obtain a clear image of an object at a range of distance.

FODDER (FODDER CROP): Plant material or a crop such as grass or clover which is grown to feed domesticated livestock such as cattle, goats, sheep, horses, chickens and pigs. It also includes hay, straw, silage, compressed and pelleted feeds, oils and mixed rations and also sprouted grains and legumes.

FODDER BEET: A root crop of sugar beet and mangolds, usually grown after cereals and used to feed stock.

FODDER RADISH: A type of brassica grown primarily for use as a green fodder crop.

FOEHN (FÖHN; FOEHN WIND, FÖHN WIND): A warm, dry wind coming down the leeward slopes of mountains due to air having lost its moisture while ascending the windward slope, then warming on its descent. The foehn effect describes warm, moist air, which crosses a mountain range, cooling as it climbs the windward side and depositing most of its moisture as rain. On its descent on the leeward side of the range, the now drier air heats up much more rapidly, giving rise to hot, dry weather downwind. Foehn winds can raise temperatures by as much as 30 °C (303 K) in just a few hours.

FOETAL MEMBRANES: Tissues produced by an animal embryo for protection and nutrition but which otherwise takes no part in its development. The four membranes are chorion, amnion, allantois and yolk sac. In the course of development, the chorion becomes the outermost and the amnion the innermost membrane surrounding the developing embryo. As the allantois increases in size, it expands and becomes closely associated, if not fused, with the chorion.

FOETUS: 1. An unborn or unhatched vertebrate that has passed through the earliest developmental stages and at a stage with the same overall shape and parts. 2. Any of the membranous structures closely associated with or surrounding a developing vertebrate embryo, including the amnion, chorion, allantois and yolk sac, when development has reached a stage at which the main features of the adult form are recognisable.

FOG: A cloud that is in contact with the ground composed of water vapour that has condensed on particles of dust in the atmosphere. Fog reduces visibility to less than 1 km, whereas mist reduces visibility to between 1 km and 2 km.

FOG HORIZON: The top of a fog layer which is confined by a low-level temperature inversion and gives the appearance of the horizon (which it actually obscures) when viewed from above, against the sky.

FOGGAGE: 1. A winter grazing of cattle on non-rye grass swards. 2. Grass which has been left standing to provide winter grazing for sheep and cattle.

FOKKER-PLANCK EQUATION: An equation in non-equilibrium statistical mechanics that describes a superposition of a dynamic friction process and a diffusion process for the evolution of variables in a system. Mathematically, the general Fokker-Planck equation for one variable x is given by, $\frac{\partial w}{\partial t}=\left[-\frac{\partial}{\partial x}D^{(1)}(x)+\frac{\partial^2}{dx^2}D^{(2)}(x)\right]$, where $D^{(1)}(x)$ is the drift coefficient and $D^{(2)}(x)>0$ is the diffusion coefficient.

FOLATES: 1. A group of heterocyclic compounds based on the pteroic acid and conjugated with one or more L-glutamate units. Folate derivatives are important in DNA synthesis and erythrocyte formation. 2. Folate, also known as folic acid or folacin is a water-soluble B vitamin that is naturally present in some foods, added to others and available as a dietary supplement. It is found in leafy green vegetables, beans, peas and lentils, liver, beets, brussel sprouts, poultry, nutritional yeast, tuna, wheat germ, mushrooms oranges, asparagus, broccoli, spinach, bananas, strawberries and cantaloupes. It works together with vitamin B12 and vitamin C to metabolise protein in the body. It is important for the formation of red and white blood cells,necessary for the proper differentiation and growth of cells and for the development of the foetus. It is also used to form the nucleic acid of DNA and RNA. Its deficiency leads to anaemia.

-FOLD: A suffix that indicates the number of members used in some repetitive process, for example, ten-fold comprises ten parts or members; or ten times as great or as much.

FOLD: A bend in a layer of sedimentary rock caused by slight differences in pressure in the Earth's crust. Folds sizes may range from only a few centimetres in width or may cover several kilometres. They usually occur in a series and look like waves. If the bend is arched in the middle it is called an anticline and if it sags downwards in the middle it is called a syncline.

FOLD DOMAIN: A portion of a polymer crystal wherein the fold planes have the same orientation. The sectors of lamellar crystals frequently represent fold domains.

FOLD MOUNTAIN: A mountain made up from sedimentary rock layers created where two of Earth's tectonic plates are pushed together. The various types of folded mountain include anticlines, where the rocks are folded upwards; synclines are where the rocks are folded downwards; nappes are where the rocks are severely folded and faulted.

FOLD PLANE: A crystallographic plane defined by a large number of stems that are connected by chain folds.

FOLDED SHEEP: Sheep kept in movable folds, as a means of controlling their grazing.

FOLDED-CHAIN CRYSTAL: A polymer crystal consisting predominantly of chains that traverse the crystal repeatedly by folding as they emerge at its external surfaces. The re-entry of the chain into the crystal is assumed to be adjacent or near-adjacent within the lattice.

FOLDER (DIRECTORY): In computer science, it is a location on a computer system, which stores multiple files and other folders (sub folders) on Apple Macintosh computers and Microsoft Windows-based graphical user interface (GUI) operating systems. A file is the common storage unit in a computer containing all programs and data; the computer reads and writes files. A folder holds one or more files and it can be empty with just a name. Folders provide a method for organising files similar to a manila file folder containing paper documents in a file cabinet.

FOLIACEOUS: Having a leaf-like appearance; relating to leaves.

FOLIAGE: 1. The above-ground plant organ or leaves specialised for photosynthesis. 2. An ornamental representation of leaves, stems and flowers, especially in architecture.

FOLIAR DIAGNOSIS: An estimation of the extent to which plants are getting certain necessary chemical elements from the soil, based on an examination of the colour and the growth habits of the foliage of the plants.

FOLIAR SPRAY: Liquid fertiliser sprayed directly on to leaves of plants.

FOLIATION: The changes of rocks such as igneous, sedimentary or any rock type into metamorphic rocks when they undergo shearing forces or differential pressure causing breaks in the rock brought about by factors such as high heat, high pressure, hot mineral-rich fluids or a combination of these factors deep within the Earth or where tectonic plates meet. Foliated rocks have distinct, repetitive layers because of breaks in the rock. Foliation is common in rocks in mountain belt areas. When foliation occurs, the minerals in the original rock are replaced by new minerals and the new mineral crystals are arranged in parallel bands between the repetitive layers.

FOLIC ACID: A vitamin of the B-complex found chiefly in liver and green leafy vegetables, citrus fruits, cereals, beans, poultry and egg yolks and also synthesised by intestinal bacteria. It is an organic compound essential to animal growth and health and needed by bacteria as a growth factor. Folic acid is necessary for synthesis of nucleic acids and formation of the haeme component of haemoglobin in red blood cells. Low intake leads to folic acid deficiency anaemia.

FOLIOSE: Leafy, pertaining to or having leaflets, for example, as seen in leafy lichens such as Peltigera and sometimes leafy liverworts such as Diplophyllum and other organisms that bear leaf like structures

FOLIUM: 1. **Folium of descartes** is an algebraic plane curve consisting of a loop whose two ends, intersects at a node and are asymptotic to the same line. Its equation is of the form $x^2 + y^2 = 3axy$, where $x + y + a = 0$ is the equation of the line. 2. A thin, leaf-like layer or stratum occurring especially in metamorphic rock.

FOLLICLE: 1. A small group of cells protecting and nourishing a cell or structure of an animal. For example, each hair is contained in a tubular pit of the epidermis called the hair follicle or the follicles in an ovary surrounding a growing ovary. 2. A dry, usually many seeded fruit that splits along one side to release the seeds within.

FOLLICLE STIMULATING HORMONE (FSH): A hormone found in humans and other animals. It is made in the pituitary gland and regulates the development, growth, pubertal maturation and reproductive processes of the body. For example, it stimulates the production of ova (egg cells) in females and sperms in males.

FOLLICULAR PHASE (PROLIFERATIVE PHASE): The phase of the oestrous cycle during which the Graafian follicles mature and the lining of the uterus thickens under the influence of oestrogen. In humans and great apes it is the menstrual cycle. The main hormone controlling this stage is oestradiol (estradiol).

FOLLOWERS: 1. Young cows in a dairy herd which are not yet in milk and are being reared to replace the older stock. 2. Cattle put to graze a pasture after it has been used by another group of animals.

FOLLOWING CROP: A crop sown by a tenant farmer before leaving the farm at the end of his tenancy. The tenant farmer is permitted to return and harvest the crop and remove it.

FOMITE: Any object or substance such as a dish, toy, book, doorknob or clothing that is capable of carrying infectious organisms, such as germs or parasites.

FONT: In computer science, it is a complete set of printed or displayed characters of the same typeface, point size and style. For example, one font may be Arial 12 point bold, while another font may be Times New Roman 14 point italic.

FONTANELLE (FONTANEL): A soft, membrane-covered space between the bones at the front and the back of an infant's skull. During birth, it enables the bony plates of the skull to flex, allowing the child's head to pass through the birth canal. Fontanels disappear as the bones in the skull fuse. The fusion is complete at around 18 months of age.

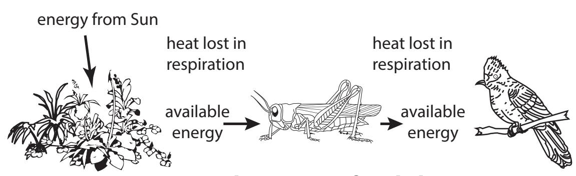
FOOD: Any substance usually of plant or animal origin, that contains essential nutrients, such as carbohydrates, fats, proteins, vitamins, or minerals, and is ingested and assimilated by an organism to produce energy, stimulate growth, and maintain life.

FOOD ADDITIVE: A substance added to manufactured or processed food, often, to improve the colour or to keep the properties of food. They include antioxidants, emulsifiers, thickeners, preservatives and colourants.

FOOD BALANCES (ECOLOGICAL BALANCE): The balance between food supply and the demand for food from a population. It is a state of dynamic equilibrium within a community of organisms in which genetic species and ecosystem diversity remain relatively stable, subject to gradual changes through natural succession. This balance may be disturbed due to the introduction of new species, the sudden death of some species, natural hazards or man-made causes.

FOOD CALORIMETER: Any of various devices used to measure thermal properties, such as calorific values or heats of chemical reactions. It can also be used to measure the energy content of food. The material is burned in the calorimeter and the heat energy produced is measured. The amount of heat can be determined from the increase in temperature, taking into account the properties of the container and the liquid.

FOOD CHAIN: A linear feeding relationship in an ecosystem in which energy and nutrients are transferred from plants through a series of organisms by each stage feeding on the preceding stage and providing food for the succeeding one. There are three types of food chains: the grazing/predatory food chain, parasitic food chain and saprophytic/detrital food chain. A food chain has a base made of autotrophs, which are organisms such as plants and chemotrophs, which manufacture their own food and the energy or carbon compounds produced are transferred through a sequence of organisms. Each organism in the food chain feeds on and hence obtains energy from the one preceding it and is in turn fed on and provides for the next organism. For example, plants are fed on by herbivores, which in turn are fed on by carnivores. The directional flow of materials and energy from one organism to another is graphically represented by arrows. Food chains are linear depictions of energy flow, while food webs show the multiple interactions among the different types of organisms. Most food chains have only about four to five links since too many links in a food chain will result in high demand, less supply of food (and therefore energy).



Energy losses in a food chain

FOOD CONSUMERS (HETEROTROPHS): Organisms which, after green plants called autotrophs, occupy the subsequent links of a food chain. They cannot manufacture their own food but need organic carbon for growth, which they obtain by consuming other organisms. An organism capable of synthesising its own organic substances from inorganic compounds is called an autotroph. Autotrophs produce their own sugars, lipids and amino acids using energy obtained from the sun and carbon dioxide and ammonia or nitrates as a source of nitrogen. Organisms such as green plants and certain algae that use light for the energy to synthesise organic compounds are called photosynthetic autotrophs; organisms such as iron bacteria, the nitrifying bacteria and the nonpigmented sulfur bacteria that oxidise such compounds as hydrogen sulfide (H_2S) to obtain energy are called chemosynthetic autotrophs.

FOOD CROP: A plant grown mainly for food.

FOOD GRAINS: Small, hard, dry seeds, with or without hull or fruit layers attached and harvested for human food or animal feed. Examples are wheat, barley or rye.

FOOD HYGIENE: The series of actions taken to ensure clean, healthy conditions for the handling, storing and serving of food.

FOOD INSECURITY: A situation that exists when people do not have reliable access to a sufficient quantity of affordable, nutritious food for normal growth and development and an active and healthy life. Its causes include the following: unavailability caused by drought and other extreme weather events, or pests/other agricultural problems; climate change; conflicts; corruption and political instability; economic reasons; or inappropriate distribution. Food insecurity may be chronic, seasonal, or transitory.

FOOD IRRADIATION: A technique used in food technology, involving the exposure of food to low level non-ionising radiation to destroy microorganisms, bacteria, viruses or insects that might be present in the food. The technique is also applied for sprout inhibition, delay of ripening, increase of juice yield and improvement of re-hydration. It is also used on non-food items, such as medical hardware, plastics, tubes for gas pipelines, hoses for floor heating, shrink-foils for food packaging, automobile parts, wires and cables (isolation), tires and gemstones.

FOOD MILE: A measure of the distance that food is transported from the time of its production to the consumer. Food mile is a factor used to assess the environmental impact of food, including the impact on global warming.

FOOD POISONING: An illness caused by eating food that is contaminated with bacteria, viruses, environmental toxins or toxins present within the food itself, such as the poisons in some mushrooms or certain seafood.

FOOD PRESERVATION: A branch of food science and technology that deals with spoilage of food and aim to control or eliminate factors capable of adversely affecting the safety, nutritive value, appearance, texture, flavour and keeping qualities of raw and processed foods. Some examples of food preservation are: (i) **Dehydration** or **food drying** by removing the water content to inhibit the growth of microorganisms (enzymes) and bacteria by the circulation of hot, dry air through the food; (ii) **Irradiation**, which is the process of exposing food to ionising radiation to destroy food-borne pathogens, such as salmonella, bacteria, viruses or insects that might be present in the food; (iii) **pH control** of low-acid canned goods to prevent botulism by adding agents such as citric acid, acetic acid and some alkalis which control acidity or alkalinity; (iv) **Salting** to remove moisture and to create an environment in which most bacteria, fungi and other potentially pathogenic organisms cannot survive; (v) **Pickling** by treatment with vinegar, which is acidic, which retards bacteria growth or kill them; (vi) **Blanching** by heating food up to 90 °C, etc. In freeze drying, food is rapidly frozen and then dehydrated, usually in a vacuum. Prepared food can be preserved by the addition of chemicals such as sodium benzenecarboxylate and sulfur dioxide, but some of these may have adverse side-effects.

FOOD PROCESSING INDUSTRY: The complex, global collective of diverse businesses involved in the treatment of raw materials to produce foodstuffs, especially one that involves value-addition.

FOOD PRODUCERS (AUTOTROPHS; PRIMARY PRODUCERS): Organisms, mainly green plants, which occupy the first level in a food chain and are capable of producing their own food. They obtain energy from the sun and carbon dioxide from the atmosphere and from these constituents they build the large organic molecules that they need to survive. Autotrophs such as green plants and certain algae that use light for the energy to synthesise organic compounds are called **photosynthetic autotrophs**. Organisms such as iron bacteria, the nitrifying bacteria and the nonpigmented sulfur bacteria that oxidise such compounds as hydrogen sulfide (H_2S) to obtain energy are called **chemosynthetic autotrophs**.

FOOD PYRAMID: A chart of food showing a healthy diet graphically, by grouping foods and the amounts of each group that should be eaten each day, based on nutritional recommendations. It was originally developed in the USA in 1992 and now adopted in many countries, with differences to allow for different national patterns of diet.

FOOD RESERVE: Reserves of fat, carbohydrate or protein in cells and tissues that function as an important store of energy that can be released and used in adenosine triphosphate (ATP) production when required by the organism.

FOOD SAFETY: A shared responsibility among producers, industry, government and consumers on issues surrounding the production, handling, storage and cooking of food that determine whether or not it is safe to eat. Safe food is food that is free not only from toxins, pesticides and chemical and physical contaminants, but also from microbiological pathogens such as bacteria, parasites and viruses that can cause illness.

FOOD SECURITY: Situation in which all people or populations at all times have physical and economic access to sufficient, safe, and nutritious food to meet their dietary needs and food preferences for an active and healthy life.

FOOD SUPPLY: The production of food and the way in which it gets to the consumer.

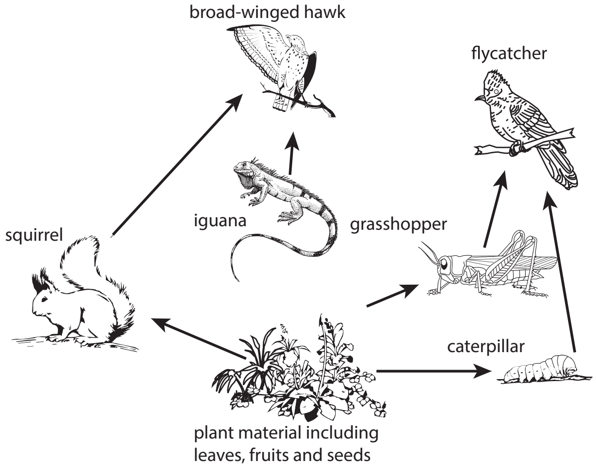
FOOD TECHNOLOGY: The application of science to commercial processing of foodstuffs. It involves the study of the physical, microbiological and chemical makeup of food and also the selection, preservation, processing, packaging and distribution.

FOOD TEST: Any of the several types of simple tests, easily performed in the laboratory for identifying the main classes of food. For example, iodine test is used to test for starch; Benedict's test for sugar; Biuret's test for protein and Fehling's test for reducing sugars.

FOOD VACUOLE: A spherical part in some single-celled organisms that engulf and digest captured food.

FOOD VALUE: The amount of energy produced by a specific amount of a type of food.

FOOD WEB (FOOD CYCLE): A complex of interconnected food chains in an ecosystem showing a greater variety of feeding or trophic relationship among organisms. It is a network of energy flows in and out of the ecosystem of interest and depicts how an ecosystem is structured and functions. Organisms of the same feeding habit occupy the same trophic level in the food chain.



A food web

FOODBORNE DISEASES (FOODBORNE ILLNESSES): Diseases which are transmitted from the consumption of contaminated food. A variety of pathogenic bacteria, viruses or parasites may contaminate food from the producer to the consumer.

FOODSTUFF: A substance used as food, especially a simple food material that is cooked and/or mixed with other foods for eating.

FOOL'S PARSLEY: A poisonous European weed, *Aethusa cynapium* with finely divided leaves, an unpleasant odour and white flowers that resemble parsley.

FOOT: 1. The part of the leg of a vertebrate below the ankle joint consisting of the heel, arch and toes that supports the rest of the body and maintains balance when standing and walking. 2. An organ or muscle surface that an invertebrate such as a mollusc uses to grip to enable it to move. 3. The lower part of the stem of a plant or the base of the spore-producing body sporophyte of mosses and liverworts. 4. An imperial unit of length, equivalent to 0.3048 metres.

FOOT OF LINE: The point of intersection of a line with a line or plane.

FOOT POUND: An imperial unit of energy or work, which is equal to the work done by a force of one pound acting through a distance of one foot in the direction of the force.

FOOT ROT: 1. A plant disease in which the root system and the bottom part of the stem rot. Foot rot of tomato is caused by *Phytophthora* *cryptogea* and *Phytophthora parasitica*. Stem tissues around soil level appear brown or blackish and the root system is found to be rotten. 2. **Footrot** also known as **pododermatitis** is a bacterial infection of hoofed animals, especially sheep and cattle that causes inflammation of the hooves. It is caused by Fusiformis necrophorus.

FOOT-AND-MOUTH DISEASE (HOOF-AND-MOUTH DISEASE;): *Aphtae epizooticae*, a highly contagious eruptive viral disease of cloven-hoofed mammals, characterised by blisters in the mouth and around the hooves. It is severe in cattle and pigs but mild in sheep and goats. In cattle, it causes decline in milk yield and abortions.

FOOT-CANDLE: A unit of measure of the intensity of light falling on a surface, which is equal to 10.764 lux.

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FOOTBATH: 1. A trough containing disinfectant through which sheep or cattle are driven to prevent or cure various diseases such as foot rot. 2. A shallow container containing disinfectant in which a person walks to disinfect shoes or boots before entering, for example, an animal enclosure to prevent diseases.

FOOTER: In computer science, it refers to a common text or information, usually containing the page number that repeats at the bottom of every page in a document or report. It is often used to display company data or copyright information. In longer documents, the footer may be used to specify the current section of the document as well. Footer also refers to the bottom section of a webpage. This area typically contains the name of the company or organisation that publishes the website, along with relevant copyright information. Some websites may also include basic navigation links, such as "About Us," "Contact," and "Help." Corporate website footers often include additional links to "Terms of Use," "Privacy Guidelines," and "Advertising" pages as well.

FOOTPRINTING: 1. A technique for detecting regions where a protein is bound to DNA, for example, where RNA polymerase is bound to the DNA of a gene, based on the principle that segments with bound protein are resistant to endonuclease activity and these can be demonstrated as missing bands in gel electrophoresis. 2. A technique that results in a DNA sequence ladder in which part of the ladder is missing due to the binding of protein to the DNA before processing. 3. In computing, footprinting is the process of finding out about computer systems and their networks, usually used for the purpose to intrude into the environment.

FORAGE: 1. Crop plants such as alfalfa hay or corn silage grown for their vegetative matter to feed animals either fresh or preserved as roughage. 2. To wander or go in search of food.

FORAGE BOX: A large movable container used mainly to transport forage from a silo to a trough.

FORAGE FEEDING: The practice of cutting herbage from a sward or foliage and feeding it to animals.

FORAGE HARVESTER (SILAGE HARVESTER; FORAGER; CHOPPER): A farm implement which cuts, chops and loads green crops such as lucerne into a trailer, to make silage. Forage harvesters can be implements attached to a tractor or they can be self-propelled units. The three main groups of forage harvester are: **Precision-chop machines** used for inert cut material which gives better clamp fermentation; **Double-chop machines** cut the crop twice; and **Single-chop machines** use a flail to cut the crop and produce a chop length of 150mm and above. Two different types of this machine are in use: the in-line, directly behind the tractor and the off-set which allows a field to be cut round and round.

FORAGE MAIZE: Maize grown for ensilage. Forage maize is drought tolerant and provides high yields with minimum input from the farmer. It is low in both protein and minerals and is sometimes difficult to fit into the cropping rotation.

FORAGE WAGON: A mobile container with a pick-up attachment used for collecting and carrying cut forage.

FORAMEN: An aperture in an animal part or organ, especially one in a bone or cartilage. For example, the **foramen magnum** is the opening at the base of the skull through which the spinal cord passes. The position of the foramen magnum is a strong indicator of the angle of the spinal column to the head and subsequently whether the body is habitually horizontal for example, as in the case of a horse or vertical as in the case of an ape or human.

FORB: A broad-leafed herbaceous plant which is not a grass.

FORBIDDEN BAND (ELECTRONIC BAND STRUCTURE; BAND STRUCTURE): The ranges of allowed energies of electrons in a solid. Certain ranges of energies between two such allowed bands are called forbidden bands, in which electrons within the solid may not possess these energies. The forbidden band of a material determines several characteristics, in particular the material's electronic and optical properties. In metals, forbidden bands do not occur in the energy range of the most energetic (outermost) electrons. Accordingly, metals are good electrical conductors. Insulators have wide forbidden energy gaps that can be crossed only by an electron having an energy of several electron volts. Because electrons cannot move freely in the presence of an applied voltage, insulators are poor conductors. Semiconductors have relatively narrow forbidden gaps, which can be crossed by an electron having energy of roughly one electron volt and so are intermediate conductors.

FORBIDDEN TRANSITIONS: Transitions between energy levels in a quantum mechanical system that are not allowed to take place because of selection rules. Forbidden transitions in rare earth atoms such as erbium and neodymium make them useful as dopants for solid-state lasing media.

FORBIDDEN ZONE: An energy separation at the highest filled electron energy band of an insulator and the next higher energy vacant band. It begins in the fourth energy level, which is a set of seven degenerate orbitals per energy level, higher in energy than s, p and d orbitals of the same energy level.

FORCE: Any influence (push or pull) that causes an object to accelerate, slowdown or change shape or any influence that tends to change the state of rest or the uniform motion in the straight line of a body. A force has both magnitude and direction, making it a vector quantity; and expressed as Force = mass x acceleration. Its symbol is F and the unit is the Newton (N), defined as the force required to cause a mass of one kilogram to accelerate at a rate of one meter per second squared in the absence of other force-producing effects which is equal to 1 kg m/s^2.

FORCE CHARGE: A property of a particle that determines how it responds to a particular force. For example, the electric charge of a particle determines how it responds to the electromagnetic force.

FORCE CONSTANT: A value which is unique to a bond which characterises the strength of the bond in a diatomic molecule. The force constant is a measure of the resistance offered to stretching and vibration of a bond, the magnitude of which indicates how strongly the atoms are held together and hence stability.

FORCE DIAGRAM: A diagram that uses arrows to represent the direction and magnitude of all the forces acting on an object.

FORCE FIELD: A non-contact force that acts on particles at various positions in space. Examples are gravitation force, electrostatic force and magnetic force.

FORCE METER: A device comprising a hook and spring that is used to measure force.

FORCE METRE (NEWTON METRE): A unit of the metre-kilogram-second system, which is equal to the energy expended or work done, by a force of one newton acting through a distance of one metre. It is equal to one joule.

FORCE MULTIPLIER: Any device designed to reduce the amount of force necessary to move an object by employing a small effort in order to move a larger load. The ratio by which a machine multiplies the effort is called its **mechanical advantage**. Examples of a force multiplier include inclined planes, most levers (such as a crowbar), wheelbarrow and nutcrackers.

FORCE OF ATTRACTION: 1. A force that pulls oppositely charged particles together. For example, the north and south poles of magnets come together by magnetic force. 2. Any type of force that pulls objects towards each other, even if those objects are not close to or touching. Force of attraction occurs between all masses in the universe; especially the attraction of the earth's mass for bodies near its surface. Forces that cause attraction are gravitational force, electric force or electrostatic force and magnetic force. Any two bodies in the universe attract each other due to gravitational force.

FORCE OF GRAVITY (GRAVITATION; GRAVITY): A force which keeps Earth and all planets in orbit around the Sun and with which objects with mass attract one another. Gravity is responsible for keeping the Moon in its orbit around the Earth; for the formation of tides; for natural convection, by which fluid flow occurs under the influence of a density gradient and gravity; for heating the interiors of forming stars and planets to very high temperatures. Together with strong force, electromagnetism and the weak force, they form the four fundamental interactions of nature. On Earth, gravity is the force of attraction between any object in the Earth's gravitational field and the Earth itself. According to Newton's Law of Gravitation, all objects fall to Earth with the same acceleration, regardless of mass.

FORCE OF MORTALITY (HAZARD RATE FUNCTION; FAILURE RATE): The probability of failure of a component or the death of an organism, within the next unit of time, Δt, given survival to time t. It is defined mathematically as $\Delta t \cdot h(t) = \mathrm{P}[\mathrm{X} \leq (t+\Delta t) : \mathrm{X} > t]$.

FORCE OF REPULSION: A force that pushes like charged particles apart. For example, the north and north poles or south and south poles of magnets do not come together because of force of repulsion.

FORCE RATIO (MECHANICAL ADVANTAGE): The magnification of a force by a simple machine such as lever, wedge, wheel and axle, pulley or screw. It is the ratio of the output force (load) of a machine to the input force (effort), expressed as Load/Effort.

FORCED CONVECTION: Heat convection in which fluid motion is maintained by some external agency.

FORCED OSCILLATIONS: Vibrations that occur at any given frequency when an external driving force is repeatedly applied to an object.

FOREARM: 1. Also known as **antebrachium**, it is the lower part of the arm between the hand and the elbow. The forearm contains two long bones, the radius and the ulna, forming the radioulnar joint. The interosseous membrane connects these bones. 2. The corresponding part of the foreleg in certain quadrupeds, such as a horse.

FOREBRAIN (PROSENCEPHALON): One of the three sections of the brain of a vertebrate embryo, which is the forward-most portion of the brain. The forebrain develops to form the cerebrum, hypothalamus and thalamus in adults and controls body temperature, reproductive functions, eating, sleeping and any display of emotions.

FORECAST: Prediction or statement of the likely occurrence of future events based on some kind of knowledge or judgement. An example is the weather forecast.

FOREDEEP: A long, narrow depression lying inland from an active mountain system and receiving sediment from the rising mountains.

FOREGUT: The anterior region of the alimentary canal of vertebrate embryo. In vertebrates it develops into the pharynx, oesophagus, stomach and anterior intestine.

FORELEG: One of the two front legs of an animal, as opposed to the hind legs.

FORENSIC COMPUTING: The process of collecting, analysing and reporting on data obtained from a computer such that it can be used in the law courts.

FORENSIC ENTOMOLOGY: A branch of forensic science, involving the study of insects and their arthropod relatives that inhabit decomposing bodies to aid in legal investigations. It is divided into three branches: medicolegal forensic entomology, urban forensic entomology and stored product forensic entomology. The **medicolegal forensic entomology** section is involved in studying the life cycle of insects and beetles on the body of corpses to help in investigations to establish the time of death. The **urban forensic entomology** deals with the insects that infest man and his immediate environment such as public spaces, private dwellings or commercial buildings with the aim of controlling the pests; or to identify the causes that led to an infestation, sometimes resulting in criminal or civil litigation. **Stored product forensic entomology** studies insects that are commonly found in foodstuffs and used to detect drugs and poisons that can be used in investigations involving food contamination to establish the chemicals present in a corpse at the time of death.

FORENSIC MEDICINE (FORENSIC PATHOLOGY; MEDICAL JURISPRUDENCE): The branch of medicine involving the application of medical knowledge to establish facts such as the cause and time of death in civil or criminal legal cases. The primary tool used is autopsy.

FORENSIC SCIENCE: The use of scientific techniques to solve criminal cases and the application of science and medicine for criminal and civil law or regulatory purposes. It includes the recognition, collection, identification, individualisation and interpretation of physical evidence.

FOREST (WOOD; WOODLAND): A large area of land thickly covered with a high density of trees. It functions as habitats for organisms, flow modulators and soil conservers, constituting one of the most important aspects of the Earth's biosphere. A person who manages woodland and plantations of trees is called a **forester**. The science of plants and caring for large areas of trees is called **forestry**.

FOREST FLOOR: The upper layer of the soil surface under forest vegetation made up of shed vegetative parts, such as leaves, branches, bark, and stems, in various stages of decomposition. It is an important part of the ecosystem as it harbours large number of decomposers and predators, mostly invertebrates, fungi, algae, bacteria, and archaea. The forest floor also stores carbon, recycles nutrients and has effects on soil moisture and temperature.

FOREST SOIL: Soil that forms under a cover of leaves and other plant remains influenced by forest vegetation and contains very little humus. It is characterised by deeply rooted trees, recycling of organic matter and nutrients, including wood, and wide varieties of soil-dwelling organisms.

FORESTRY COMMISSION: A government agency responsible for the management of state-owned forests.

FORGE: 1. The part of a factory containing a large apparatus that produces great heat inside itself, used for melting metals, making irons, etc. A forge is both the workplace and hearth of a smith or a blacksmith. 2. The process of shaping metal and increasing its strength by hammering or a more gradual application of pressure. In most forging an upper die is forced against a heated work piece positioned on a stationary lower die. To increase the force of the blow, power is sometimes applied to augment gravity.

FORGETFUL FUNCTOR (STRIPPING FUNCTOR; UNDERLYING FUNCTOR): A functor obtained by considering a category of algebraic forms such as groups, Abelian groups, modules, rings and vector spaces, as another category of sets with simpler objects. For example, the functor from the category of groups to the category of sets that preserves all mappings forgets the structure of the groups.

FORK: 1. A common hand implement usually having four square prongs each sharpened to a point for turning over soil and lifting out weeds. Forks with flat prongs are used for lifting potatoes. Larger, five-pronged forks with round curved prongs are used for spreading manure. 2. A branch or flash of the type of lightning that splits into two branches. 3. A part of a machine or device that has prongs or is fork-shaped.

FORM: 1. A category used in the classification of organisms into different types in which a variety may be placed. 2. Shape, develop, bring gradually into existence or make or produce especially by combining parts.

FORM GENUS: 1. Any genus in which imperfect states of fungi are placed. Similarly a form species is a species of imperfect fungus. When the connection between an imperfect state and a perfect state is established then the name of the perfect state takes precedence, although both names are still valid. 2. A group of fossil species whose affinities are unknown.

FORMAL CHARGE: The difference between the valence electrons in an isolated atom and the number of electrons assigned to that atom in a Lewis structure. Lewis structure is a structural formula in which electrons are represented by dots, for example, 2 dots between atoms represent a covalent bond.

FORMAL EQUIVALENCE: In logic, it is the relationship etween two open sentences that holds when their universal closures are materially equivalent, as between the two sides of a mathematical idetity such as $a + b = b + a$.

FORMAL LANGUAGE: Any language designed for use in situations in which natural language is unsuitable for the required precision, as in formal logic or in computer programs. The symbols and formulae of such a language stand in precisely defined syntactic and semantic relationships. In logic, the difference between a formal language and a formal calculus is that the semantics enable us to regard the former as about some subject matter. In computing, it is a set of strings of symbols that are constrained by rules and used in the specification of systems. These languages may be defined by their own grammar.

FORMAL LOGIC: 1. The study of formal systems. 2. Also known as **symbolic logic**, it is the study of deductive argument and of the structure and relations of statements, which includes primitive symbols, combinations of these symbols, axioms and rules of inference. 3. Any specific formal calculus that can be interpreted as representing natural argument and determines the validity or invalidity of a conclusion or inference deduced from two or more statements called premises. The basic principles are (i) principle of identity, in which if a statement is true then it is true; (ii) principle of excluded middle that a statement is either true or false; (iii) principle of contradiction that no statement can be both true and false at the same time.

FORMAL METHOD: A mathematically based method for the specification, development and verification of software and hardware systems.

FORMAL SYSTEM (FORMAL CALCULUS; FORMAL THEORY): A mathematical formalism in which statements can be constructed and manipulated with logical rules as between the two sides of a mathematical identity such as $a+b=b+a$. Some formal systems, such as Euclidean geometry, are built around a few basic axioms and can be expanded with theorems that can be deduced through proofs.

FORMALDEHYDE (METHANAL; FORMOL; METHYL ALDEHYDE; METHYLENE GLYCOL; METHYLENE OXIDE): A toxic colourless gaseous compound belonging to the family of organic compounds called aldehydes. Its chemical formula is CH_2O, density 0.8153 g/cm³ (-20°C, 253 K), melting point -92°C (181 K) and a boiling point of -19°C (254 K). It is used for manufacturing melamine and phenolic resins, fertilisers, dyes and embalming fluids and in aqueous solution as a preservative and disinfectant.

FORMALIN: An aqueous solution of methanal or formaldehyde in water with methanol as a stabiliser. It is used to preserve animal specimen and also as a disinfectant and preservative for biological specimens.

FORMALISE (FORMALIZE): In logic, to extract the logical form of an expression, to express in the symbols of some formal system.

FORMALISM: 1. The philosophical doctrine that mathematical statements have no extrinsic meaning but that their symbols themselves, regarded as physical objects, exhibit a structure that has useful applications. 2. A formal language, especially one intended as a formalisation of some fragment of natural language. 3. The mathematical or logical structure of some theory or argument, as distinguished from its subject matter.

FORMAT: 1. The general physical appearance such as the typeface, binding, quality of paper, margins, etc., of a publication such as a book, magazine or newspaper. 2. The medium in which something such as video or audio recording is made available and played back, for example, DVD format. 3. The arrangement of data for computer input or output, such as the number and size of fields in a record or the spacing and punctuation of information in a report. 4. A defined structure for the processing, storage or display of data, for example, a data file in binary format. 5. **Disk formatting** is initialising a storage media, such as a hard drive or flash drive to be recognised by a computer. This enables the computer to read the disk's directory structure, which defines the way files and folders are organised on the disk. 6. A **file format** is the way data is saved within a file. For example, some files are saved in plain text format, while others are saved as binary files. This prevents the files from being used by other applications. 7. **Full format** is a method of removing files from a disk partition. In this process, files are erased and the hard disk is scanned for bad sectors. It takes a longer time to complete. The scan for bad sectors is responsibe for the majority of the time that it takes to format the volume. 8. **Page format** is the page layout or style of text in a word processing document, which can modify the page size, page margins, alignment, line spacing, etc. Formatting the text enables the choice of the font and font size, text styles such as bold, underlined and italics. 9. **Quick format** is a method of removing all data or files from a disk or a partition so that it can be written on. It erases files from the partition but does not scan the disk for bad sectors. It takes a very short time to complete. This option can be used if the hard disk has been previously formatted and you are sure that the disk is not damaged.

FORMATION CONSTANT (STABILITY CONSTANT; BINDING CONSTANT): The equilibrium constant for the formation of a complex ion from its components in solution. It is a measure of the strength of the interaction between the reagents that come together to form the complex.

FORMATION RULES: In logic, the set of rules that specify the syntax of a formal calculus. It is the algorithm that generates all and only the well-formed formulae (wff) of the system. The rules are: (i) an expression consisting of a predicate variable of degree n followed by n individual variables is a wff; (ii) if α is a wff, so is $\sim\alpha$; (iii) if α and β are wffs, so is $(\alpha \vee \beta$; and (iv) if α is a wff and a is an individual variable, then $(\forall a)\alpha$ is a wff, where α is said to be the scope of the quantifier.

FORMATIVE ASSESSMENT: A wide variety of methods that teachers use to conduct in-process evaluations of student comprehension, learning needs, and academic progress during a lesson, unit, or course. This enables students and teachers to identify the students' strengths and weaknesses so students may improve their achievement and better manage their own learning.

FORMAZANS: The parent compound $H_2NN=CHN=NH$ and its derivatives formed by substitution at carbon and/or nitrogen.

FORMIC ACID (METHANOIC ACID): The simplest carboxylic acid, which is a colourless liquid having a highly pungent, penetrating odour at room temperature. Its chemical formula is CH_2O_2, density 1.22 g/ml, melting point 8.4 °C (281.4 K) and boiling point 100.8 °C (374 K). It is miscible with water and most polar organic solvents. It is secreted by some insects, especially red ants. In hydrocarbons and in the vapour phase, it consists of hydrogen-bonded dimers rather than individual molecules. Owing to its tendency to hydrogen-bond, gaseous formic acid does not obey the ideal gas law. It is used in textile and leather manufacture, as an industrial solvent and as an intermediate.

FORMICA: A trademark for Formica Corporation for a heat-proof plastic laminate, which is widely used as a veneer on kitchen surfaces and children's furniture.

FORMULA: 1. A set of symbols and numbers that expresses a fact, a rule or a relationship. For example, $A = \pi r^2$ is the formula for calculating the area of a circle; and $C = \pi D$, expresses the relationship between the circumference and diameter of a circle. 2. A formal expression of some rule or other result, for example, Stirling's formula or Frenet formulae. 3. A representation of a molecule, ion or radical in which the compound's chemical elements are represented by the chemical symbol. 4. Functions within computer programs often contain formulas, which are used to process data that is passed through the function and output the result. For example, a basic formula might convert miles to kilometres.

FORMULA BAR (FORMULA BOX): A section in Microsoft Excel and other spreadsheets that displays the data or formula stored in the active cell and allows users to enter formulas instead of static data. In Microsoft Excel, it is located above the work area of the spreadsheet.

FORMULA UNIT: The lowest reduced ratio of ions in a compound. It is the empirical formula of an ionic or covalent network compound used as an independent entity for stoichiometric calculations. Examples include ionic sodium chloride (NaCl), Potassium oxide (K_2O) and covalent networks such as silicon dioxide (SiO_2) and C (as diamond or graphite).

FORMULA WEIGHT: The sum of the atomic weights of all atoms in a molecule. The term is generally applied to a substance that consists of ions rather than individual molecules. An example of such a substance is sodium chloride (NaCl). Such a substance's chemical formula describes the simplest ratio of the number of atoms of the constituent elements.

FORMULATION: 1. Putting together of components in appropriate relationships or structures, according to a formula. 2. The particular mixture of base chemicals and additives required for a product. In pharmacy, it is a mixture or a structure such as a capsule, a pill, tablet or an emulsion, prepared according to a specific procedure. 3. Formulation science is concerned with the knowledge and practice of blending and mixing of various components (chemical molecules) in a way that they do not react, but interact to provide a final product with very specific desirable properties or functions. Formulation science covers the core of physical and colloid chemistries, as well as analytical chemistry, chemical engineering, pharmaceutical and biological sciences.

FORMYL TRICHLORIDE (CHLOROFORM; METHANE TRICHLORIDE; METHYL TRICHLORIDE; METHENYL TRICHLORIDE; TRICHLOROMETHANE): A colourless volatile sweet-smelling liquid haloform with the chemical formula $CHCl_3$, relative density 1.48, melting point -63.5 °C (209.5 K) and boiling point 61.7 °C (334.7 K). It can be made by chlorination of methane followed by separation of the mixture of products or by the haloform reaction. It is an effective anaesthetic but can cause liver damage and it has now been replaced by other halogenated hydrocarbons. It is used as a solvent and raw material for making other compounds.

FORMYLATION: A chemical reaction that introduces a formyl group (HCO), a chemical group containing carbon, hydrogen and oxygen, into an organic molecule.

FORNIX: One of a set of small crests or scales in the throat of a corolla, as occurs in many of the Boraginaceae.

FORSTERITE (WHITE OLIVINE): A whitish or yellowish, magnesium-rich variety of olivine, with the chemical formula Mg_2SiO_4. It is the end-member of the olivine solid-solution series. At the other end of the olivine series is the iron-rich mineral fayalite (Fe_2SiO_4). On Earth, forsterite is found in ultramafic igneous rocks or in marbles made from the metamorphism of dolomitic limestones. When it occurs as crystals it is called **peridot**. Some forsterite also occurs in certain types of meteorites in which olivine is common. The olivine in most stony-iron meteorites ranges in composition from 60% to 85% forsterite with the remainder being fayalite.

FORTIFIED FOOD: Food containing added vitamins and minerals and proteins to make it more nutritional.

FORTIN BAROMETER: A type of cistern barometer, in which provision is made to increase or decrease the volume of the cistern so that when a pressure change occurs, the level of the cistern can be maintained at zero of the barometer scale (the ivory point).

FORTRAN (FORMULA TRANSLATION): A high level computer programming language designed for mathematical and scientific use by John W. Backus and others at IBM in 1957. It has been revised several times and now includes capabilities for handling structured data, dynamic data allocation, recursions (procedures that call themselves) and other features.

FORTRESS: A programming language developed by Sun Microsystems for high performance computing that can also serve as a general purpose programming language.

FORWARD BIAS: A p-n junction is said to be forward biased if the positive terminal of the battery is connected to the p-type semiconductor and the negative terminal of the battery is connected to the n-type semiconductor. Majority charge carriers are made to move across the p-n junction, decreasing the depleting layer and the barrier potential and hence resulting in the flow of large current.

FORWARD CREEP GRAZING: A grazing method in which grassland which has been allocated to ewes and lambs during the fattening period is divided into paddocks, each separated by portable fencing. As one area is finished, the fencing is moved to allow the animals to move to the next.

FORWARD DIFFERENCE: A series of quantities obtained from a function whose values are known at a series of equally spaced points by repeatedly applying the forward difference operator, Δ to these values. The forward difference operator, Δ is defined as $\Delta f(x) = f(x + h) - f(x)$. The expression $f(x+h) - f(x)$ gives the **first forward difference** of $f(x)$ and the operator Δ is called the **first forward difference operator**. The **second forward difference operator**, denoted by Δ^2, is defined as $\Delta^2 f(x) = \Delta[\Delta f(x)] = \Delta f(x+h) - \Delta f(x)$. It is used in interpolation or numerical calculation and integration of functions.

FORWARD ERROR ANALYSIS: The analysis, for a given algorithm, of the error between an exact quantity and its computed approximated quantity. The aim is to determine a measure of the error that if small, it validates the calculation and does not distinguish rounding error from the truncation error in numerical analysis.

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FORWARD GENETICS: The traditional approach to genetic investigation in which the aim is to identify the gene that governs a particular function. Forward genetics is a counter measure to reverse genetics, which seeks to alter genes in order to illuminate their multiple phenotypes.

FORWARD SCATTERING: Scattering of photons (particles of light or other forms of electromagnetic radiation) by a medium in such a way that most of the photons end up travelling in roughly the direction they started. More generally, it is scattering in which the angle between the initial and final directions of motion of the scattered particles is less than 90°. Forward scattering is always accompanied by some backscattering (scattering in a generally backward direction), the proportion of the former increasing with particle size.

FOSS: Abbreviation for Free and Open-Source Software, which refers to free and open-source software. Examples of mathematics software are Sage, licensed under the GPL; Axiom, a general purpose Computer Algebra system; and SINGULAR, a Computer Algebra System for polynomial computations with special emphasis on the needs of commutative algebra, algebraic geometry and singularity theory.

FOSSIL: The remains of an animal or plant of a past geologic age preserved in sedimentary rock, asphalt deposits and coal and sometimes in amber and certain other materials. The scientific study of fossils is known as **palaeontology**. The data recorded in fossils, known as the **fossil record**, constitute the primary source of information about the history of life on the Earth.

FOSSIL EMISSIONS: Emissions of carbon dioxide (CO_2) into the atmosphere caused by the combustion of fuels from fossil carbon deposits such as oil, natural gas, and coal. This can be detrimental to health, as well as causing potentially irreversible global warming.

FOSSIL FUEL: Any of a class of materials of biologic origin occurring within the Earth's crust such as the remains of plants and animals that lived and died hundreds of millions of years ago, which have been transformed into oil, coal and natural gas. Fossil fuels contain carbon and include coal, petroleum and natural gas that can be used as a source of energy.

FOSSORIAL: A term used to describe a burrowing animal.

FOUCAULT PENDULUM: A simple pendulum suspended from a long wire and set into motion along a meridian. As a result of the Earth's rotation on its axis, the plane of the pendulum's wings slowly turns. At the poles of the Earth it makes one complete revolution in 24 hours. The plane of motion appears to turn clockwise in the Northern Hemisphere and anticlockwise in the Southern Hemisphere, demonstrating the axial rotation of the earth.

FOUCAULT TEST: A test that uses interference patterns produced by a knife edge to determine the deviation of a mirror from its ideal shape.

FOUL OF THE FOOT: A disease of cattle and sheep marked by inflammation and ulceration of the feet caused by damage to the cleft of the hoof and invasion by the bacterium *Fusifonnis necro-phorus*.

FOULING: A process resulting in loss of performance of a membrane due to the deposition of suspended or dissolved substances on its external surfaces, at its pore openings or within its pores.

FOULING AGENT (MECHANICAL INHIBITOR IN CATALYSIS): Inhibitory substances bound by neither covalent nor other strong bonds to the active centres: the interaction is usually of the van der Waals, H-bond or sometimes ionic type. They form protective layers or block pores, thus physically impeding access of reactants to the active centres. The fouling agents, which cause real problems are those which have a long standing effect and do not disappear spontaneously. Carbon deposits act, partially or totally, this way. Other examples are vanadium and nickel sulfide deposits in hydrotreating catalysts.

FOUNDATIONS OF MATHEMATICS: The study of the logical basis or justification of mathematics. It attempts to establish an axiomatic basis upon which mathematics could be built.

FOUNDER CROP (NEOLITHIC FOUNDER CROPS; PRIMARY DOMESTICATES): A crop such as wheat, barley, lentils and chickpeas that was one of the earliest to be used and developed by humans.

FOUNDER PRINCIPLE (FOUNDER EFFECT): The principle that genetic drift occurs when a small pioneer community establishes itself in genetic isolation from the main population. The founder effect can be observed in geographically isolated areas and among populations that practice intermarriage based on circumstances, customs, beliefs or economics. Consequently, the community that develops may take a different evolutionary path from the parental population. Such a community has relative uniformity among its members and frequent occurrence of distinct or unusual characteristics.

FOUNDRY: A factory which produces metal castings from ferrous or non-ferrous alloy. Molten metal is poured into a mould and the mould material is removed after some time to reveal the metal casing.

FOUNTAIN EFFECT: The effect that occurs when two containers of superfluid helium are connected by a capillary tube and one of them is heated, the expansion pushes up the free surface of the liquid forming a fountain. This causes liquid helium to flow up the sides of open containers.

FOUR TOOTH SHEEP: A sheep which is 18-21 months of age.

FOUR SQUARES THEOREM (LAGRANGE'S FOUR-SQUARE THEOREM; BACHET'S CONJECTURE): The theorem that every positive integer can be expressed as the sum of not more than squares of four positive integers, exactly four non-negative integers. For example, $1 = 1^2 + 0^2 + 0^2, 5 = 2^2 + 1^2 + 0^2 + 0^2$ and $15 = 3^2 + 2^2 + 1^2 + 1^2$.

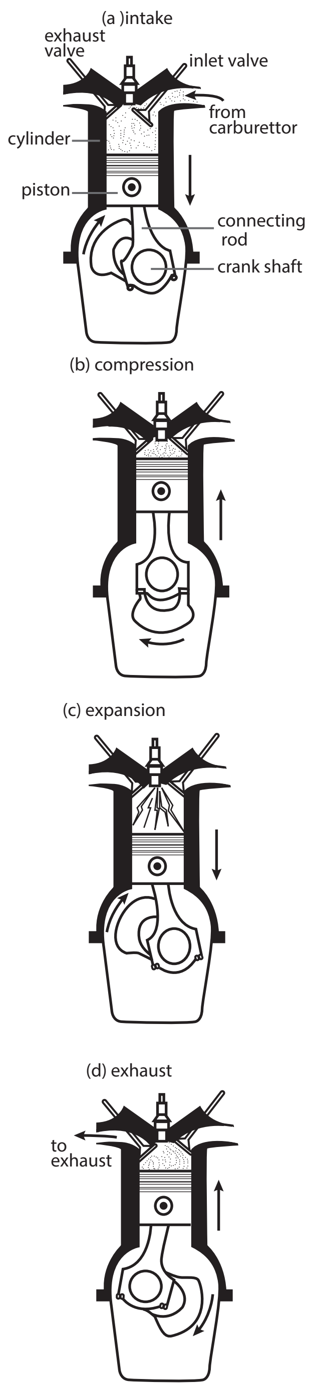
FOUR-CIRCLE DIFFRACTOMETER: An instrument used in x-ray crystallography to automatically determine the shape and symmetry of the unit cell of a crystal.

FOUR-COLOUR PROCESS: A colour printing using four printing plates based on the principle that any colour is made up of differing properties of the primary colours: blue, red and green. The process utilises four ink colours (black, magenta, cyan and yellow) to produce a printed image that matches the colouring of the original image. Four-colour process is used in gravure, letterpress, offset and screen printing.

FOUR-COLOUR THEOREM: The theorem that any planar map can be coloured using at most four colours in such a way that no two adjacent areas are of the same colour. Two American mathematicians, Kenneth Ira Appel (1932-2013) at the University of Illinois and his colleague Wolfgang Haken (1928 -) provided a mathematical proof in 1978 that solved the long-standing four-colour map problem.

FOUR-LEVEL LASER: A laser in which the lowest level for a laser transition is an excited state rather than the ground level.

FOUR-STROKE ENGINE (OTTO ENGINE): An internal combustion engine in which the operation of each working cycle is in four stages or strokes, which include suction stroke also known as intake stroke, compression stroke, ignition stroke also known as power stroke and exhaust stroke. It was first designed by Nikolaus Otto (1832-1891), German inventor. A stroke is the full travel of the piston in a cylinder from Top Dead Centre (TDC), when the piston is farthest away from the axis of the crankshaft, to Bottom Dead Centre (BDC). Most internal combustion engines such as those found in cars, including diesel engines, use the four-stroke system. Each of the four stages corresponds to one movement of a reciprocating piston along a cylinder. In the suction stroke also known as intake stroke, the piston goes down from the top of the cylinder to the bottom, reducing the pressure inside the cylinder. It then draws a mixture of fuel and air into the cylinder through the intake port. In the compression stroke, the piston rises, compressing the fuel-air mixture. In the ignition stroke, also known as power stroke, the mixture is ignited by a spark or compression and expanding gases from the explosion force the piston downward. The cycle ends with the exhaust stroke, in which the piston rises to open valves to expel the spent gases from the cylinder and another cycle begins.



Four-stroke petrol engine

FOUR-WAVE MIXING: The production of a photon in a medium as a result of the interaction of three photons. In four-wave mixing, the photon produced has a wavelength that is different from the wavelengths of any of the three original photons. Four-wave mixing can be compared to the intermodulation distortion in standard electrical systems. The effect is important in optical fibres, which can lead to loss of signal when a number of wavelengths are transmitted along the same fibre.

FOURIER ANALYSIS: The study and application of Fourier series and related integrals, especially in the study of differential equations, mathematical physics and approximation. It is a method of defining periodic waveforms in terms of trigonometric functions. It was named after French mathematician and physicist Jean Baptiste Joseph, Baron de Fourier (1768-1830). Fourier analysis is used in electronics, acoustics and communications.

FOURIER COEFFICIENTS: The coefficients

$$a_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\cos(\mathrm{nx})\mathrm{dx}\,(n \geq 0),$$

$$b_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\sin(\mathrm{nx})\mathrm{dx}\,(n \geq 1),$$ and in complex terms

$$C_n = \frac{a_n - ib_n}{2} = \frac{1}{2\pi}\int_0^{2\pi} f(x)\exp(-\mathrm{inx})\mathrm{dx}$$ that can be used to express a function formally in terms of its Fourier series.

FOURIER METHOD (SEPARATION OF VARIABLES): Any of several methods for solving ordinary and partial differential equations, using algebra to rewrite the equation with each of the two variables occurring on a different side of the equation.

FOURIER SERIES: An expression of a periodic function $f(x)$, which is periodic in x, as an infinite series of sine and cosine functions. It is defined as follows:

$$g(t) = a_o + \sum_{m=1}^{\infty} a_m \cos(\frac{2\pi mt}{T}) + \sum_{n=1}^{\infty} b_n \sin(\frac{2\pi mt}{T}) = \sum_{m=0}^{\infty} a_m \cos(\frac{2\pi mt}{T}) + \sum_{n=1}^{\infty} b_n \sin(\frac{2\pi nt}{T})$$

where

$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)dx;\ a_m = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\cos(mx)dx, 1 \leq m;$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(\pi)\sin(nx)dx, 1 \leq n$$

FOURIER TRANSFORM: 1. The integral transform $F(y) = \int_{-\infty}^{\infty} f(x)\exp(-iyx)\mathrm{dx}$, that sends a function f to another function F. The Fourier transform decomposes any function into a sum of sinusoidal basis functions. Each of these basis functions is a complex exponential of a different frequency. It provides a way of viewing any function as the sum of simple sinusoids. 2. The function, F that is so related to a given function; the image of the given function under Fourier transformation.

FOURIER TRANSFORM INFRARED: Infrared spectroscopy in which computers are part of the spectroscopic apparatus and used in Fourier transforms to enable the curve of intensity against wave number to be plotted with very high sensitivity.

FOURIER-TRANSFORM SPECTROSCOPY: A measurement technique whereby spectra are collected based on measurements of the temporal coherence of a radiative source, using time-domain measurements of the electromagnetic radiation or other type of radiation.

FOURIER, JEAN BAPTISTE JOSEPH (1768-1830): French mathematician, best known for his contribution to the mathematical analysis of heat flow and for showing that all periodic oscillations can be reduced to a series of simple, regular wave motions.

FOURIER'S LAW OF CONDUCTION (LAW OF HEAT CONDUCTION; FOURIER'S LAW): An empirical law based on observation. It states that the rate of heat flow, dQ/dt, through a homogeneous solid is directly proportional to the area, A, of the section at right angles to the direction of heat flow and to the temperature difference along the path of heat flow, dT/dx.

FOURTH DIMENSION: An abstract concept derived by generalising the rules of three-dimensional space. It is time in relativity theory modified mathematically and used in combination with the usual three spatial dimensions to specify the location in space and time of events.

FOURTH GENERATION COMPUTER: Computers based on Very Large Scale Integrated (VLST) circuits, with microchips that contain thousands of transistors. These computers are powerful, compact, reliable and affordable.

FOURTH GENERATION LANGUAGE: A type of higher-level programming language designed for rapid programming of applications but often lacking the ability to control the individual parts of the computer.

FOURTH-GENERATION (4G) MOBILE COMMUNICATIONS: Broadband mobile communications that succeeds the third generation (3G). Potential and current applications include amended mobile web access, IP telephony, gaming services, high-definition mobile TV, video conferencing, and 3D television.

FOURTH HARMONIC POINT (HARMONIC CONJUGATE): The fourth point, collinear with a pair of points, A and B and another collinear point, X, that can be found by constructing the complete quadrangle corresponding to a trapezoid ABCD, for which X is the point of intersection of the extensions of AB and CD. If Y is the sixth point of the complete quadrangle and the diagonals of the trapezoid intersect at Z, then the intersection, H, of YZ with the line XAB is the fourth harmonic of X with respect to A and B. These four collinear points are then a harmonic set as their cross ratio is a harmonic ratio in which A, B and X, H are conjugate pairs. Iterating the process leads to a harmonic net of rationality. Dually, a complete quadrilateral produces an harmonic pencil of four lines.

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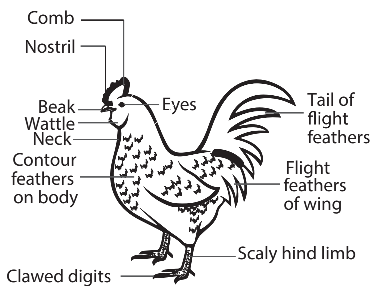
FOURTH ROOT OF UNITY: A complex number z such that $z^4 = 1$ has 4 fourth roots of unity, which are are $1, i, -1$ and $-i$.

FOVEA (FOVEA CENTRALIS; YELLOW-SPOT): A part of the eye, located in the centre of the macula region of the retina, which is a shallow depression in the retina of the eye, opposite the lens, that is present in the eyes of some vertebrates, specialised for acute vision and containing numerous cones but no rods. The fovea is responsible for sharp central vision, which is necessary in humans for reading, watching television or movies, driving, etc.



Domestic fowl

FOWL: Any of various birds of the order Galliformes, especially the common, widely domesticated chicken *(Gallus domesticus)*. Some types such as domesticated chicken, duck, goose, turkey or pheasant are used as food or hunted as game.

FOWL PEST: An acute and usually fatal viral disease of domestic fowl, characterised by high temperature and discolouration of the comb and wattles.

FOWL POX: 1. A viral disease of birds characterised by wart-like nodules on the comb, wattles, eyelids and openings of the nostrils of fowls. 2. Also known as **Newcastle disease, avian pneumoencephalitis, avian pseudoplague, Philippine fowl disease**, it is a contagious acute viral disease of turkeys and many wild birds, occasionally infecting humans. It is caused by avian paramyxovirus-1, in the genus *Avulavirus* which affects the respiratory, gastrointestinal and central nervous system. It may be transmitted to human beings as a mild conjunctivitis.

FOWL SICK: A land which has become infested with parasites as a result of having been used for free-range hens for too long a period of time.

FOX: A carnivorous canine predator, of about 37 species belonging to the Canidae family with red fur, a long narrow snout and a large bushy tail. Foxes are slightly smaller than the median-sized domestic dog. The most common and widespread species of fox is the red fox (*Vulpes vulpes*).

FOXGLOVE: A common weed (*Digitalis purpurea*) of about 20 species of herbaceous plants of the genus *Digitalis*, in the family Plantaginacea, which is poisonous and harmful to animals. It is cultivated as the source of the heart-stimulating drug digitalis.

FOXTAIL MILLET: The first cereal (*Setaria italic*) to be cultivated in China. It is an annual grass with slim, vertical, leafy stems which can reach a height of 120-200 cm. The small seeds, around 2 mm in diameter, are encased in a thin, papery hull which is easily removed in threshing. Seed colour varies greatly between varieties. It is used for silage, hay, brewing and flour in many parts of the world.

FRACTAL: Irregular or fragmented geometrical objects generated by a process involving successive subdivision. They are self-similar under a change of scale, for example, magnification. A snowflake curve is an example which is formed by dividing an equilateral triangle into three segments. The middle segments are then replaced by two equal segments to form the sides of a smaller equilateral triangle. The result is a 12-sided star-shaped figure whose sides are subdivided continuously to get a snowflake. Fractals are used in computer modelling of natural structures that do not have simple geometric shapes such as clouds, mountainous landscapes and coastlines.

FRACTILE: The value of the item that is a given fraction of the way through a distribution. An item in the distribution that is hundredths of the way through the distribution is called a percentile; for example, 85th percentile is $\frac{85^{th}}{100}$ of the way through the distribution. Its location is $\frac{85}{100}\sum f$. The item in the distribution that divides the distribution into four quarters is called the **quartile**. The first quartile, Q_1 is the item that is one-quarter of the way through the distribution; the second quartile, Q_2 is the item that is the median of the distribution; the third quartile, Q_3 is the item that is three-quarters of the way through the distribution. The locations in the distribution are $Q_1 = \frac{\sum f}{4}, Q_2 = \frac{\sum f}{4}$, and $Q_3 = \frac{3\sum f}{4}$.

The following formula can be used to calculate any required fractile:

$$L + \frac{\text{fractile location} - \text{sum of frequencies above fractile class}}{\text{frequency of fractile class}},$$

where L is the lower class boundary of the fractile class. An item that is tenths of the way through a distribution is called a **decile**, for example, the third decile is $\frac{3}{10}$th of the way through the distribution. The location of the third decile is $\frac{3\sum f}{10}$ and $\sum f + 1$ for odd sum of frequencies. The fractiles can be determined from the cumulative frequency curve (the ogive).

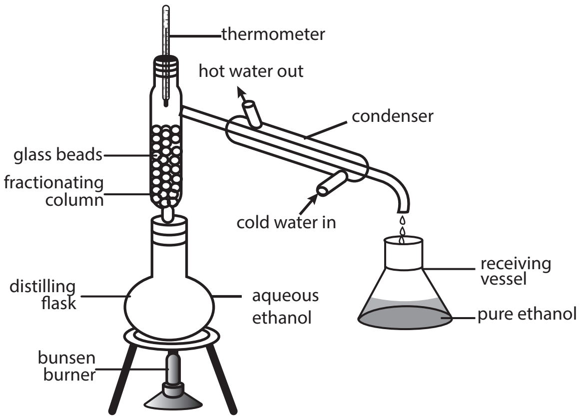
FRACTION: 1. A number that indicates one or more equal parts of a unit. In other words, it is a ratio of two integers or any number that can be expressed as such a ratio, $\frac{a}{b}$, where a is not a multiple of b and b is not zero or one. In this case, a is the number of equal parts the unit have taken, called the **numerator**; and b represents the total number of parts into which the unit has been divided, called the **denominator**. The rules for addition/subtraction, division and multiplication of fractions are $\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm cb}{bd}, \frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$. 2. A group of similar compounds, the boiling point of which fall within a particular range and which are separated during fractional distillation.

FRACTION COLLECTOR: A device for the automatic collection of consecutive, commonly equal-sized, fractions of a flowing liquid, such as the eluate from a chromatographic column. The size of the fractions may be predetermined by volume, mass, time or number of drops.

FRACTIONAL: 1. Relating to or involving fractions. 2. Relating to the process of separating individual components from a mixture on the basis of the chemical or physical properties that make them different from other components.

FRACTIONAL CRYSTALLISATION: A process of refining substances by which a mixture of soluble solids is separated by dissolving them in a suitable hot solvent and then slowly lowering the temperature.

FRACTIONAL DISTILLATION: A method of separating a mixture of several volatile components of different boiling points by heating them to a temperature at which several fractions of the compound will evaporate. The distillate is collected as one fraction until the temperature of the vapour rises, showing that the next higher boiling component of the mixture is beginning to distil; this component is then collected as a separate fraction. Generally, the component parts boil at less than 25 °C (298 K) from each other under a pressure of one atmosphere (atm). If the difference in boiling points is greater than 25 °C (298 K), a simple distillation is used.



Apparatus for fractional distillation

FRACTIONAL LINEAR TRANSFORMATION (MÖBIUS TRANSFORMATION): A rational invertible transformation of the complex plane of the form, $w = \frac{az+b}{cz+d}$ with $ad \neq bc$. These are conformal mappings.

FRACTIONAL PART: The difference between the given real number and its integral part. For example, the fractional part of 3.42 written as {3.42}, is 0.42 and that of -3.42 is 0.58.

FRACTIONAL PRECIPITATION: The process of removing ions from a solution by precipitation while leaving other ions with similar properties in the solution.

FRACTIONAL QUANTUM HALL EFFECT: The physical phenomenon in which the Hall conductance of 2D electrons becomes precisely equal to e^2/h. It is a property of a collective state in which electrons bind magnetic flux lines to make new quasi particles and excitations have a fractional elementary charge and possibly also fractional statistics. The 1998 Nobel Prize in Physics was awarded to Dan Tsui and Horst Stormer, who first observed it, with Robert Laughlin who explained the result in terms of new quantum states of matter.

FRACTIONATING COLUMN: A long narrow vertical tube often equipped with internal plates used in fractional distillation, in which many separate distillations can occur, so that a liquid mixture can be separated into its components. On heating, high and low boiling point fractions separate, those of the lowest boiling point rising highest. It contains trays or glass balls that cause the ascending vapour to mix thoroughly with the descending liquid.

FRACTIONATION: The separation of a mixture using fractional distillation or fractional crystallisation. In each stage some proportion of one of the substances is removed from the mixture.

FRACTURE: A complete or incomplete break in the continuity of a bone, with or without displacement of any fragments. It may be pathological, the result of a relatively mild injury to an already diseased bone caused by diseases such as osteomalacia (a disease occurring mainly in women, causing softening of the bones and resulting in pain and weakness) or, more often, it is the result of an injury to healthy bone such as the application of excessive force. **Comminuted fracture** is a fracture of a bone in which the separated parts are splintered or fragmented into more than two pieces. A **complete fracture** is a type of bone fracture in which the bone is split completely across. A **compound fracture** is a type of fracture in which the broken bone is exposed through a break in the skin. A **simple** or **closed fracture** is a type of fracture in which the bone does not pierce the skin.

FRAGILE X SYNDROME (FRAGILE X; MARTIN-BELL SYNDROME; MARKER X SYNDROME; FRAXA SYNDROME): An X-linked hereditary mental disorder, partially explained by genomic imprinting and the addition of nucleotides to a triplet repetition near the end of an X chromosome. It is is characterised by moderate to severe mental retardation, by large ears, chin and forehead and by enlarged testes in males and that often has limited or no effect in heterozygous females.

FRAGMENT ANTIGEN-BINDING (FAB FRAGMENT): The part of an antibody that binds to antigens. It is one of the fragments produced by partial digestion of immunoglobulin with the proteolytic enzyme, papain. It consists of an arm of the Y-shaped molecule and an entire light chain paired with part of a heavy chain, complete with antigen binding site.

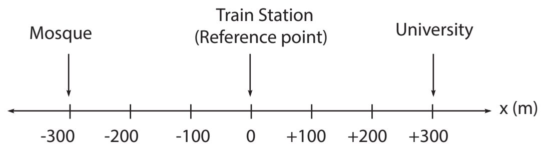
FRAGMENT ION: An electrically charged dissociation product of an ionic fragmentation. Such an ion may dissociate further to form other electrically charged molecular or atomic moieties of successively lower formula weight.

FRAGMENTATION: 1. An operating systems process of writing different parts of a file to discontinuous sectors on a computer storage medium when contiguous space that is large enough to contain the entire file is not available. When data is thus fragmented, the time that it takes to access the data may increase because the operating system must search different tracks for information that should be in one location. 2. Also known as **IP fragmentation**, it is the breaking of an IP datagram (packet) into pieces in order to be sent across a transmission link with a frame size smaller than the datagram. Performed in a router, the header of the original IP packet is replicated with minor changes to each of the fragments. If one of the fragments is dropped, the original datagram must be fragmented again and retransmitted. Fragmentation also describes storing data in non-contiguous areas on disk. As files are updated, new data are stored in available free space, which may not be contiguous. Fragmented files cause extra head movement, slowing disk accesses. A defragger program is used to rewrite and record all the files. 3. Also known as **clonal fragmentation**, it is a method of asexual reproduction or cloning occurring in certain coelenterates and annelids, moulds and sea stars in which parts of the organism is split into fragments by breaking off and subsequently differentiating and developing into new individuals. Fragmentation is caused by mitosis. 4. The breakdown of a radical into a diamagnetic molecule or ion and a smaller radical.

FRAME: 1. A structural system that supports other components of a physical construction. 2. In statistics, a **sampling frame** is an enumeration of a population for the purpose of sampling, especially as the basis of a stratified sample. It includes a numerical identifier for each individual, plus other identifying information about characteristics of the individuals, to aid in analysis and allow for division into further frames for more in-depth analysis. 3. In video display, it is a single photograph in a sequence representing motion or movement. 4. In telecommunications, it is data that is transmitted between network points as a unit complete. 4. In word processing or desktop publishing, it is rectangular marked-out area on a page that can contain text or graphics. 5. In HTML, it refers to a browser display page that have been divided into separate sections, each consisting of an independent Web page.

FRAME BUFFER: In computing, it is an area of random access memory (RAM) that contains the complete bit-mapped image that is continuously being sent to the monitor.

FRAME OF REFERENCE (REFERENCE FRAME): Any set of planes or curves, such as the three coordinate axes, taken as being for practical purposes at rest that enables the position of a point or body in space to be defined at any instant in time. As a point moves, its velocity can be described in terms of changes in displacement and direction. For example, if a person is sitting in a moving train, the description of the person's motion depends on the chosen frame of reference. If the frame of reference is the train, the person is considered not moving relative to the train. If the frame of reference is the Earth, the person is moving relative to the Earth.



Frame of reference

FRAMES AND ARROWS: Diagrams consisting of frames connected by arrows used to represent number sequences. Each frame contains a number, and each arrow represents a rule that determines which number goes in the next frame. There may be more than one rule, represented by different-colour arrows.

FRAMESHIFT: The addition or deletion of one or more bases from a DNA molecule normal sequence during transcription. This causes a change in translational reading that allows the production of multiple distinct proteins from overlapping genes. The process can be programmed by the nucleotide sequence of the mRNA and is sometimes also affected by the secondary or tertiary mRNA structure. It has been described mainly in viruses, retrotransposons and bacterial insertion elements and also in some cellular genes.

FRAMESHIFT MUTATIONS: Insertions or deletions of genetic material that lead to a shift in the translation of the reading frame. This occurs when the number of nucleotides inserted or deleted is not a multiple of three, so that every codon beyond the point of insertion or deletion is read incorrectly during translation. The mutation usually leads to non-functional proteins.

FRANCIUM: A radioactive metallic element that decays into astatine, radium and radon. Its symbol is Fr, atomic number 87 and relative atomic mass 223. It belongs to the alkali - metal groups and it is obtained from uranium ore or made artificially from thorium and actium. It is the second rarest naturally occurring element after astatine.

FRANCK-CONDON PRINCIPLE: A principle governing the intensity of transitions in the vibrational structure during an electronic transition in a molecule. The principle states that since nuclei are heavier and move much more slowly than electrons, an electronic transition occurs much more rapidly than the time required for the nuclei to respond to it.

FRANGIPANI: A tropical American deciduous tree or shrub with clusters of strongly perfumed, pink, yellow or white flowers. It belongs to the genus Plumeria in the Apocynaceae family.

FRANK-KASPER PHASE: A phase that occurs in certain complex alloys involving both icosahedra and tetrahedral packed closely together.

FRANKLIN, BENJAMIN (1706-90): American scientist and statesman who held various government posts and was also a first-class scientist and inventor. As an amateur scientist he experimented with electricity and made major contributions to the theory of electricity, including the first recognition of conservation of charge, a theory of static electricity and proof that lightning is an electrical phenomenon and also introduced the concepts of 'positive' and 'negative'. In 1752 he carried out the extremely dangerous experiment of flying a kite during a thunderstorm and proved the electrical nature of lightning. He also invented the lightning conductor and also recognised the importance of ocean currents and had the Gulf Stream mapped for the first time. His inventions include the Franklin stove and bifocal spectacles.

FRASCH PROCESS: A method for the extraction of sulfur from deep underground deposits, using a tube consisting of three concentric pipes, which are sunk to the bottom of the sulfur bed. Superheated steam at 180 °C (453 K) is passed down the outer pipe to melt the sulfur which is forced up through the middle pipe by compressed air fed through the inner tube. The steam in the outer casing keeps the sulfur molten in the pipe.

FRASER SPIRAL: An optical illusion in which overlapping black arc segments appear to form a spiral but are in reality a series of overlapping concentric circles. It was first described by the British psychologist, James Fraser in 1908.

FRATERNAL TWINS (DIZYGOTIC TWINS): Two individuals that result from a single pregnancy, because they develop from two separate eggs that are fertilised by two separate sperm. Fraternal twins have different genetic makeup (they share about 50% of their genetic traits, the same as any other siblings born at different times.) and may be of the same or opposite sex. About 70-80% of twins are fraternal twins. Monozygotic or identical twins develop from one zygote that splits and forms two embryos. Monozygotic or identical twins have the same sex, identical genetic makeup and closely similar appearance.

FRATTINI SUBGROUP: The subgroup, Φ(G), of a given group G defined to be the intersection of all maximal subgroups of G. If G has no maximal subgroups, its Frattini subgroup is defined as Φ(G) = G.

FRAUNHOFER DIFFRACTION (FAR-FIELD DIFFRACTION): A form of wave diffraction named after Joseph von Fraunhofer (1787-1826), a German optician and physicist, in which the light source and the receiving screen are in effect at infinite distances from the diffracting object, so that the wave fronts can be treated as planar rather than spherical.

FRAUNHOFER LINES: Dark lines in the solar spectrum produced by absorption of light by the cooler gases in the Sun's outer atmosphere at frequencies corresponding to the atomic transition frequencies of these gases. It was named after Joseph von Fraunhofer (1787-1826), a German optician and physicist.

FRAUNHOFER, JOSEF VON (1787-1826): German optician and physicist, who made many improvements in the manufacture of optical glass, the grinding and polishing of lenses and the construction of telescopes and other optical instruments. His name is associated with the fixed, dark lines in the solar spectrum, called Fraunhofer lines, which he was the first to describe in detail. His investigations into the refraction and dispersion of light led to the invention of the spectroscope and the science of spectroscopy. Fraunhofer's other great achievement concerned the measurement of wavelengths in the optical spectrum. He transformed the spectroscope into a precision instrument, but his finest precision instrument was the micrometer, described in his memoir of 1824 to the Munich Academy.

FRÉCHET DIFFERENTIAL: Named after Maurice Fréchet, French mathematician, the function $\delta f(x;)$, derived from a given function f between normed spaces and defined on an open domain for which

$$\lim_{|\Delta x|\to 0} \frac{f(x+\Delta x)-f(x)-\delta f(x;\Delta x)}{|\Delta x|}=0.$$

If the limit SF(x:) is continuous and linear in Δx, then the function is Fréchet differential at x and the associated linear operator SF(x:) is the Fréchet derivative f at x, often written ?$f(x)$.

FRÉCHET SPACE (F-SPACE): Named after Maurice Fréchet, French mathematician, it is a complete metrizable locally convex space sometimes also with the restriction that the space be locally convex. The topology of Fréchet space is defined by a countable family of seminorms. For example, the space of smooth functions on [0,1] is Fréchet space.

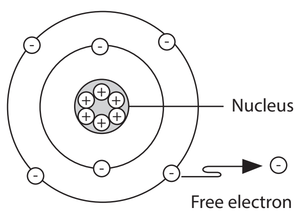
FREDHOLM ALTERNATIVE: In mathematics, the alternative theorem, holding for a continuous linear operator, A, with closed range or its representation as a matrix, that either A is surjective or else the adjoint A^* has a non-trivial kernel. Hence, either the inhomogeneous equation $Ax = b$ is always solvable or that the homogeneous equation $A^*y = 0$ has a non-zero solution.

FREE: 1. In mathematics, of an algebraic structure, consisting of all formal objects satisfying the requisite algebraic conditions without additional relations being imposed; for example, the symmetric group on a set of three elements is not free as a group. 2. Of a tree, not having a root or origin. 3. Of a variable, not bound, so that it is at best a place-holder in an open sentence; in some formulations of logic these are interpreted as universally quantified. 4. Of a vector, determined only up to translation; unattached. 5. Of an element of a group, not of finite period.

FREE ACIDITY: 1. The portion of the total acidity that exists in the form of acid, both ionised and un-ionised. 2. The titratable acidity (a measure of the amount of acid present in a solution) that is present in the aqueous phase of a soil. It may be expressed in milliequivalents per unit mass of soil.

FREE CONVECTION (NATURAL CONVECTION): A convection or type of heat transport in which fluid motion results entirely from the presence of a hot body in the fluid, causing temperature and hence density gradients to develop, so that the fluid moves under the influence of gravity.

FREE ELECTRON: An electron that is not attached to an ion, atom or molecule and is free to move under the influence of an applied electric or magnetic field.



Free electrons of an atom/ion/ molecule

FREE ELECTRON APPROXIMATION: The approximation resulting from the assumption that electrons in metals can be analysed using the kinetic theory of gases, without taking the periodic potential into account.

FREE ELECTRON LASER: A type of laser, invented by John Madey in 1976 at Stanford University that works by the stimulated emission of electromagnetic radiation produced by the motion of electrons moving rapidly in external magnetic fields. The free-electron laser has the widest frequency range of any laser type and can be widely tunable, ranging in wavelength from microwaves, through terahertz radiation and infrared, to the visible spectrum, to ultraviolet and to X-rays.

FREE ELECTRON THEORY (THE DRUDE–SOMMERFELD MODEL): The theory that treats electrons as freely moving within the body of a metal crystal or fluid. It was developed principally by Arnold Sommerfeld who combined the classical Drude model with quantum mechanical Fermi-Dirac statistics. The free electron Empty Lattice Approximation forms the basis of the band structure model known as nearly-free electron model. Given its simplicity, it is surprisingly successful in explaining many experimental phenomena, especially the Wiedemann-Franz law which relates electrical conductivity and thermal conductivity; the temperature dependence of the heat capacity; the shape of the electronic density of states; the range of binding energy values; electrical conductivities; and thermal electron emission and field electron emission from bulk metals.

FREE ENERGY (GIBBS FREE ENERGY): A measure of a system's thermodynamic state function in order to determine the available energy to do work.

FREE FALL: Any motion, determined solely by gravitational forces and which is unimpeded by a medium that will provide a frictional retarding force. A body is said to be in free fall if the only outside force acting on it is gravity. Examples are the Moon's motion around the Earth and an object dropping to the Earth's surface. A person in free fall experiences weightlessness.

FREE GROUP: 1. In mathematics, it is a group with a generating set such that the only products of generators and their inverses that equal identities are of the form aa^{-1} or $a^{-1}a$. 2. The set, F_x, of equivalence classes, $[u]$, of words in a non-empty set, X, where a pair of elements are equivalent if and only if there is a finite sequence of words starting with one and finishing with the other, such that each is obtained from the preceding by elementary reduction. Multiplication is defined on F_x by $[u]\,[v] = [uv]$.

FREE METAL (NATIVE METAL): Any of the pure or alloy metallic elements found in nature. They include bismuth, cadmium, chromium, indium, iron, nickel, tellurium, tin, titanium and zinc, as well as two groups of metals: the gold group and the platinum group. The gold group consists of gold, copper, lead, mercury and silver. The platinum group consists of platinum, iridium, osmium, palladium, rhodium and ruthenium. Only gold, silver, copper and the platinum metals occur in nature in larger amounts. Alloys found in native state are brass, bronze and silver-mercury and gold-mercury amalgam.

F

FREE MODULE: In mathematics, it is a module that has as a basis a subset $\{a_1, a_2, ..., a_n\}$ in terms of which every non-zero element can be written uniquely in the form $\sum_i u_i a_i$, where the u_i are elements of the ring over which the module is a left module. Every vector space is a free module and an Abelian group is free if and only if it has no element of finite period. Submodules of a free module are not in general free unless the underlying ring is a principal ideal domain.

FREE OXYGEN: Oxygen in its molecular forms, O_2 (normal diatomic oxygen) or O_3 (ozone), uncombined with other elements. Free oxygen is a requirement of all aerobic organisms and its copious presence in a planetary atmosphere is diagnostic of the presence of life.

FREE RADICAL (RADICAL): A highly reactive molecule that has one or more unpaired electrons that do not interact with each other, which makes them easy to form a chemical bond. They are usually short lived and highly reactive. Processes involving free radicals are used in the production of rubber and plastics. Radicals are common in chain reactions such as the ones that produce fire. They also occur in body chemistry, in processes such as the destruction of invading organisms by white blood cells. Free radicals might play a role in various diseases such as arthritis, heart disease and Alzheimer's disease. When natural enzyme controls fail, free radicals in the body attack lipids, proteins and nucleic acids.

FREE ROTATION (HINDERED ROTATION, RESTRICTED ROTATION): In a stereochemical context the rotation about a bond is called 'free' when the rotational barrier is so low that different conformations are not perceptible as different chemical species on the time scale of the experiment. The inhibition of rotation of groups about a bond due to the presence of a sufficiently large rotational barrier to make the phenomenon observable on the time scale of the experiment is termed hindered rotation or restricted rotation.

FREE SOFTWARE: Software that respects users' freedom, thus giving users the freedom to run, copy, distribute, study, change and improve the software.

FREE SOFTWARE FOUNDATION (FSF): A non-profit organisation founded in 1983 by Richard M. Stallman (1953), US computer scientist that is concerned with the creation and dissemination of free software.

FREE SPACE: A region in which there is no matter, electromagnetic or gravitational fields. It has a temperature of absolute zero, unit refractive index and the speed of light is at its maximum value.

FREE SPACE PROPAGATION MODEL: In teleommunications, it refers to the loss in signal as it travels through space. It is defined as $P_L = \left(\frac{4\pi d}{\lambda}\right)^2$, where P_L = path loss, λ = signal wavelength and d = distance from transmitter.

FREE-BOUND TRANSITION: The capture of a free electron by an ion, in which a photon is emitted. This process is known as recombination. The energy of the photon given off is the difference between original energy of the electron and its new energy in whatever level of whatever atom it ends up in. Since this difference can have any value, the result of many free-bound transitions is a continuous spectrum.

FREE-CENTRAL PLACENTATION: A type of placenta arrangement in which ovules are attached to the ovary wall on a central column by partition.

FREE-LIVING: It describes an organism living inpendently and not parasitic or symbiotic with any other organism.

FREE-PISTON ENGINE: A type of diesel engine in which the power output is extracted from a turbine driven by the exhaust gases and not via any direct rod and crank drive. The combustion chamber is formed between two opposed pistons in a closed metal tube, the piston oscillating in and out in phase with each other, regularly exposing the exhaust and air-inlet ports. Fuel is injected at the midpoint. The engine, has a good power-to-weight ratio and is used in some ships. It has the advantage that the gasifier and turbine units need not be sited adjacently.

FREEHOLD: The absolute right to hold land or property for an unlimited time without paying rent.

FREELY DRAINING: An adjective referring to a chain macromolecule with segments that produce such small frictional effects when moving in a medium that the hydrodynamic field in the vicinity of a given segment is not affected by the presence of other segments. Thus, the solvent can flow virtually undisturbed through the domain occupied by a freely draining macromolecule.

FREELY JOINTED CHAIN: A hypothetical linear chain molecule consisting of infinitely thin rectilinear segments uniform in length. Each segment can take all orientations in space with equal probability, independently of its neighbours. **Random-walk chain** describes models in which the segments are not all uniform in length, the name has been used.

FREELY ROTATING CHAIN: A hypothetical linear chain molecule, free from short-range and long-range interactions, consisting of infinitely thin rectilinear segments (bonds) of fixed length, jointed at fixed bond angles. The torsion angles of the bonds can assume all values with equal probability.

FREEMARTIN: A sterile female calf produced when a male and female embryo share one uterus. The production of testosterone from the testes of the male embryo causes the reproductive system of the female embryo to take on masculine features.

FREEWARE: A type of software that can be downloaded and used without payment for non-commercial purposes. The author retains the copyright and it may have some restricted usages. Minor program updates and small games are commonly distributed as freeware.

FREEZE DRYING (LYOPHILISATION): A process used in dehydrating food, blood plasma and other heat-sensitive substances by deep freezing and the trapped ice is removed by placing it in a vacuum to remove moisture before returning it to room temperature, leaving an undamaged dry product. The low processing temperature and absence of liquid water help to retain colour, flavour and texture.

FREEZE FRACTURE: A method of preparing material for electron microscopy that allows the visualisation of the interior of plasma membranes and organelles. Cells are frozen at -196 °C (77 K) and cracked so that the plane of fracture runs through the middle of lipid bilayers, separating the two halves.

FREEZE-THAW (FROST-SHATTERING): A form of physical weathering of rock caused by the expansion of water by 9% as it freezes. The water, which has penetrated the joints and cracks, freezes. Freeze-thaw is most active where there is a maximum number of temperature oscillations around 0 °C (273.15 K). It is, therefore, common in mountainous and glacial environments.

FREEZER (DEEP FREEZER): A mechanically refrigerated room or cabinet for the storage of perishable items, especially food. The compartment, cabinet or room is thermally insulated. It is usually maintained at a temperature of 0 °C or below. The basic components of a freezer include a compressor, a condenser, an evaporator, a capillary tube and a thermostat, which determines the operating temperature. The compressor, which is controlled by a thermostat compresses a refrigerant gas to a hot, high pressure gas. The hot, high pressure gas flows to condenser coils, which are bent tubes that run through thin pieces of metal known as fins. During this flow, the gas changes into a liquid and the tubes radiate heat. The fins conduct this heat away from the coils. The capillary tube, which connects the condenser to the evaporator coils regulates the pressure of the refrigerant as it enters the coils. A cold, low pressure gas is then produced, when the liquid refrigerant expands, boils and evaporates into the evaporator coils. The gas absorbs heat while cooling down air that flows past the coils, which cools down the compartment.

FREEZING: The conversion of a liquid to a solid by lowering the temperature and/or the application of pressure. An example is when water becomes ice.

FREEZING MIXTURE: A mixture of substances that produces low temperatures below the freezing point of water. For example, a mixture of ice and sodium chloride gives a temperature of -20 °C (253 K).

FREEZING OUT: A term used in combustion for the analysis of incinerator flue gas components (largely the organic fraction) in which a series of traps at progressively lower temperatures is employed. In more general use, the term implies the removal of a condensable gas or liquid by condensation in a trap at low temperatures.

FREEZING POINT: The temperature at which a pure liquid changes into a solid. For example, the freezing point of pure water at standard atmospheric pressure is 0 °C (273 K). When the pressure surrounding the liquid is increased, the freezing point is raised. The addition of some solids can lower the freezing point of a liquid, a principle used when salt is applied to melt ice on frozen surfaces. For pure substances, the freezing point is the same as the melting point. In mixtures and certain organic compounds, the early solid formation changes the composition of the remaining liquid, usually steadily lowering its freezing point, a principle that is applied in mixture separation.

FREEZING POINT DEPRESSION: Lowering of a solution's freezing point below that of the pure solvent as a result of the presence of impurities in them. The depression is proportional to the active mass of the solute in a given amount of solvent.

FREGE, FRIEDRICH LUDWIG GOTTLOB (1848- 1925): German mathematician and philosopher, who is considered to be one of the founders of modern logic and developed a set of logic symbols. He made the distinction between sense and reference and logic of quantification, in which the quantifiers are treated as properties of properties. He wrote on the foundations of mathematics, setting out axioms for set theory and defined the cardinal numbers, using set theory.

FRENCH CHALK: A soft white variety of steatite or soapstone used to make tailoring marks on cloths and to remove grease stains from clothes.

FRENCH CURVE: A stencil or template used for drawing smooth curves of various radii through predetermined points. The curvatures vary throughout the instrument.

FRENET FORMULAE (SERRET-FRENET FORMULAE): The fundamental formulae that describe the kinematic properties of a particle moving along a continuous, differentiable curve in three-dimensional Euclidean space or the geometric properties of the curve itself irrespective of any motion. In other words, the formulae are for a space curve that recaptures the unit vector tangent (to the curve, pointing in the direction of motion) denoted T, the normal unit vector (the derivative of T with respect to the arclength parameter of the curve, divided by its length), denoted N and the binormal unit vector B (the cross product of T and N), from the curvature κ and torsion τ of the curve. The formulae are:

(i) $N' = -\kappa T + \tau B$, (ii) $B' = -\tau N$ and (iii) $T' = \kappa N$.

FRENKEL DEFECT (FRENKEL PAIR; FRENKEL DISORDER): A type of point defect in a crystal lattice that occurs when an ion occupies a vacant interstitial site, leaving its proper site empty. It was named after the Soviet physicist Yakov Frenkel, who discovered it in 1926.

FRENKEL-KONTOROWA MODEL: A one dimensional model of atoms, such as xenon, adsorbed on a periodic substrate such as graphite, which can be used to investigate the nature of the lattice formed by the adsorbed gas. It can also be used to investigate the phase transition between a commensurate lattice and an incommensurate lattice.

FREON: The trade name for certain fluorocarbons and chloroflourocarbons (CFCs) derived from methane and ethane and used as refrigerants, as blowing agents for plastic foams, as fire-extinguishing agents and in aerosol sprays. They are nonflammable, nontoxic and noncorrosive; they have low boiling points (-29.8 °C, 243.35 K), which make them useful as refrigerants. They are now known to destroy the ozone layer, so most of their uses have been banned. International agreements signed by most of the industrialised countries have called for the phasing out of CFCs.

FREQUENCY: 1. A property of waves that describes how many wave patterns or cycles that pass a fixed reference point in one second. It describes the number of cycles of a periodic motion such as sound pressure, electrical intensity, radio or other quantities per unit time. It is named in honour of German physicist, Heinrich Rudolf Hertz (1857 - 1894). The unit of frequency (f) is the hertz (Hz), which is equal to 1 cycle per second (1/s). Its symbol is Hz. It is expressed as $f = \omega/2\pi$, where f is the frequency of the vibrating article and ω is the angular speed.

FREQUENCY BANDS: The division of the radio spectrum into bands. It ranges from extremely low frequency (longest wavelengths) with a wavelength of 10,000 – 100 km with a frequency of 30 Hz – 3 kHz to extremely high frequency (shortest wavelengths) with a wavelength of 1 cm – 1 mm with a frequency of 30 – 300 GHz. Only a small part of this spectrum, from about 1 GHz to a few tens of GHz, is used for practical radio communications. International agreements control the usage and allocation of frequencies within this important region.

FREQUENCY CHART: An illustration of the frequency of a given event. The frequency is the number of times the event occurs.

FREQUENCY DENSITY: Frequency density of a frequency distribution is given by $\frac{Class\ frequency}{Class\ width}$. The frequency density is used when drawing histograms, finding the modal class, etc. of a frequency distribution. It allows comparisons between classes of unequal width.

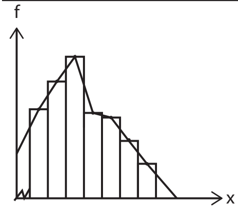
FREQUENCY DISTRIBUTION: In statistics, it is a way of organising raw data into an ordered form usually by a list showing the number of times each event occurs. It allows for easy comparison of the results in each category or classification of data and is particularly useful in summarising large data sets and assigning probabilities. If the range of observed measurement values is subdivided into a regular sequence of smaller intervals, this distribution is a plot of the frequency of occurrence of values falling into each interval.

FREQUENCY DIVISION MULTIPLEXING (FDM): A technique used to derive a number of separate data channels from a single transmission medium. Each data channel is assigned a portion of the total available bandwidth. Hence, different ranges of the bandwidth are used for each signal.

FREQUENCY GRAPH: A graph showing how often each value occurs in a data set; it shows how data is spread out.

FREQUENCY MODULATION (FM): A method by which a radio wave is altered by varying the frequency of a carrier wave in accordance with the signal being broadcast, used for the transmission of broadcasting signal. Compared to amplitude modulation (AM), FM offers important advantages such as extra bandwidth occupancy and negative feedback, to minimise noise problems and receiver distortion.

FREQUENCY MODULATION SYNTHESIZER (FM SYNTHESIZER): In audio and music, a method for generating sounds based on techniques used to transmit FM radio signals.



A frequency polygon on the Cartesian plane

FREQUENCY POLYGON: A graph of a frequency distribution with values of the variable on the x-axis and the number of observations on the y-axis that is obtained by joining the midpoints of the top of rectangles to form a histogram. It is created in exactly the same way as the frequency curve but the points are joined by straight lines.

FREQUENCY TABLE: A table that lists the frequency counts for a categorical variable or relative frequency and uses tally marks to record and show the number of times they occur.

FREQUENCY-DEPENDENT SELECTION: A decline in the reproductive success of a morph resulting from the morph's phenotype becoming too common in a population. It is a cause of balanced polymorphism in populations and occurs when the fitness of a genotype depends on whether it is rare or common. An individual's fitness is affected by allele or genotype frequencies of other individuals in the same population. Pathogen-host coevolution, sex ratio, self-sterility alleles and mimicry may often result from frequency-dependent selection. **Negative frequency-dependent selection** favours rare genotypes and promotes genetic variation in populations. **Positive frequency-dependent selection** favours common alleles, accelerates the loss of rare alleles and decreases genetic variation.

FRESHWATER: A water body that does not contain salt or water that is not salty, stale or bitter. Freshwater is stored in glaciers, lakes, streams and rivers. It is also stored as groundwater in the soil and rocks. There are about 36 million cubic kilometres (cubic km) of fresh water on the Earth. The atmosphere holds about 12,000 cubic kilometres (cubic km) of water at any time, while all the world's rivers and freshwater lakes hold about 120,000 cubic kilometres (cubic km). The world's two main reservoirs of fresh water are the great polar ice caps, which contain about 28 million cubic kilometres (cubic km) and the ground, which contains about 8 million cubic kilometres.

FRESHWATER BIOME: The aquatic biome consisting of water containing fewer salts than the waters in the marine biome. It is divided into two zones: running waters consisting of rivers and streams; and standing waters (lakes, ponds).

FRESHWATER LENS: A convex-shaped fresh groundwater body that lies below small coral or limestone islands and atolls; the lens lies above saline water.

FRESNEL: A unit of frequency, which is equal to 10^{12} hertz (10^{12} cycles per second).

FRESNEL BRIGHT SPOT (ARAGO'S SPOT; POISSON SPOT): A bright spot at the centre of the shadow of a circular opaque obstacle occurring as a consequence of the wave theory of light.

FRESNEL DIFFRACTION: A form of diffraction in which the light source or the receiving screen or both, are at finite distances from the diffracting object so that the wave fronts are not plane as in Fraunhofer diffraction.

FRESNEL INTEGRALS: The two definite integrals, denoted $S(x)$ and $C(x)$ used in optical theory and given as: $S(x) = \int_0^x \sin(t^2)\,dt$, and $C(x) = \int_0^x \sin(t^2)\,dt$.

FRESNEL LENS: A type of compound lens designed by Augustin Jean Fresnel (1788-1827), a French physicist, with one face cut into a series of steps, which enables a relatively light and robust lens of short focal length. It has poor optical quality and used in projectors as condenser lenses, as searchlights, spotlights, car headlights and in lighthouses.

F

FRICTION: A force that resists sliding or rolling of one solid object over another or a force that opposes the relative motion of two bodies in contact or sliding across each other. Frictional force depends the following: (i) the nature of the surface and contact, μ; and (ii) the normal reaction, R. Frictional force acts in a direction parallel to the area of contact and opposes the motion. However, frictional force is independent of the surface area.

FRICTION FACTOR (COEFFICIENT OF FRICTION; FRICTION COEFFICIENT): 1. Quotient of frictional force and normal force, for a sliding body. Its symbol is μ. This is given by $\mu = \frac{F}{R}$, where F is the frictional force and R is the reaction. 2. A tensor correlating the frictional force, opposing the motion of a particle in a viscous fluid and the velocity of this particle relative to the fluid. In the case of an isolated spherical particle in a viscous isotropic fluid, it is a constant.

FRIDGE (REFRIGERATOR): An appliance or cabinet fitted with electrical devices such as a motor/compressor and thermostat for controlling temperature which circulates specially compressed gas in a specially constructed pipe. It is used especially in the home to store food and drink at a low temperature without being frozen. The basic components of a freezer include a compressor, a condenser, an evaporator, a capillary tube and a thermostat, which determines the operating temperature. The compressor, which is controlled by a thermostat compresses a refrigerant gas to a hot, high pressure gas. The hot, high pressure gas flows to condenser coils, which are bent tubes that run through thin pieces of metal known as fins. During this flow, the gas changes into a liquid and the tubes radiate heat. The fins conduct this heat away from the coils. The capillary tube, which connects the condenser to the evaporator coils regulates the pressure of the refrigerant as it enters the coils. A cold, low pressure gas is then produced, when the liquid refrigerant expands, boils and evaporates into the evaporator coils. The gas absorbs heat while cooling down air that flows past the coils, which cools down the compartment. The optimum temperature range for perishable food storage in a refrigerator is 3 to 5 °C.

FRIEDEL-CRAFTS REACTION: A set of chemical reactions developed by Charles Friedel and James Crafts in 1877, in which an alkyl or acyl group is introduced into a benzene ring, in the presence of catalysts such as aluminium(III) chloride, boron triflouride or hydrogen fluoride. If an alkyl group is attached to benzene, the reaction is called **Friedel-Craft alkylation**. Attaching an acyl group to benzene is called **Friedel-Craft acylation**, also known as **Friedel-Crafts alkanoylation**.

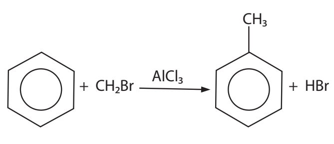


Fig. I: Friedel-Craft Alkylation of Benzene

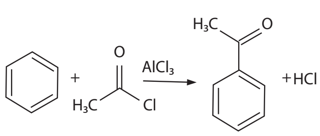


Fig. II
Friedel-Craft Acylation (Friedel-Crafts Alkanoylation) of Benzene

FRIEDMANN-LAMAITRE-ROBERTSON-WALKER MODEL: A model of the universe corresponding to solutions of the Einstein field equations of general relativity theory. It describes a homogeneous, isotropic expanding or contracting universe that may be simply connected or multiple connected.

FRIENDLY URL: A Web address that is easy to read and includes words that describe the content of the webpage. User-Friendly URLs are short and easy to remember and pages can be revisited by simply typing in the URL address bar.

FRIESIAN: A large black and white cow originating from North Holland and Friesland belonging to a breed known for its abundant milk production. The milk has relatively low butterfat content. As a dairy breed, they rank high for beef and veal production.

FRINGED: Having the edge or margin cut; having bristles along the margin.

FRINGED-MICELLE MODEL: A model of crystallinity in which the crystallised segments of a macromolecule belong predominantly to different crystals.

FRITZ JOHN CONDITIONS/THEOREM: A theorem that asserts the existence of Lagrange multipliers for constrained optimisation problems. For a constrained minimisation, one adds an additional non-negative multiplier, λ_0, for the objective function and asserts that not all the multipliers are simultaneously zero. A constraint qualification can be viewed as ensuring that λ_0 is nonzero and so can be scaled to unity.

FROBENIUS, FERDINAND GEORG (1849-1917): German group theorist and analyst, who developed the theory of abstract groups, made contributions to the theory of linear differential equations and analytic functions in series.

FROBENIUS GROUP: Named after Ferdinand Frobenius, it is a group that has a proper subgroup, H, such that, for every x in the relative complement G \ H, the intersection of H with x^{-1} is the identity element.

FROBENIUS METHOD: The method of solving ordinary differential equations near a regular singular point, a, by positing a solution of the form $(x - a)^{\alpha}$ P$(x - a)$ for some power series P and index α that are then found by iteration by substituting the potential solution into the equation.

FROBENIUS NORM (FROBENIUS TRACE): The norm on matrices that arises by treating a matrix as a vector and using the Euclidean norm of that vector: $\|A\|^2 = \sum_{i=1}^{n}\sum_{j=1}^{m}\left|a_{ij}\right|^2$. This quantity is also the trace of AA^*.

FROG: A vertebrate belonging to the order Anura (tail-less amphibian) of the class Amphibia, especially of the family Ranidae. It is usually brownish-green, hairless, tailless animal that croaks. It has long hind legs with webbed toes for swimming and jumping and protruding eyes. It lays its eggs in puddles, ponds or lakes. The larvae, called tadpoles, have gills and develop in water. Adult frogs follow a carnivorous diet, mostly of arthropods, annelids and gastropods. Frogs are most noticeable by their call, which can be widely heard during the night or day, mainly in their mating season.

FROM ABOVE (FROM THE RIGHT): A real function f is bounded from above if there is a number k such that for all x from the domain $D(f)$ one has $f(x) \leq k$. The limit of a function from above is the one-sided limit where x is restricted to values greater than a, that is, the right-hand limit.

FROM BELOW (FROM THE LEFT): A real function f is bounded from below if there is a number k such that for all x from the domain $D(f)$ one has $f(x) \geq k$.

FROND (MEGAPHYLL): A type of foliage leaf in ferns and seed plants that have branched or parallel vascular bundles running through the lamina. The term is also applied to the leaves of palms and less commonly to the plant bodies of certain seaweeds, liverworts and lichens.

FRONT: In meteorology, it is the boundary between two air masses of different temperatures and humidities. They are regions of uncertain weather such as cloud and rain, variable humidity and wind and low air pressure. In a **warm front** warm air advances, displacing cooler air. A **cold front** is the leading portion of a cold atmospheric air mass moving against and eventually replacing a warm air mass. An **occluded front** occurs often toward the end of a system of storms, when air mass is forced upwards when cold front overtakes warm front.

FRONT END PROCESSOR (FEP; COMMUNICATIONS PROCESSOR): A small-sized computer used to coordinate and control the communication between a large mainframe computer and its input and output device. Data is transferred between the host computer and the front end processor using a high-speed parallel interface. The front end processor communicates with peripheral devices using slower serial interfaces, usually also through communication networks. Examples of FEP are the IBM 3705 Communications Controller and the Burroughs Data Communications Processor.

FRONTAL BONE: A cranial bone consisting of a vertical portion, called the squama frontalis, corresponding to the forehead and a horizontal portion called the pars orbitalis, comprising the forehead and roofs over most of the orbits and nasal cavity and that at birth consists of two halves separated by a suture. The frontal bone contains several air sinuses, known as parasinuses.

FRONTAL CHROMATOGRAPHY: A procedure in which the sample (liquid or gas) is fed continuously into the chromatographic bed. In frontal chromatography no additional mobile phase is used.

FRONTAL LOBE: The lobe of the cerebral cortex of humans and other mammals that is responsible for motor activity, speech and thought processes.

FRONTIER (BOUNDRY): 1. The set of points (frontier- points) that are members both of the closure of a given set and of the closure of its complement. 2. The set of points in the closure but not in the interior of the given set, usually written FrA, for example, Fr⍰(0, 1)⍰ = {0, 1}. The frontier of the rationals is the set of all real numbers.

FRONTIER ORBITAL: Any one of two orbitals in a molecule determining predominantly chemical and spectroscopic characteristics of molecules. They are highest occupied molecular orbital (HOMO) and the lowest unoccupied molecular orbital (LUMO).

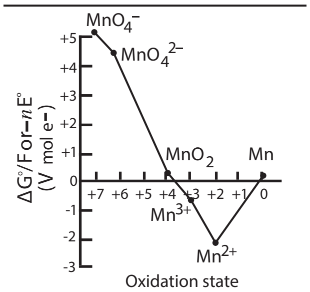
FRONTIER ORBITAL THEORY: A theory put forward to explain reactions involving π-bonds such as Diels-Alder reactions and pericyclic reactions emphasising the symmetry as well as the energies of the frontier

FRONTING: Asymmetry of a peak such that, relative to the baseline, the front is less steep than the rear. In paper chromatography and thin-layer chromatography, it refers to the distortion of a spot showing a diffuse region in front of the spot in the direction of flow.

FRONTSIDE BUS (FSB): The main route for data transfer in a PC. It connects all of a computer's major components such as memory, AGP socket and chipset.

FROST: A condition of the weather that occurs when the air temperature is below freezing (below 0 °C, 273 K) and also below the frost point, resulting in the deposit of ice crystals such as bits of ice that form on the outside of something left in the freezer for a very long time. It also refers to crystals of frozen water deposited on a cold surface.

FROST DIAGRAM: A diagrammatical representation of the variation of standard electrode potentials with variations in oxidation states for an element which exhibits variable oxidation states.



The Frost diagram (for manganese)

FROST HOLLOW: A depression or steep-sided valley in which cold night air collects resulting in frequent and severe frosts.

FROST POINT: Dew point temperature, below 0°C. At this temperature, moisture in the air will condense as a layer of frost on exposed surfaces that are also at a temperature below the frost point.

FROST POINT HYGROMETER: An instrument for measuring the frost point of the atmosphere, in which the sample is passed over a cooled surface. The temperature at which frost forms on it is a function of the water content of the gas passing over the surface.

FROST-SHATTERING (FREEZE-THAW): A form of physical weathering of rock caused by the expansion of water by 9% as it freezes. The water, which has penetrated the joints and cracks, freezes. Freeze-thaw is most active, where there is a maximum number of temperature oscillations around 0 °C (273.15 K). It is therefore, common in mountainous and glacial environments.

FROST WEDGING: A form of physical weathering that occurs when water seeps into the cracks of a rock and the water repeatedly expands and contracts in the cracks of the rock, eventually breaking the rock. This occurs in areas with extremely cold weather.

FRUCTOSE (FRUIT SUGAR): A monosaccharide with the chemical formula C6H12O6 that occurs with glucose in sweet fruits and fruit juices. It is 173% as sweet as sucrose.

FRUCTOSE 1,6-BIPHOSPHATE (HARDEN-YOUNG ESTER): An intermediate product, with the chemical formula $C_6H_{14}O_{12}P_2$ which lies within the glycolysismetabolic pathway formed in the initial stage of glycolysis by the phosphorylation of glucose using adenosine triphosphate (ATP), which is further broken down into two compounds: glyceraldehyde 3-phosphate and dihydroxyacetone phosphate. It is an allosteric activator of pyruvate kinase. Most of the glucose and fructose entering a cell are converted to fructose 1,6-biphosphate at some point.

FRUGIVORE: An animal that mainly eats fruit. Many bats and birds are frugivores. Frugivores can either benefit fruit-producing plants by dispersing seeds or they can negatively affect plants by digesting seeds along with the fruits.

FRUIT: The fleshy or dry ripened ovary part of a flowering plant that contains and protects its seeds or the ripened ovary of flower enclosing seeds that develops from one or more carpels or the structure formed from the ovary of a flower usually after the ovules have been fertilised. Fruits are important sources of dietary fibre and vitamins especially vitamin C. There are two broad categories of fruit: fleshy and dry. Fleshy fruits include berries, such as tomatoes oranges and cherries; and dry fruits include the legumes, cereal grains, capsules and nuts.

FRUIT AND SEED DISPERSAL: The methods by which most flowering plants spread seeds far away from the parent plant. The seedlings become more dispersed and do not compete for resources ensuring a wider area of colonisation.

FRUIT DROP: The premature shedding of fruit from a tree before fully ripe. It is a normal process and in many fruits certain peak periods of fruit drop can be identified. For example, apple fruits are lost immediately following pollination (post-blossom drop), when the embryos are developing rapidly (June drop) and during ripening (preharvest drop). Like leaf fall, fruit drop is associated with low auxin levels and auxin sprays have been used to prevent excessive fruit drop.

FRUITING SEASON: The time of year when a particular tree bears fruit.

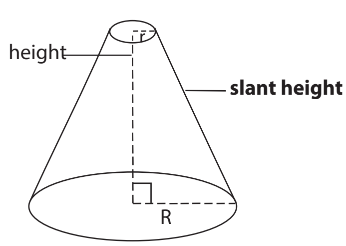
FRUITWALL (PERICARP): The outer and often edible layer of a fruit, which is the tissue that develops from the ovary wall of the flower and surrounds the seeds. The fruitwall is made up of three layers: the **exocarp**, which is the outer layer or peel; the **mesocarp**, which is the middle layer or pith; and the **endocarp**, which is the inner layer surrounding the seeds.

FRUITWASTE: Residue left after juice has been extracted from fruit, used as an animal feed.

FRUSTATION (GEOMETRICAL FRUSTATION): A situation in which there are competing interactions in a system. It is a phenomenon in which the geometrical properties of the crystal lattice or the presence of conflicting atomic forces forbid simultaneous minimisation of the interaction energies acting at a given site. This may lead to highly degenerate ground states with a nonzero entropy at zero temperature. In other words, it is a substance that can never be completely frozen, because the structure it forms does not have a single minimal-energy state, so motion on a molecular scale continues even at absolute zero and even without input of energy.

FRUSTULE: The hard, siliceous bivalve shell of a diatom (microscopic alga). Frustules are intricately patterned on the surface with spots, ridges and furrows, with each pattern being characteristic of a particular species.

FRUSTUM: A solid figure produced when two parallel planes cut a larger solid or when one plane parallel to the base cuts it or a slice taken out of a solid figure by a pair of parallel planes. A conical frustum, for example, resembles a cone with the top cut off.



Side view of a frustum

FRUTESCENT: Resembling a shrub; shrubby

FRUTICOSE: A term used to describe organisms that have shrub-like, bushy, branching stems rather than a single trunk, especially the erect branching lichen thallus.

FTP (FILE TRANSFER PROTOCOL): In computing, it is a standard method of transferring files through the Internet from one computer to another.

FUBINI'S THEOREM: Named after the Italian mathematician, Guido Fubini (1879-1943), it is the result that gives conditions under which it is possible for the evaluation of multiple integrals as iterated integrals and changing the order of integration of an iterated integral.

FUC- (FUCO-): Prefixes used to denote seaweeds. For example, fucin and fucoidin are colloids found only in the walls of brown algae (Phaeophyta). Fucoxanthin is a carotenoid pigment found in the Bacillariophyta, Phaeophyta and Chrysophyta. Fucosterol is the main type of sterol found in seaweeds.

FUCALES (WRACKS): An order in the brown algae, whose members are fucoids. It is the only order of the Cyclosporae, a subclass of the Phaeophyta. It consists solely of marine forms with the commoner species such as Fucus, Pelvetia and Ascophyllum making up a large proportion of the vegetation of the coastal littoral zone. The thallus is differentiated into holdfast and lamina and often contains air bladders. Reproduction is oogamous with the gametangia developing in specialised invaginations, the conceptacles.

FUCOXANTHIN: The major carotenoid pigment present with chlorophyll in brown algae and most other heterokonts, giving them a brown or olive-green colour, with the molecular formula $C_{42}H_{58}O_6$. Fucoxanthin absorbs light primarily in the blue-green to yellow-green part of the visible spectrum.

FUEL: Any material that can be used to generate heat and energy or any substance that burns and can be used as a source of heat energy to produce mechanical work in a controlled manner. Examples include hydrocarbons, charcoal and wood. The processes used to convert fuel into energy include chemical reactions such as combustion and nuclear reactions such as nuclear fission or nuclear fusion. Fuels are also used in the cells of organisms in a process known as **metabolism**.

FUEL CELL: An electrochemical cell for converting chemical energy from a fuel, such as liquid hydrogen and an oxidant, such as liquid oxygen directly into electrical energy.

FUEL CYCLE: The nuclear fuel cycle is the sequence of steps, such as utilisation, reprocessing and refabrication, through which nuclear fuel may pass.

FUEL ELEMENT: A rod, tube, plate or other geometrical form into which nuclear fuel is fabricated for use in a reactor.

FUEL INJECTION: Injecting fuel directly into the cylinders of an internal combustion engine, instead of by way of a carburettor. On diesel engines, without spark plugs, the heat created by compressing air in the cylinders ignites the fuel, which has been pumped in as a spray. In engines with spark ignition, fuel-injection pumps are often used instead of conventional carburettors. Fuel injection distributes the fuel more evenly to the cylinders than does a carburettor; more power can be developed and undesirable emissions are reduced. In engines with continuous combustion, such as gas turbines and liquid-fuelled rockets, which have no pistons to create suction, fuel-injection systems are necessary.

FUEL OIL: A viscous liquid containing hydrocarbons with 30 to 40 carbon atoms per molecule, used as a fuel in ships and power stations. Fuel oil is obtained from petroleum distillation, either as a distillate or a residue.

FUEL REPROCESSING: Nuclear fuel reprocessing refers to the processing of nuclear fuel, after its use in a reactor, to remove fission products and recover fissile and fertile material.

FUEL ROD: A long, slender tube, often made of a zirconium alloy and containing nuclear fuel (fissionable material), such as uranium-235, which is inserted into certain types of nuclear reactor. Fuel rods are assembled into bundles of about 200, called **fuel elements** or **fuel assemblies**, which are loaded individually into the reactor core. The rods must be surrounded by a coolant, otherwise temperatures can rise to levels hot enough to melt metallic components over a prolonged period. Some other examples of nuclear fuels (fissionable materials) used in fuel rods are uranium-233, plutonium-239 and plutonium-241.

FUEL SWITCHING: The substitution of one energy source for another to reduce carbon dioxide emissions by changing to lower carbon-content fuels, such as from coal to natural gas.

FUGACIOUS: Withering; fading off quickly or lasting a very short time.

FUGACITY: A thermodynamic function used in place of partial pressure in reactions involving real gases and mixtures. The fugacity of a gas is equal to the pressure, if the gas is ideal and for liquids and solids, the fugacity is the fugacity of vapour with which it is in equilibrium. **Fugacity coefficient** is the ratio of the fugacity of a gas to its pressure.

FUGITIVE SPECIES: A species that is able to coexist with a competitively superior species due to its better dispersal capabilities to seek new habitats. The fugitive species is quicker to exploit any vacant patches in the environment that becomes available, have high level of reproduction and capable of withstanding difficult conditions.

FULCRUM (PIVOT): The point on which a lever rests and where it turns.

FULL ADDER: A combinational circuit that forms the arithmetic sum of three inputs. It consists of three inputs and two outputs. Two of the input variables, denoted by x and y, represent the two significant bits to be added. The third input, z, represents the carry from the previous lower significant position.

FULL ENERGY PEAK (TOTAL ABSORPTION PEAK): In a gamma-ray spectrum, it is the part of the spectral response curve corresponding to the total absorption in a detecting material of the energy of the detected radiation.

FULL FORMAT: This is a method of removing files from a disk partition. In this process, files are erased and the hard disk is scanned for bad sectors. It takes a longer time to complete. The scan for bad sectors is responsibe for the majority of the time that it takes to format the volume.

FULL LINEAR GROUP (GENERAL LINEAR GROUP): The group, often written $\mathrm{GL}(V)$, of all linear transformations of a finite-dimensional vector space, V, that are invertible. Where V is over a field F, it is defined as the group of all non-singular $n \times n$ matrices over F, written $\mathrm{GL}(n, F)$.

FULL MEASURE: In mathematics, as modifier, of a set in a measure space, having a set of null measure as complement.

FULL MOON: The lunar phase that occurs when the Moon is on the opposite side of the Earth from the Sun. It occurs when the geocentric apparent (ecliptic) longitudes of the Sun and Moon differ by 180 degrees; the Moon is then in opposition with the Sun. As seen from Earth, the hemisphere of the Moon that is facing the Earth (the near side) is almost fully illuminated by the Sun and appears round.

FULL RANK: In mathematics, as modifier, of a matrix, having maximum possible rank. In other words, having rank equal to the number of rows or the number of columns, whichever is less.

FULL-DUPLEX (DUPLEX): In telecommunication, it is a type of communication in which data can flow two ways at the same time. **Full duplex** devices, such as telephones can communicate back and forth simultaneously. Most network protocols are duplex, enabling hardware devices to send data back and forth simultaneously. For example, two computers connected via an Ethernet cable can send and receive data at the same time. Wireless networks also support full-duplex communication.

FULL-MOTION VIDEO (FMV): Any digital image that is transmitted or stored on a video disk for display on a computer or other multimedia system at a rate of not less than 25 frames per second.

FULL-MOUTHED: An animal which has a complete set of permanent teeth.

FULL-WAVE RECTIFIER: A double-element rectifier that provides full-wave rectification; one element functions during positive half cycles and the other during negative half cycles. Rectifiers may be made of solid state diodes, vacuum tube diodes, mercury arc valves and other components.

FULLER'S EARTH: A soft, greenish-grey absorbent rock resembling clay, but without clay's plasticity that decolourises mineral and vegetable oils and has high sorbent capacity for water and oil. It was formerly used in fulling of wool to remove oil and grease but is now used chiefly in bleaching and clarifying petroleum and secondarily in refining edible oils.

FULLERENE: Any of a class of all-carbon molecules whose atoms are arranged in closed hollow shells, discovered in 1985. It is made up of up to 500 carbon atoms arranged in a sphere or tube. The molecules of the most symmetrical of the fullerenes are called buckminster fullerenes and have 60 carbon atoms arranged in a pattern like that on a standard soccer ball.

FULLERING: The groove on the lower surface of a horse's shoes, into which the heads of the nails will fit.

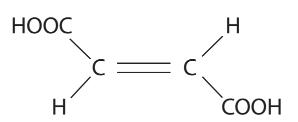
FULLERITE: Any of a class of all-carbon molecules whose atoms are arranged in closed hollow shells. The best known and most stable fullerene, buckminsterfullerene (C_{60}, nicknamed buckyball), has 60 carbon atoms arranged in a pattern like that on a standard soccer ball. It is named after R. Buckminster Fuller, whose geodesic dome designs its structure resembles. It was first prepared in 1985 by Harold Kroto, James Heath, Sean O'Brien, Robert Curl and Richard Smalley at Rice University. Kroto, Curl and Smalley were awarded the 1996 Nobel Prize in Chemistry for their roles in the discovery of buckminsterfullerene and the related class of molecules, the fullerenes.

FULMINATES: Any salt of fulminic (cyanic) acid (HOCN), with the chief ores being silver and mercury. It is a volatile, poisonous, explosive liquid, which is isomeric with cyanic acid. They are readily explosive when heated or struck and used as an explosive. An example is fulminate of mercury, $Hg(ONC)_2$.

FULVIC ACID: A major organic acid found in humus obtained from soil, sediment, peat or aquatic environments. It is a colloid that diffuses easily through membranes whereas all other colloids do not. A related acid, humic acid, is of higher molecular weight than fulvic acid. Both are commonly used as a soil supplement in agriculture, and also as a human nutritional supplement.

FULVOUS: A colour described as reddish-yellow, yellowish brown or tawny.

FUMARIC ACID (TRANS-BUTENEDIOIC ACID; 2-BUTENEDIOIC ACID; ALLOMALEIC ACID; LICHENIC ACID; BOLETIC ACID): A colourless crystalline fatty acid, with the molecular formula $C_4H_4O_4$, relative density 1.6, melting point 185 °C (458 K) and boiling point 235 °C (508 K). It is a weak carboxylic acid, prepared by catalytic oxidation of benzene or by bacterial action on glucose. It is also found in small amounts in a variety of plants and is essential to the respiration of animal and vegetable tissue. It occurs in living organisms as an intermediate in metabolism, especially in the Krebs cycle (tricarboxylic acid cycle). Fumaric acid is used as a substitute for tartaric acid in beverages and baking powder. It is also used as a mordant in dyeing and in the manufacture of synthetic resins and polyhydric alcohols.



Chemical structure of Fumaric Acid

FUME CHAMBER: An enclosed and ventilated chamber in which reactions which involve toxic by-products are performed to prevent direct contact with them.

FUMES: Fine solid particles (aerosol), predominantly less than 1 μm in diameter, which result from the condensation of vapour from some types of chemical reaction. It is usually formed from the gaseous state generally after volatilisation from melted substances and often accompanied by chemical reactions such as oxidation. Fume also describes gas, smoke or vapour that smells strongly or is dangerous to inhale.

FUMIGANT: A pesticidal chemical or chemical formulation that functions in a gaseous state and gives off fumes, especially one used as a disinfectant or to kill pests. They are designed to increase toxicity, reduce flammability, give off warning odours and provide for sorption at different rates. Types of fumigants include halogenated hydrocarbons, such as carbon tetrachloride and ethylene dibromide; sulfur-containing compounds, such as carbon disulfide and sulfur dioxide; cyanides, such as hydrogen cyanide and calcium cyanide.

FUMIGATE: To treat something with fumes or be treated with fumes, especially to kill microorganisms or pests.

FUMIGATION: 1. An atmospheric phenomenon in which pollution, retained by an inversion layer near its level of emission, is brought rapidly to ground level as the inversion breaks up. 2. The process of removing unwanted pests such as insects from an environment by the application of smoke, vapour or gas. Fumigation may be used to remove pests from grains, soil or foods or from items being shipped, from a building environment, etc. The most widely used chemicals include sulfuryl fluoride, chloropicrin, methyl bromide and telone (dichloropropene).

FUMITORY: A common name for a family of flowering sprawling herbaceous plant with deeply divided leaves and acrid-smelling roots with pink or white, pouch-shaped flowers belonging to the family Fumariaceae. Fumitory is classified in the division Magnoliophyta, class Magnoliopsida order Papaverales.

FUNCTION: 1. The natural or usual duty or purpose of a person or something or the way something works or operates. 2. Function rule is a rule that maps each element in a given set (the domain) onto just one element (or image) in another set (the range) or any operation or procedure that relates one variable to one or more other variables. If y is a function of x, written $y = f(x)$, a change in x produces a change in y and if x is known, y can be determined. 3. A small part of a program in a computer that supplies a specific value. An example is the square root of a specified value or the current date. 4. A characteristic role or action of a structure or process in the normal metabolism or behaviour of an organism.

FUNCTION KEY: Any of a sequence of keys, usually labelled "F1" up to "F12" on most keyboards when pressed alone or in combination with a second key such as the shift key, alt key or control key performs a specific task, such as ending a computer program.

FUNCTION MACHINE: A diagram that represents a machine that takes an input, applies a rule such as a set of operations and delivers the answer as an output.

FUNCTIONAL: 1. A term applied to certain scalar-valued functions in mathematics, computer science and Lagrangian mechanics. In mathematics, it is a function whose domain is itself a set of functions and whose range is another set of functions that may be numerical constants. The term is used for linear functional, a type of functional often simply called a functional in the context of functional analysis. 2. Functional group is certain atomic combinations that occur in various molecules, for example, carbonyl and amine. 3. Functional model is a structured representation of functions, activities or processes in systems and software.

FUNCTIONAL ANALYSIS: 1. A branch of modern mathematical analysis that studies linear and non-linear functions in terms of the underlying linear spaces on which the functions are defined and the duals of those spaces. Its areas of study include algebra, real analysis, numerical analysis, calculus of variations and differential equations, through the application of general theorems such as the Hahn-Banach theorem, the uniform boundedness principle, the open mapping theorem and the Riesz representation theorem. 2. A method that is used to explain the workings of a complex system as computing a function or generally, as solving an information processing problem such as addressing the activities that a system, a software or an organisation must perform to achieve its desired outputs.

FUNCTIONAL CALCULUS: 1. The branch of mathematics that studies the properties of functions and operations upon functions. 2. Any theory that guarantees that it is possible to mimic familiar constructions on the complex numbers with matrices or operators, for example, with the existence of square roots of operators.

FUNCTIONAL DETERMINANT: The determinant of the differential coefficients of n functions in n variables.

FUNCTIONAL EQUATIONS: 1. The branch of mathematics that studies equations in which the variables are functions and that attempts to establish the properties of functions from the equations that they satisfy. 2. The study of equations of the form A = 0, where A is a term containing a finite number of independent variables, a finite number of known functions and a finite number of unknown functions. 3. **Functional equation** is an equation of this form $f(x + y) = f(x) + f(y)$, the condition for a function to be additive.

FUNCTIONAL FOOD (MEDICINAL FOOD; NUTRACEUTICAL): A food designed to be medically beneficial or to have a health-promoting or disease-preventing property beyond the basic function of supplying nutrients. It helps to protect against serious conditions such as diabetes, cancer or heart disease.

FUNCTIONAL GROUP: An atom present in a molecule which is responsible for the characteristic properties of that molecule or any of numerous combinations of atoms that undergo characteristic chemical reactions themselves and in many cases influence the reactivity of the rest of the molecule. Common functional groups include hydroxyl (-OH) in alcohols and phenols; carboxyl (-COOH) in carboxylic acids; carbonyl (-C=O) in aldehydes, ketones, amides, carboxylic acids, esters and quinones; and nitro (-NO_2) and amino (-NH_2) in certain organic nitrogen compounds.

FUNCTIONAL PARENT: A structure whose name implies the presence of one or more characteristic groups and which has one or more hydrogen atoms attached to at least one of its skeletal atoms or one of its characteristic groups or in which at least one of its characteristic groups can form at least one kind of functional modification. A parent hydride bearing a characteristic group denoted by a suffix, for example, cyclohexanol, is not considered to be a functional parent, but may be described as a 'functionalised parent hydride'.

FUNCTIONAL PROGRAMMING: In computing, it is a programming method that performs a definite service characterised by values and functions rather than by the execution of commands. Examples of functional programming languages are ML (MetaLanguage) and Erlang (ErlangLanguage).

FUNCTIONAL RESPONSE: In ecology, it is the relationship between prey and predator, or the number of prey successfully attacked per predator as a function of prey density. It is the deviation in the rate of exploitation of prey by an individual predator due to change in prey density.

FUNCTION SPACE: A topological vector space including metric spaces whose elements are functions, especially continuous or bounded functions. For example, C[0, 1] is the set of functions continuous on the interval [0,1].

FUNCTIONAL TESTING: A form of testing based upon typical extreme and invalid data values that are representative of those covered by the specification. It is a software testing process done to ensure that a developed software conforms to all functional requirements.

FUNCTIONAL TRAIT: Any characteristic of an organism that influences its performance, for example, growth rate and competitive ability and so ultimately affects its ability to survive and reproduce. For example, in plants large seeds provision robust seedlings able to succeed where competition is intense and resources are scarce, but can only be produced in relatively small numbers. In contrast, tiny seeds fail where competition is intense, but can be produced in copious numbers that increase the chance of dispersing to ephemeral sites with few competitors and high resource availability.

FUNCTIONALITY: The number of covalent bonds that a monomer molecule or monomeric unit in a macromolecule or oligomer molecule can form with other reactants.

FUNCTOR: A function between categories that maps objects to objects and morphisms to corresponding morphisms.

FUNDAMENTAL CONSTANTS (UNIVERSAL CONSTANTS; ATOMIC CONSTANTS): The parameters that do not change throughout the universe or the physical constants which play a fundamental role in the basic theories of physics. Examples include the charge of an electron, the speed of light in free space and the gravitational constant, Planck's constant.

FUNDAMENTAL COUNTING PRINCIPLE (COUNTING RULE; MULTIPLICATION COUNTING PRINCIPLE): A way of determining the total number of possible outcomes for two or more separate choices. It is determined by multiplying the events together to get the total number of outcomes. For example, if the first event has 3 choices of occurring, the second event can occur 4 ways, and the third event can occur 5 ways, then you can find out the number of unique combinations by multiplying: 3 × 4 × 5 = 60 unique combinations.

FUNDAMENTAL FORCE: In physics, any of the four most basic ways in which subatomic particles interact in the physical universe. They include the strong, weak, electromagnetic and gravitational interactions. The gravitational force or gravity keeps the planets in orbit around the Sun and acts between all particles that have mass; the electromagnetic force stops solids from falling apart and acts between all particles with electric charge. The weak force is responsible for the reactions that fuel the Sun and for the emission of beta particles by some particles; the strong nuclear force binds together the protons and neutrons in the nuclei of atoms.

FUNDAMENTAL FORM: **First fundamental form** is the quadratic equation of the form $ds^2 = Adu^2 + 2Bdudv + Cdv^2$, where $A = \left(\frac{\partial x}{\partial u}\right)^2, B = \left(\frac{\partial x}{\partial u}\right)\left(\frac{\partial x^2}{\partial v}\right), \text{and } C = \left(\frac{\partial x^2}{\partial v}\right)^2$ that determines the metric and arc length on a given surface. The **second fundamental form** is quadratic equation of the form $Ddu^2 + 2D'dudv + D''dv^2$, where $D = \sum_i x_i \frac{\partial^2 x_i}{\partial u^2}, D' = \sum_i x_i \frac{\partial^2 x_i}{\partial u \partial v} \text{ and } D'' = \sum_i x_i \frac{\partial^2 x_i}{\partial v^2}$, with x_i is the direction cosine of the normal to the surface. This enables points of the surface to be classified as flat elliptic, parabolic or hyperbolic.

FUNDAMENTAL FREQUENCY: A term used in physics to describe a component of lowest frequency of a periodic wave or quantity; or relating to the lowest possible frequency of a vibrating element or system.

FUNDAMENTAL INTERACTIONS (INTERACTIVE FORCES; FUNDAMENTAL FORCES): The effect of any of the four different types of forces that can occur between bodies. These are gravitational, electromagnetic and strong and weak interactions. The **weak interaction** are short-range, the range of the force being of the order of 10^{-18} m is responsible for the beta decay of particles and nuclei. The **strong interaction**, which is 10^2 times stronger than the electromagnetic interactions, binds quarks together in protons, neutrons and other hadrons and also holds the protons and neutrons of an atomic nucleus together, overcoming the repulsion of the positively charged protons for each other. The **gravitational interaction**, which is 10^{40} times weaker than the electromagnetic interaction is the weakest and it is the attractive force between any two objects that have mass. It causes objects to fall to the ground and maintains the orbits of planets around the Sun. The strength of the force falls off with the square of the distance between the interacting bodies. The electromagnetic interaction is responsible for the forces that control atomic structure, chemical reactions for the forces between charged particles. It can be either attractive or repulsive. Some neutral particles decay by electromagnetic interactions. These fundamental interactions can take place even when the bodies are not in physical contact and together they account for all the observed forces that occur in the universe.

FUNDAMENTAL MATRIX: 1. The matrix whose columns are the linearly independent fundamental sets of solutions of a linear homogeneous system of ordinary differential equations or matrix equation, $y' = A(t)y$, where y has dimension n. 2. In computer vision, a 3×3 matrix that relates corresponding points in stereo images. It is a relationship between any two images of the same scene that constrains where the projection of points from the scene can occur in both images.

FUNDAMENTAL PARTICLE: It is a term for elementary particle, which is the basic indivisible constituent of matter and anyone of the subatomic constituents of which matter and energy are made of. It includes electrons, photons, leptons and hadrons.

FUNDAMENTAL QUANTITY: One of the quantities that in a system of quantities are conventionally accepted as functionally independent of one another. It is a quantity whose magnitude is defined without reference to other quantities. In the SI, these are length, mass, time, electric current, temperature, amount of substance and luminous intensity. All other physical quantities (and units) are regarded as being derived from these base quantities and base units.

FUNDAMENTAL SEQUENCE (CAUCHY SEQUENCE): An infinite sequence of points or values whose distances between them tend to zero as their indices tend to infinity; $\{a_i\}$ is a Cauchy sequence in a metric space if, for every □ > 0, there is an N such that $d(a_i, a_j) <$ □ for all $ij > N$. An example of Cauchy sequence is $\{1/n\}$.

FUNDAMENTAL SET OF SOLUTIONS: Any basis for the vector space of all solutions of a homogeneous system of linear equations. Such a treatment is possible since every linear combination of solutions of the system of equations is also a solution.

FUNDAMENTAL SYSTEM OF SOLUTIONS: Any set of n linearly independent solutions of a homogeneous linear ordinary differential equation of order n. A set of n solutions is a fundamental system if and only if their Wronskian is non-zero. The general solution of the differential equation is a linear combination of any fundamental set of solutions.

FUNDAMENTAL THEOREM OF ALGEBRA: The theorem states that every polynomial equation having complex coefficients and degree ≥ 1 has at least one complex root. This theorem was first proven by Gauss. It is equivalent to the statement that a polynomial $P(z)$ of degree n has n values z_i (some of them possibly degenerate) for which $P(z_i) = 0$. Such values are called polynomial roots. An example of a polynomial with a single root of multiplicity > 1 is $z^2 - 2z + 1 = (z - 1)(z - 1)$, which has $z = 1$ as a root of multiplicity 2.

FUNDAMENTAL THEOREM OF ARITHMETIC (UNIQUE FACTORISATION THEOREM): The theorem stating that any positive integer can be expressed as a unique product of prime numbers.

FUNDAMENTAL THEOREM OF CALCULUS: A theorem that links the concept of the derivative of a function with the concept of the integral. The first part of the theorem, sometimes called the **first fundamental theorem of calculus** is that an indefinite integration can be reversed by a differentiation. This part of the theorem is also important because it guarantees the existence of antiderivatives for continuous functions. The second part, sometimes called **the second fundamental theorem of calculus** states that the definite integral of a function can be computed by using any one of its infinitely many antiderivatives. This part of the theorem has key practical applications because it markedly simplifies the computation of definite integrals.

FUNDAMENTAL THEOREM OF PROJECTIVITY: The theorem that three distinct corresponding pairs of points uniquely determine a projectivity.

FUNDAMENTAL UNITS: A fundamental unit within a system of measurement from which other units in the system are derived. Such a set requires three mechanical units, usually of lengths, mass and time and one electrical unit.

FUNDAMENTAL VIBRATION: In a vibrating object such as a stretched string or air column, it is the vibration that produces the lowest resonant frequency (lowest-pitched note). This frequency is called its **fundamental frequency**. The fundamental vibration of a string has a stationary node at each end and a single antinode at the centre where the amplitude of vibration is greatest.

FUNGAL DISEASE (PATHOGENIC FUNGI): Diseases caused by the growth of fungi on or in the body or fungi that cause disease in humans or other organisms. In most healthy people these are mild, involving only the hair, skin, nails or other superficial sites. They include the familiar athlete's footrot and ringworm.

FUNGAL ENDOPHYTES: Microscopic plants growing within plant tissue.

FUNGI: Relatively simple organisms which lack chlorophyll and are not photosynthetic. Most of them have filaments called **hyphae** and they reproduce by spores. Examples are mushrooms and bread mould (*Rhizopus*).

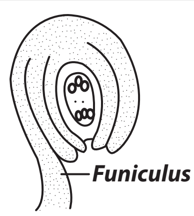
FUNGI IMPERFECTI (IMPERFECT FUNGI; DEUTEROMYCOTA): A class of fungi which do not fit into the commonly established taxonomic classifications of fungi, in which the asexual reproduction is known. They are generally similar to ascomycetes and are consequently thought to be ascomycetes that have lost the ability to produce ascospores.

FUNGICIDE: A chemical substance used for destroying fungus or preventing fungal growth. Fungicides are used both in agriculture and to fight fungal infections in animals.

FUNGISTASIS: Suppression of the growth of new fungal cells without destroying them. This may be caused by excessive competition for nutrients, or due to the presence of excessive inhibitory compounds in the soil.

FUNGIVOROUS (MYCOPHAGOUS): Organisms that feed on fungi. Many different organisms feed on fungi, for example, birds, mammals, insects, plants, amoebas, gastropods, nematodes, and bacteria. Some derive energy for their metabolism from consuming only fungi; whilst some others eat it only sometimes.

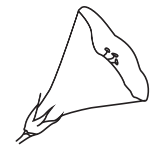
FUNICLE: 1. The stalk that attaches an ovule to the placenta in the ovary of a flowering plant. 2. A cord-shaped part of the body, for example, the umbilical cord or a bundle of nerve fibres in the spinal cord. 3. The segment connecting the club with the base in insect antennae.



FUNICULUS: The stalk of a seed by which the ovule is attached to placenta.

FUNNEL: 1. A tube or pipe with a wide, often conical mouth and a narrow stem used to transfer liquids or solids into containers with small openings or a cone-shaped utensil with a large opening at the top and a small opening or tube at the bottom used to guide liquids and other substances into containers. 2. A vertical pipe or the chimney or smokestack from which smoke and exhaust gases escape, especially one on a steamship or steam engine.

FUNNELFORM: Having the shape of a funnel. Gradually widening from base to apex, as shown in the diagram.



Funnelform flower

FUNTUMIA: A genus of species of tropical trees in the family Apocynaceae in the major group Angiosperms (Flowering plants), which grow between 15 and 35 m tall. The well known ones are *Funtumia Africana* commonly known as bastard wild rubber, bush rubber, Lagos rubber and male funtum; *Funtumia elastica* with the common name Silkrubber tree; and *Funtumia latifolia*. *Funtumia Africana* and *Funtumia elastica* are very similar morphologically and in distribution. *Funtumia elastica* and *Funtumia Africana* have a straight and cylindrical bole and narrow crown. The bark is thin, dark grey to almost black, glabrous, slightly yellowish to brown, astringent with large amount of white to creamy latex. When *Funtumia elastica* latex is rubbed between the fingers the latex coagulates into balls, whereas *Funtumia africana* does not. In the case of *Funtumia Africana* the latex is sticky. The leaves are opposite and very similar in shape in all species, which is elliptic to elongated elliptic, tapering to the apex or shortly acuminate at apex, shortly cuneate at base, leathery, dark green glossy above, with 8 -12 pairs of rather straight lateral nerves, reaching the margin. *Funtumia elastica* has acorn-like protuberances in the axils of the nerves beneath. In *Funtumia elastica,* the flowers are very small with lobes of 4-5 mm length and about half as long as the tube, the calyx forming a cup with well rounded overlapping lobes. In *Funtumia Africana* the lobes are more than twice the size as those in *Funtumia elastica*. The flower buds in *Funtumia Africana* are more or less cylindrical; in *Funtumia elastica* they are sharply conical and acute. The fruit of *Funtumia Africana* is grey-brown, fusiform and an acute or acuminate apex with hairy wind borne seeds. In *Funtumia elastica*, the fruits are smaller and thicker, widening upwards and not markedly tapering towards the apex but blunt or almost rounded and lasting until after the new flowers. The seeds of both species are at one side prolonged into a slender stalk with white silky hairs along the beak in the direction of the seed. The bark of *Funtumia elastica* is used traditionally as medicine for the treatment of anus haemorrhoids; respiratory ailments including asthma, menstrual cycle problems; skin diseases; mucosae; venereal diseases; vermifuges; etc. The leaves are also used for the treatment of diarrhoea, dysentery; genital stimulants/depressants; and naso-pharyngeal affections. *Funtumia africana* is used to treat urinary incontinence and burns. The leaf and bark are used as enema. The principle alkaloids of *Funtumia africana*, funtumine and funtumidine are hypotensive. The wood is white, even textured and is used for cheap joinery, furniture and matchstick manufacture.

FUR: 1. Soft thin hair that grows densely over the body of some mammals such as bears, rabbits and cats. 2. An unhealthy, greyish covering on the tongue. 3. A hard covering on the inside of objects such as pots and pipes, formed by calcium carbonate in heated water.

FURAN (FURFURAN; DIVINYLENE OXIDE; 1,4-EPOXY-1,3-BUTADIENE; 1,4-EPOXYBUTA-1,3-DIENE): A heterocyclic organic compound, consisting of a five-membered aromatic ring with four carbon atoms and one oxygen. Its molecular formula is C_4H_4O, density 0.936 g/ml, melting point -85.6 °C (187.55 K), boiling point 31.3 °C (304.45 K) and flash point -69 °C (204 K). It is colourless, flammable and highly volatile liquid. It is toxic and may be carcinogenic.

FURANOSE: A simple sugar having a five-membered ring such as tetrahydrofuran skeleton containing four carbon atoms and one oxygen atom.

FURCATE: Forked or divided into branches.

FURFURACEOUS: 1. Covered with scale; flaky. 2. Looking like a bran.

FURFURAL (FURFURALDEHYDE; FURAN-2-CARBALDEHYDE; FURFURAL; FURAN-2-CARBOXALDEHYDE; FURAL; FURFURALDEHYDE; 2-FURALDEHYDE; PYROMUCIC ALDEHYDE): A colourless, viscous liquid with a pleasant aromatic odour, with the molecular formula $C_5H_4O_2$, density 1.16 g/ml (20 °C), melting point -37 °C (236 K), boiling point 162 °C (435 K) and flash point 62 °C (335 K), which darkens on standing in air. It is soluble in ethanol and ether and somewhat soluble in water. It is prepared commercially by dehydration of pentose sugars obtained from cornstalks and corncobs, husks of oat and peanut and other waste products. It is the aldehyde derivative of furan and occurs in various essential oils and in fusel oil. It is used as a solvent for extracting mineral oils and natural resins and by itself forms resins with some aromatic compounds. It is also used in the manufacture of pesticides, phenolfurfural resins and tetrahydrofuran.

FURLONG: A unit of measurement in the imperial and U.S. customary units, which is 220 yards (660 feet) and equivalent to 201.168 metres.

FURNACE: An enclosed structure in which fuel such as coal, coke, gas or oil is burnt to produce heat for various purposes. Examples are the hot-air furnace and the hot-water furnaces.

FURNACE BLACK: A type of carbon that is produced industrially in a furnace by incomplete combustion in an adjustable and controllable process that yields a wide variety of properties within the product. The most widely employed industrial process for carbon black production is the furnace process.

FUROCOUMARINS: Photosensitising substances which occur normally in some plants such as *Ammi majus* and *Cymopterus watsonii* and in others only if they are infected with bacteria or fungi. The furocoumarins are endogenous photodynamic agents which have the capacity to cause photosensitive kerato conjunctivitis.

FURROW IRRIGATION: A type of surface irrigation in which water is allowed to flow along small, parallel ditches or channels (called furrows) dug between crop rows in a field. This type of irrigation is suitable for row crops such as maize, sunflower, sugarcane, soybean; it is also suitable for crops like tomatoes, vegetables, potatoes, beans as the water does not touch them directly to flood them.

FURROW PRESS: A special type of very heavy ring roller attached to a plough and used to press the furrow slices.

FURROW SLICE: The soil which is displaced by the mould board of a plough when it creates a furrow.

FURROWS: Grooves made in the soil or a narrow trench in soil made by a plough.

FURZE (GORSE): A common shrub, belonging to the subfamily Papilionoideae of the family Fabaceae, found in wasteland and formerly often cut and used as fodder after chaffing. It is native to the U.K. and Western Europe. It is a thorny evergreen shrub with sharply pointed leaves and solitary flowers having a two-lipped, shaggy, deep yellow calyx subtended by two small bracts. It contains a small amount of a poisonous alkaloid called ulexine.

FUSARIUM EAR BLIGHT, FEB (FUSARIUM HEAD BLIGHT, FHB, SCAB): A serious fungal disease of wheat and oats that can cause significant loss in yield and quality and also represents a serious threat to human health, because it is responsible for dangerous mycotoxin-infected grain and food items.

FUSCOUS: Dark, brownish grey colour.

FUSE: 1. A length of wire made of a metal alloy with low melting point that is designed to melt at a specified current loading to break a circuit in order to protect an electrical device or circuit from overloading. The components of a typical low-voltage high-power fuse are a fuse element or wire, an insulating material support and housing, two metal end fittings and a filler. 2. A mechanical or electrical mechanism used to detonate an explosive charge or device such as a bomb or grenade.

FUSEL OIL: An oily, colourless liquid with a characteristic unpleasant smell, obtained as a by-product of the distillation of any alcoholic fermentation and used in paints, varnishes, essential oils and plastics. It has a detrimental effect on the human system.

FUSIBLE ALLOYS: Alloys that melt at relatively low temperatures (around 100 °C, 273 K). They have a number of uses, including constant-temperature baths, pipes bending and automatic sprinkles to provide a spray of water to prevent fires from spreading.

FUSIFORM: Tapering at both ends, like a spindle; broadest in the middle and tapering towards both ends.

FUSION: 1. The process of liquefying or melting by the application of heat, which can join two separate metals together as, for example, in welding. 2. In physics, it is the integration of two or more atomic nuclei to make a more massive one with the simultaneous release of energy. This process can be achieved by using extremely high temperature plasmas, which may eventually lead to a new readily available source of energy.

FUSION REACTOR: A proposed device in which controlled, self-sustaining nuclear fusion reactions would be carried out in order to produce useful power.

FUTURES: Stocks of produce which are bought or sold for shipping at some later date and which may not even have been produced when they are on the market.

FUTURES WHEEL: A graphical decision making tool created in 1971 by Jerome C. Glenn (1945-), a futurist who serves as the Director of the Millennium Project. It is used to organise thoughts about a future development or trend. It uses interconnecting lines to visualise interrelationships of causes and effects resulting changes.

FUZZY LOGIC: A term used in mathematics and computing in which a form of logic allows for imprecise or ambiguous answers to questions, forming the basis of computer programming designed to mimic human intelligence. For example, a jug of water may be described as too hot or too cold, depending on whether it is to be used to wash one's face or to make tea. Variables have degrees of truthfulness or falsehood represented by a range of values between 1 (true) and 0 (false). For example, in addition to being either true or false, an outcome might have such meanings as probably true, possibly true, possibly false and probably false.

G

3GP: As defined by the 3GPP (Third Generation Partnership Project), it is a compression file format for audio, video and other media for use with 3G mobile phones but can also be played on some 2G and 4G phones. There are two standards for this format: 3GPP, with the file extension .3gp and 3GPP2, with the file extension .3g2.

3G: Abbreviation for **third generation**, which is the third generation of mobile telecommunications technology and offers higher data transfer rates, enabling users to utilise full-motion video and other higher bandwidth applications.

3GPP (THIRD GENERATION PARTNERSHIP PROJECT): A collaboration between groups of telecommunications associations, known as the Organisational Partners. The initial scope of 3GPP was to make a globally applicable third-generation (3G) mobile phone system specification based on evolved Global System for Mobile Communications (GSM) specifications within the scope of the International Mobile Telecommunications-2000 project of the International Telecommunication Union (ITU).

4G: Abbreviation for **fourth generation** of mobile telecommunications technology that offers IP based data and voice transfer at much higher speeds than 3G.

-GON: A suffix that denotes a polygon, for example, a 6-gon is a polygon with six sides known as a hexagon.

g: 1. The symbol for the gravitational constant. 2. The symbol for gram (g).

G: G: 1. The symbol used to denote that a substance is a gas. For example, $CO_{2(g)}$, $O_{2(g)}$, $H_{2(g)}$. 2. The symbol for electricity conductance, a measure of a material's ability to conduct electric charge; the reciprocal of the resistance. 3. The symbol for guanine, purine base ($C_5H_5ON_5$) that is an essential constituent of both RNA and DNA. 4. The symbol for gauss, centimetre-gram-second (cgs) unit for magnetic flux density, which is equal to one maxwell per square centimetre. The cgs system has been formally superseded by the SI system, which uses the tesla (T) as the unit for magnetic flux density **(**B**)**. One gauss equals 1×10^{-4} tesla (100 μT). 5. Symbol for giga in the International System of Units (SI). One giga represents 10^9. It is used as a prefix, for example, one gigabyte with the symbol GB is 1 000 000 000 bytes.

G FACTOR (GENERAL FACTOR): The number of specified chemical events in an irradiated substance produced or destroyed for each 100 electronvolts absorbed by a substance from ionising radiation.

G PROTEIN (GUANINE NUCLEOTIDE-BINDING PROTEINS): Any one of a group of proteins involved in transmitting chemical signals outside the cell and causing changes inside the cell. They function as molecular switches and transmit signals from many hormones, neurotransmitters and other signalling factors on the outer surface of the cell to adenylate cyclase, which catalyses the formation of the second messenger, cyclic AMP, inside the cell. G proteins bind to guanosine triphosphate (GTP) and guanosine diphosphate (GDP). When they are attached to a complex with three phosphate groups (GTP), they turn on and when attached to a complex with only two phosphate groups (GDP), they turn off. The cholera toxin exerts its effects by changing the G protein in the epithelial cells of the intestine so that it is continually activated, which causes an abnormal increase in cellular adenylate cyclase levels. One consequence of this is that sodium ions are actively pumped into the intestine, causing water to follow by osmosis, resulting in diarrhoea and dehydration.

G PROTEIN COUPLED RECEPTOR, GPCR (SEVEN-TRANSMEMBRANE DOMAIN RECEPTORS; 7TM RECEPTORS; HEPTAHELICAL RECEPTORS; SERPENTINE RECEPTOR; G PROTEIN-LINKED RECEPTORS; GPLR): Any one of a large family of proteins that are located in cell membranes and act as metabotrobic receptors transmitting signals from the exterior to the interior of the cell via associated G proteins, which activate second messengers which regulate the activity of the cell. G protein-coupled receptors are found only in eukaryotes, including yeast, choanoflagellates and animals.

G-DELTA SET (G_δ SET): A set of a topological space that is a countable intersection of open sets expressible as the intersection of countably many open sets. It is the complement of an F-sigma set.

G-FORCE (G FORCE): Gravitational force that pilots and astronauts experience when their craft accelerates or decelerates rapidly. The g-force, on something is its acceleration relative to free-fall. This is due to the vector sum of non-gravitational forces. Accelerations not produced by gravity are termed proper accelerations and cause stresses and strains on objects. Because of these strains, large g-forces may be destructive.

G-SCALE: A scale for measuring force by comparing it with g-force (force due to gravity, g).

G-SUIT: A tight-fitting suit that covers parts of the body below the heart and is designed to retard the flow of blood to the lower body in reaction to acceleration or deceleration. Bladders or other devices are used to inflate and increase body constriction as g-force increases. It is worn by astronauts and high-speed pilots to avoid blackouts during rapid acceleration.

G1 PHASE (GAP PHASE ONE): The first growth phase of the cell cycle, consisting of the portion of interphase before DNA synthesis begins.

G²: The symbol for the goodness-of-fit test statistics based on the likelihood ratio. It is often used when using the log-linear models. This is given by $G^2 = 2\sum O \ln\left(\frac{O}{E}\right)$, where O and E represent the observed and expected frequencies respectively.

G2 PHASE (GAP PHASE TWO): The second growth phase of the cell cycle, consisting of the portion of interphase after DNA synthesis has occurred.

GABBRO: Basic igneous rock formed deep in the Earth's crust. It is a dark medium- or coarse-grained rock that consists primarily of calcium-rich plagioclase feldspar and pyroxene.

GABOR, DENNIS (1900-79): Hungarian-born British physicist and electronics engineer, who worked as a research engineer from 1927 until 1933, when he joined the British Thomson-Houston company. In 1948, he joined the staff of Imperial College, London. In that same year, while working on electron microscopes, he invented holography, for which he was awarded the 1971 Nobel Prize.

GABRIEL REACTION: A procedure of making a primary amine from a haloalkane using potassium phthalamide. It was named after Siegmund Gabriel (1851–1924).

GADFLY: A fly such as the botfly and the horsefly especially of the family Tabanidae that attacks livestock and sucks their blood. They are most common from late May onwards and cause considerable trouble to cattle.

GADOLINIUM: A rare silvery-white metallic element belonging to the lanthanide series of the Periodic Table, which becomes strongly magnetic at low temperatures. Its symbol is Gd, atomic number 64,